

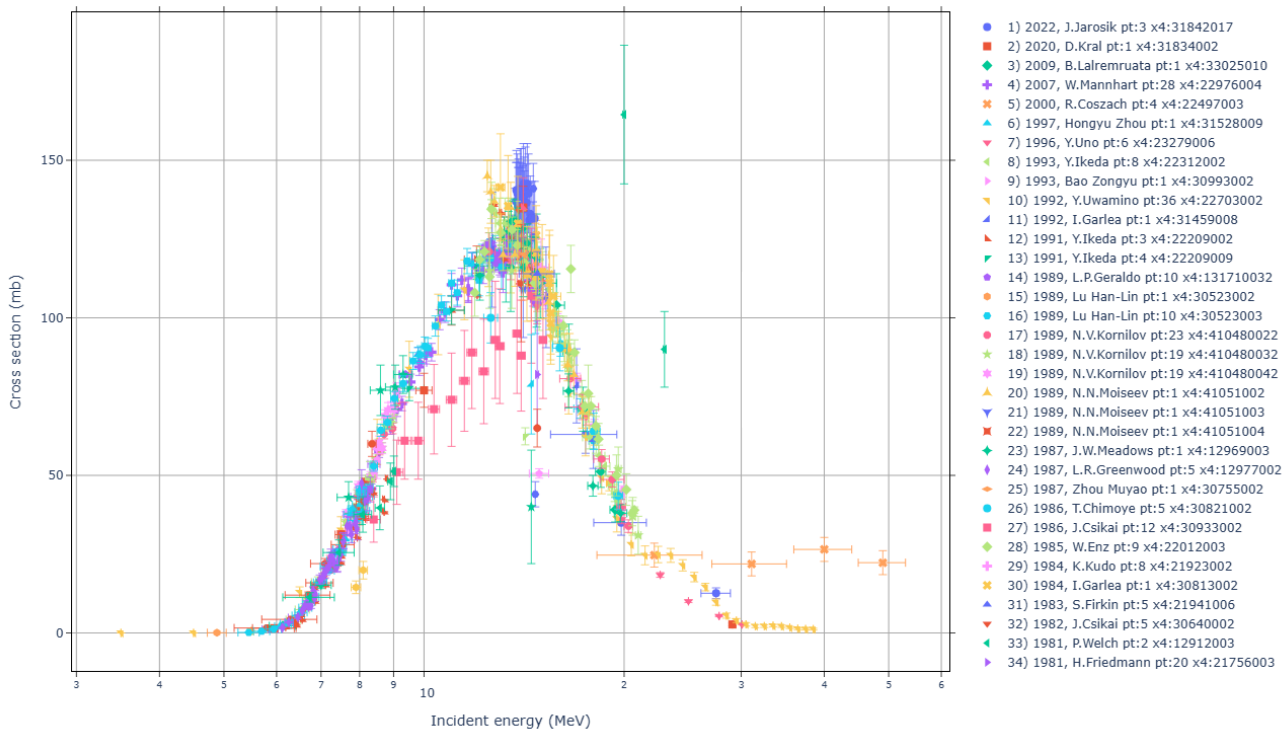
X4Pro. Plotting via Plotly.

Generated: 2026-09-29,18:34:06

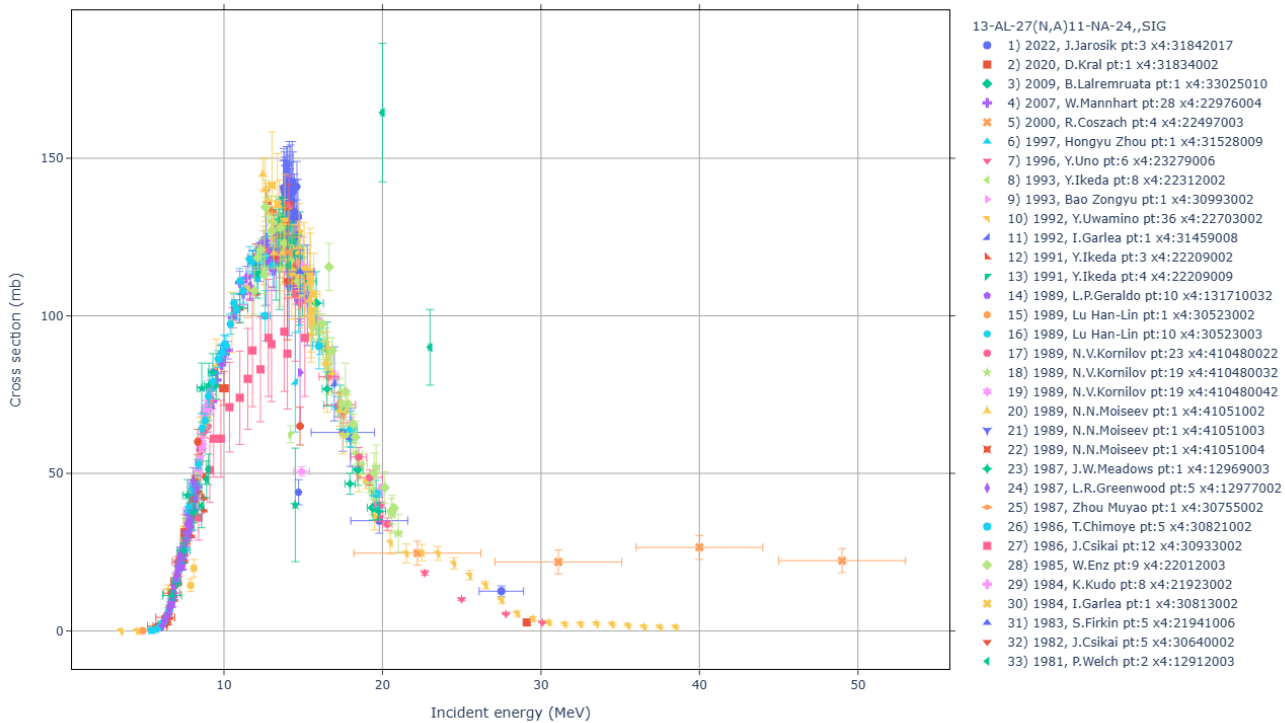
part1-1-sig/ Retrieve and plot EXFOR cross sections (modular Python-program)

1) part1-1-sig/out00/Al27na-1.html.pngEXFOR cross sections $\sigma(E)$: Al-27(n,a),sig

X4Pro, by V.Zerkin, IAEA-NDS, 2021-2022, ver.2022-11-09 //running:2026-09-27 21:39:11

**2) part1-1-sig/out00/Al27na-2.html.png**EXFOR cross sections $\sigma(E)$: Al-27(n,a),sig

X4Pro, by V.Zerkin, IAEA-NDS, 2021-2022, ver.2022-11-09 //running:2026-09-27 21:39:15

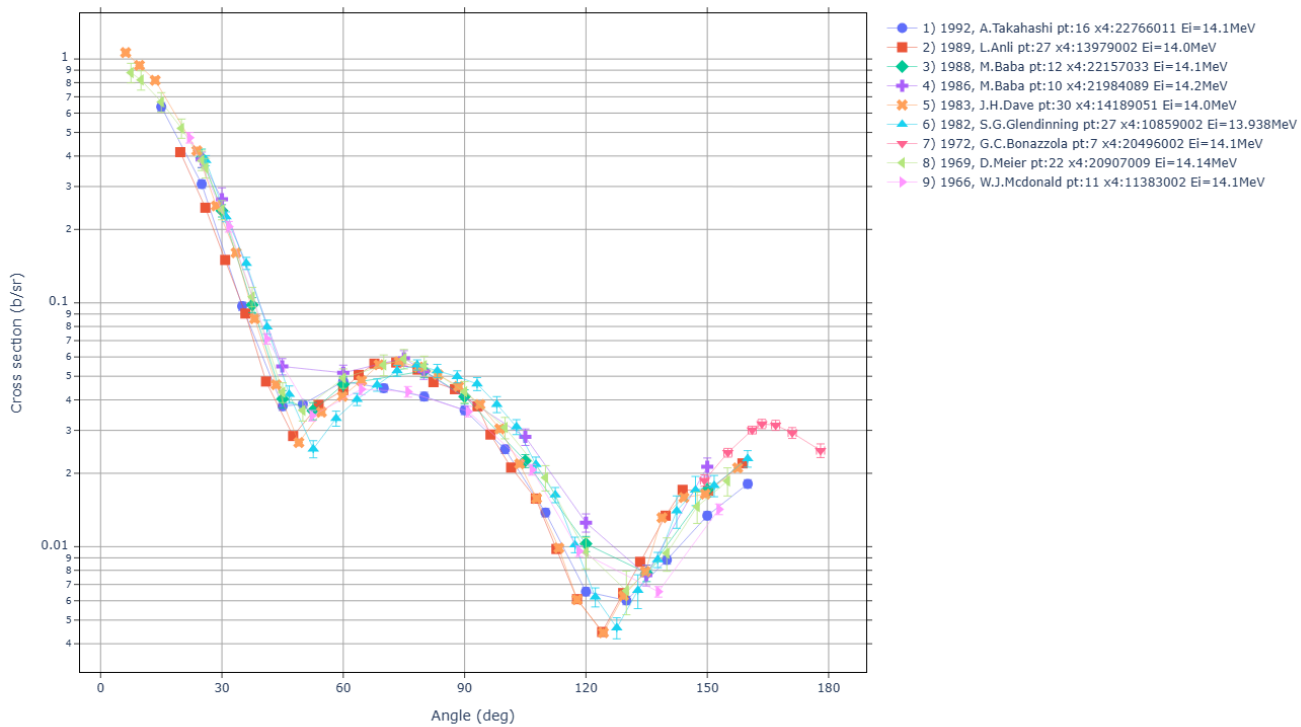


part1-2-da/ EXFOR angular distributions

3) part1-2-da/out00/da0an.html.png

Plot EXFOR angular distributions $d\sigma/d\Omega(E,\theta)$: O-16(n,el),da

X4Pro, by V.Zerkin, IAEA-NDS, 2021-2022, ver.2022-12-20 //running:2026-09-27 21:39:18

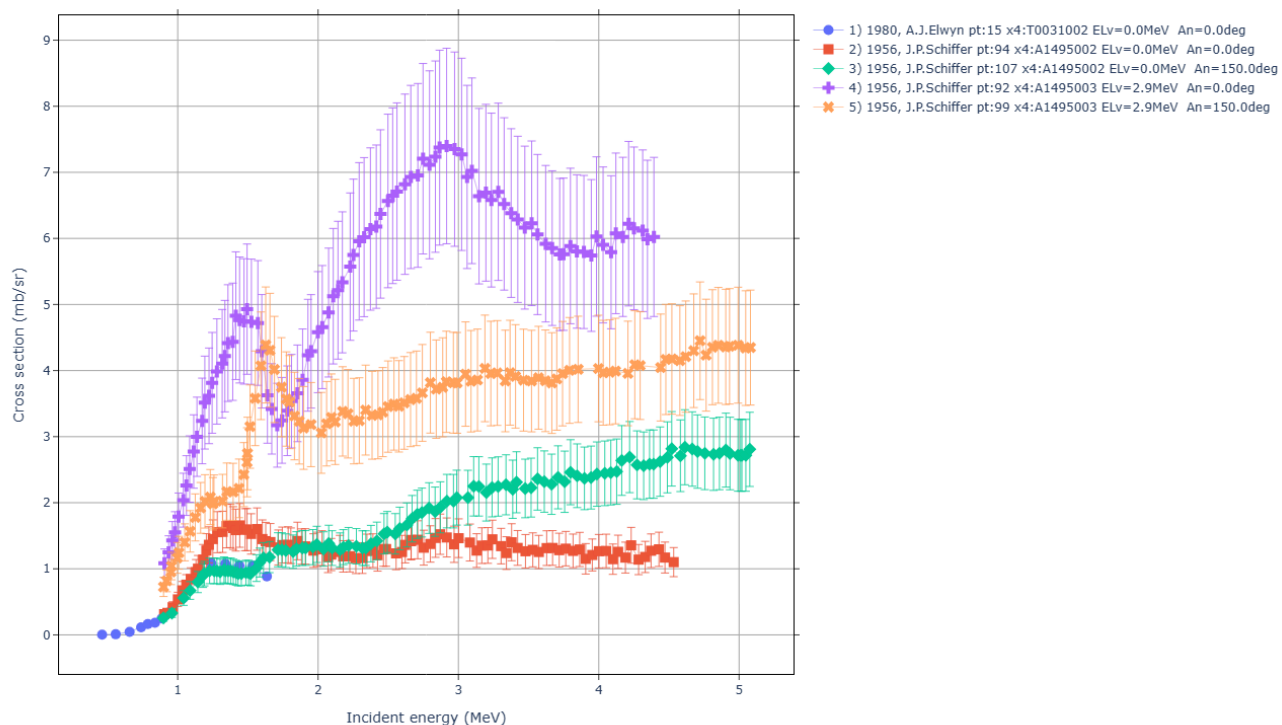


part1-3-dap/ EXFOR partial angular distributions

4) part1-3-dap/out00/dap0ei.html.png

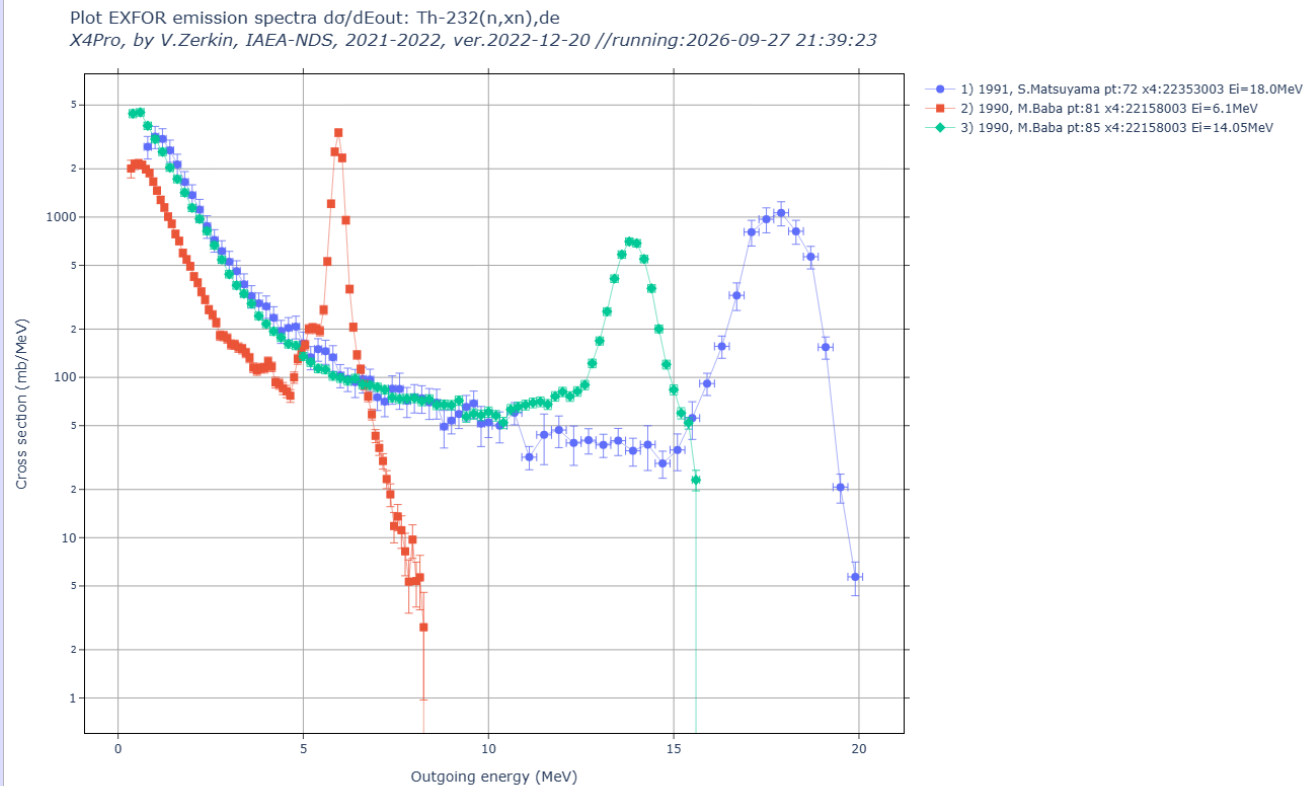
Plot EXFOR angular distributions $d\sigma/d\Omega(E,\theta)$: Li-6(he3,p)par,da

X4Pro, by V.Zerkin, IAEA-NDS, 2021-2022, ver.2022-11-09 //running:2026-09-27 21:39:20



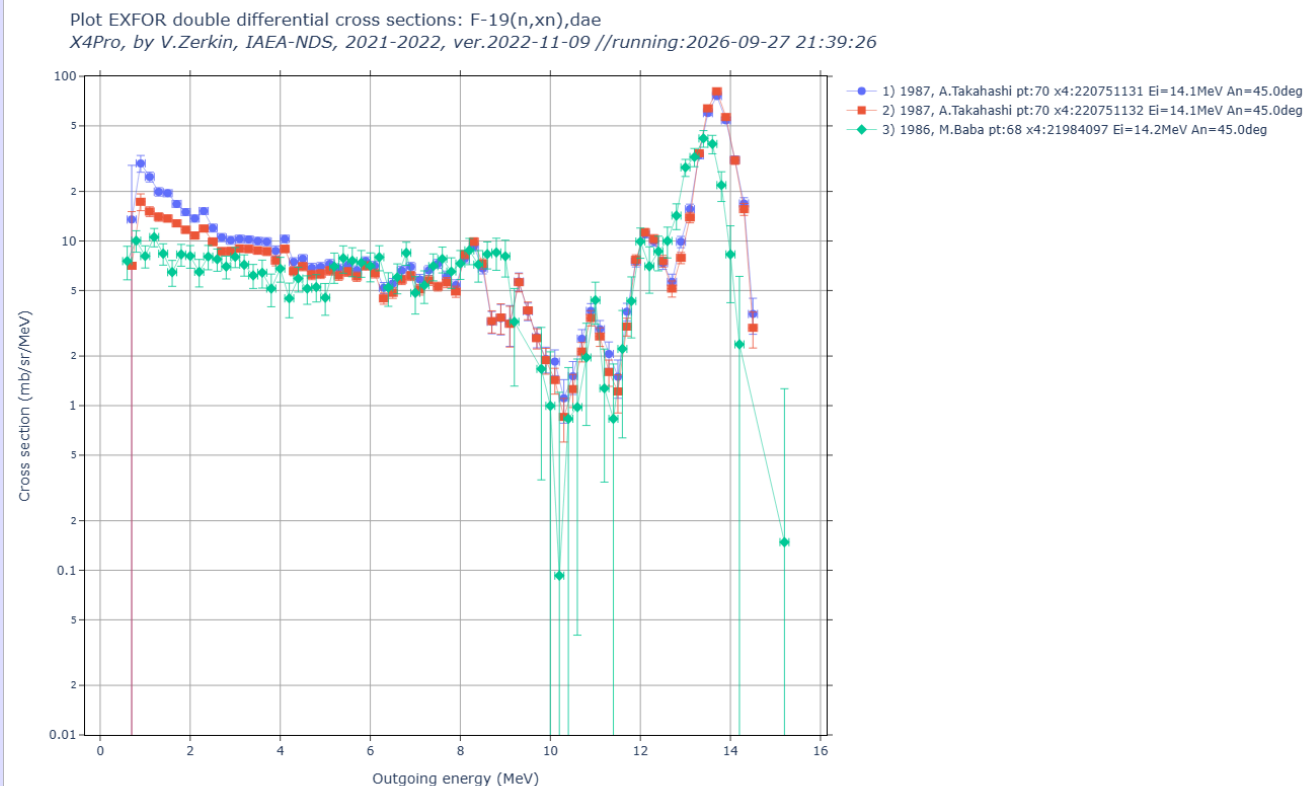
part1-4-de/ EXFOR energy spectra of outgoing particles

5) part1-4-de/out00/de0eo.html.png



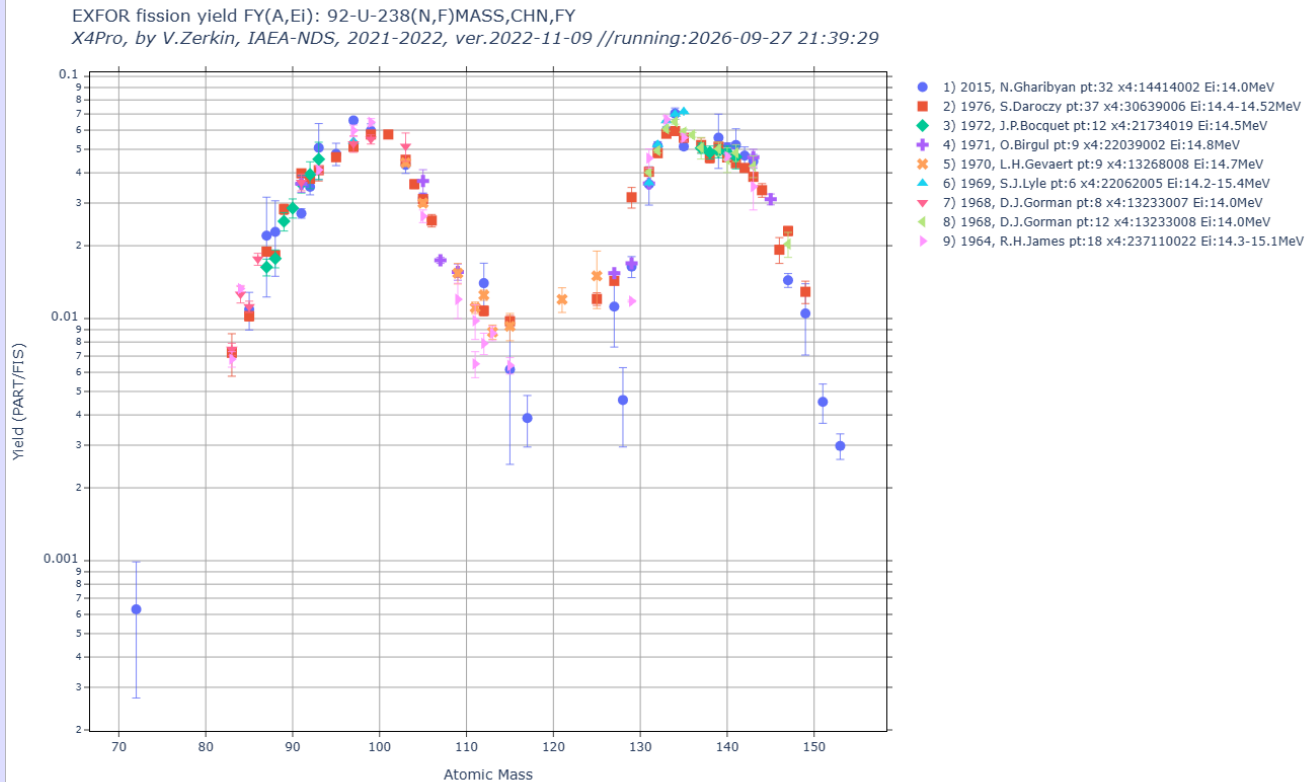
part1-5-dae/ EXFOR double differential cross sections

6) part1-5-dae/out00/dae0eo.html.png



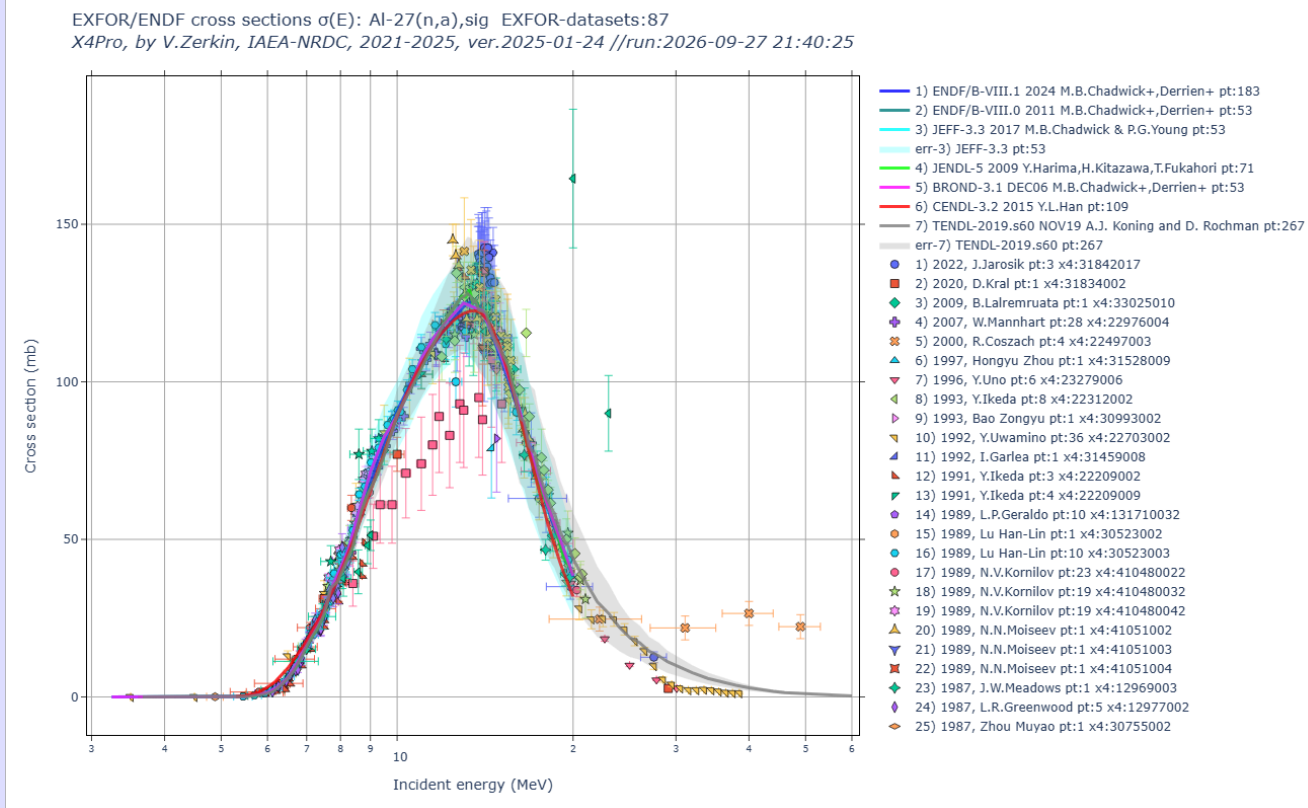
part1-6-fy/ EXFOR fission yield

7) part1-6-fy/out00/U238nf-fy.html.png



part2-1-sig1/ Retrieve and plot EXFOR/ENDF cross sections

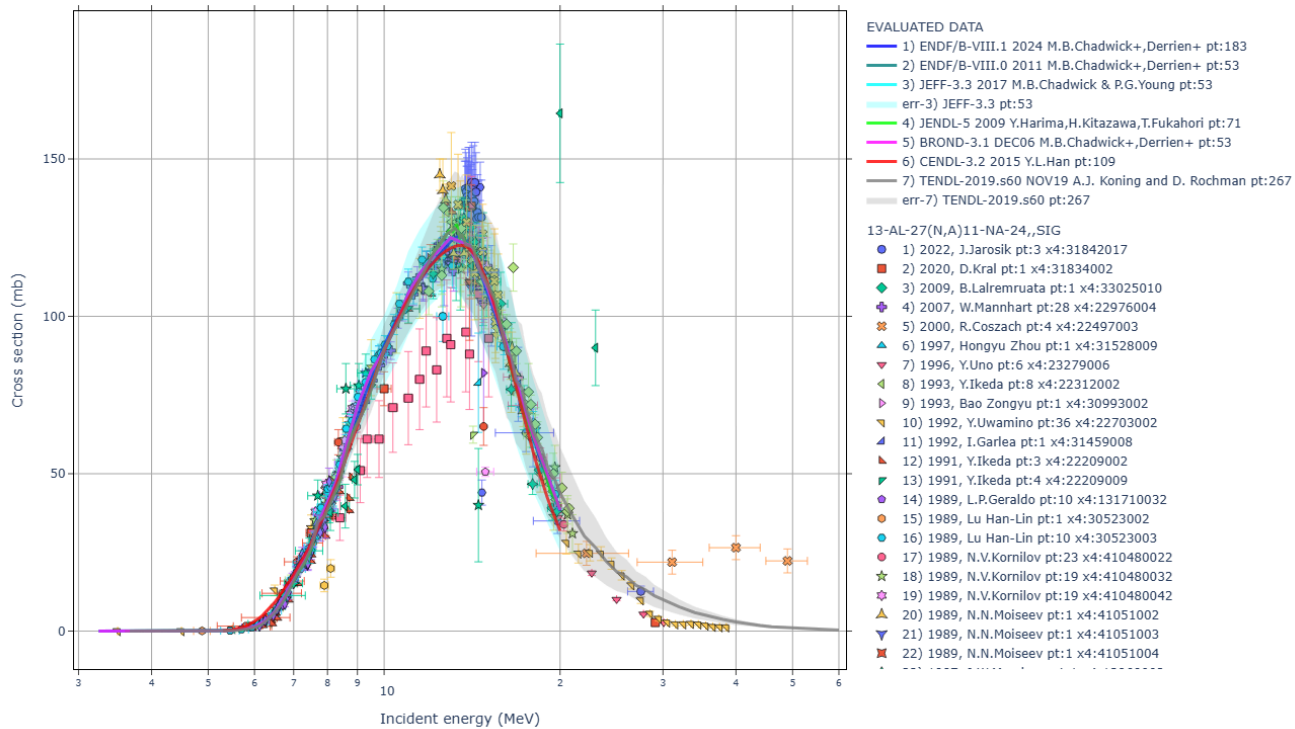
8) part2-1-sig1/out00/Al27na.html.png



9) part2-1-sig1/out00/Al27na2.html.png

EXFOR/ENDF cross sections $\sigma(E)$: Al-27(n,a),sig EXFOR-datasets:93

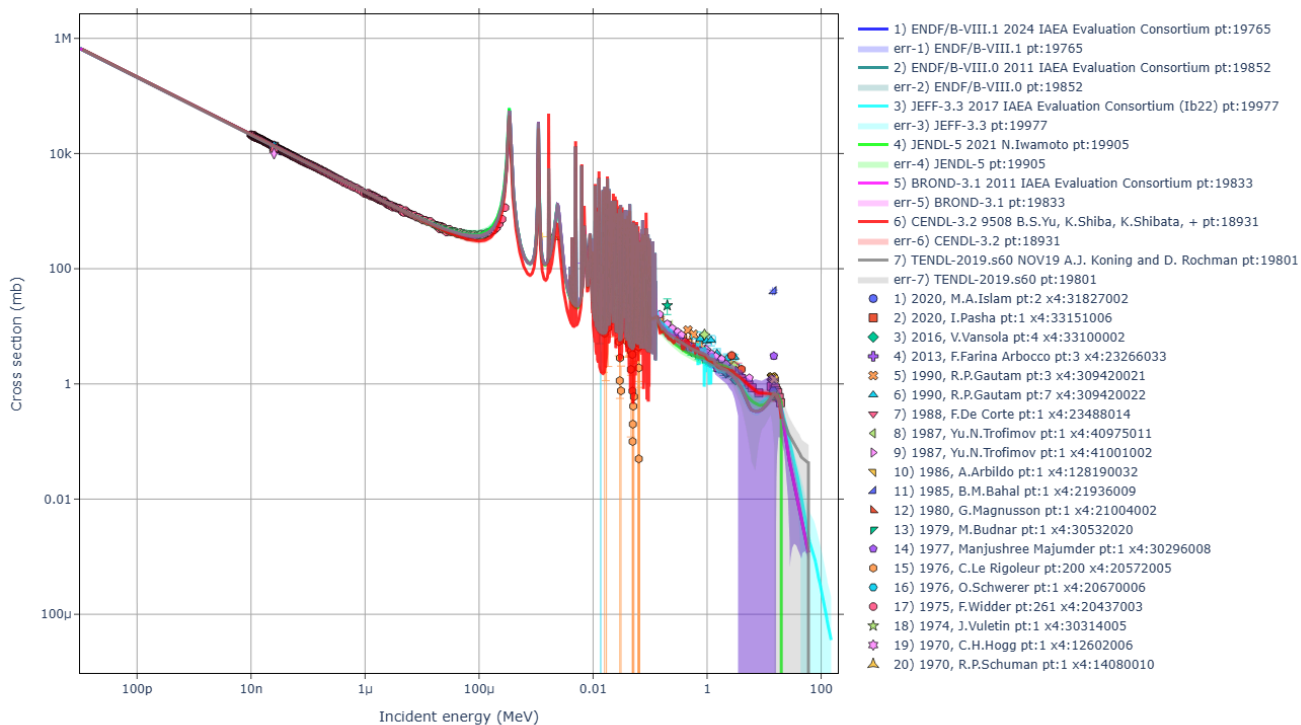
X4Pro, by V.Zerkin, IAEA-NRDC, 2021-2025, ver.2025-01-24 //run:2026-09-27 21:41:23



10) part2-1-sig1/out00/mn55ng.html.png

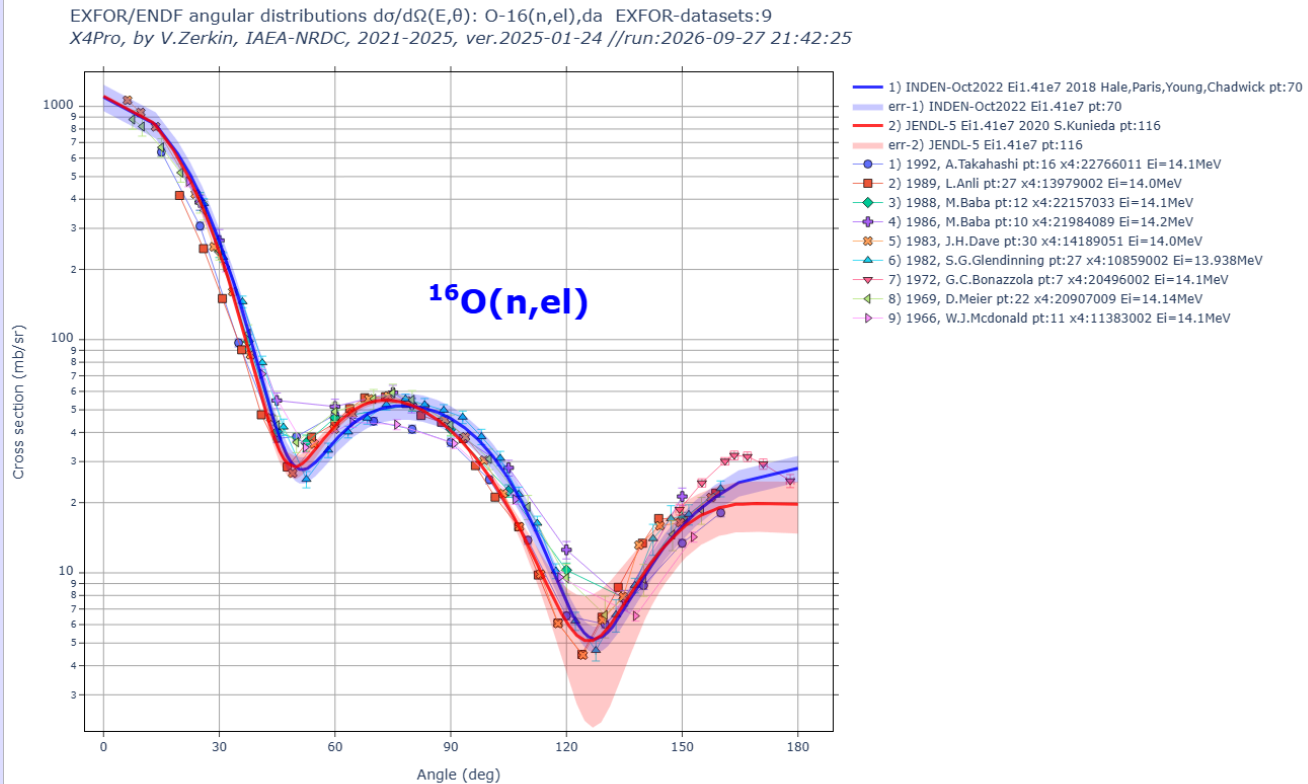
EXFOR/ENDF cross sections $\sigma(E)$: mn-55(n,g),sig EXFOR-datasets:49

X4Pro, by V.Zerkin, IAEA-NRDC, 2021-2025, ver.2025-01-24 //run:2026-09-27 21:41:26



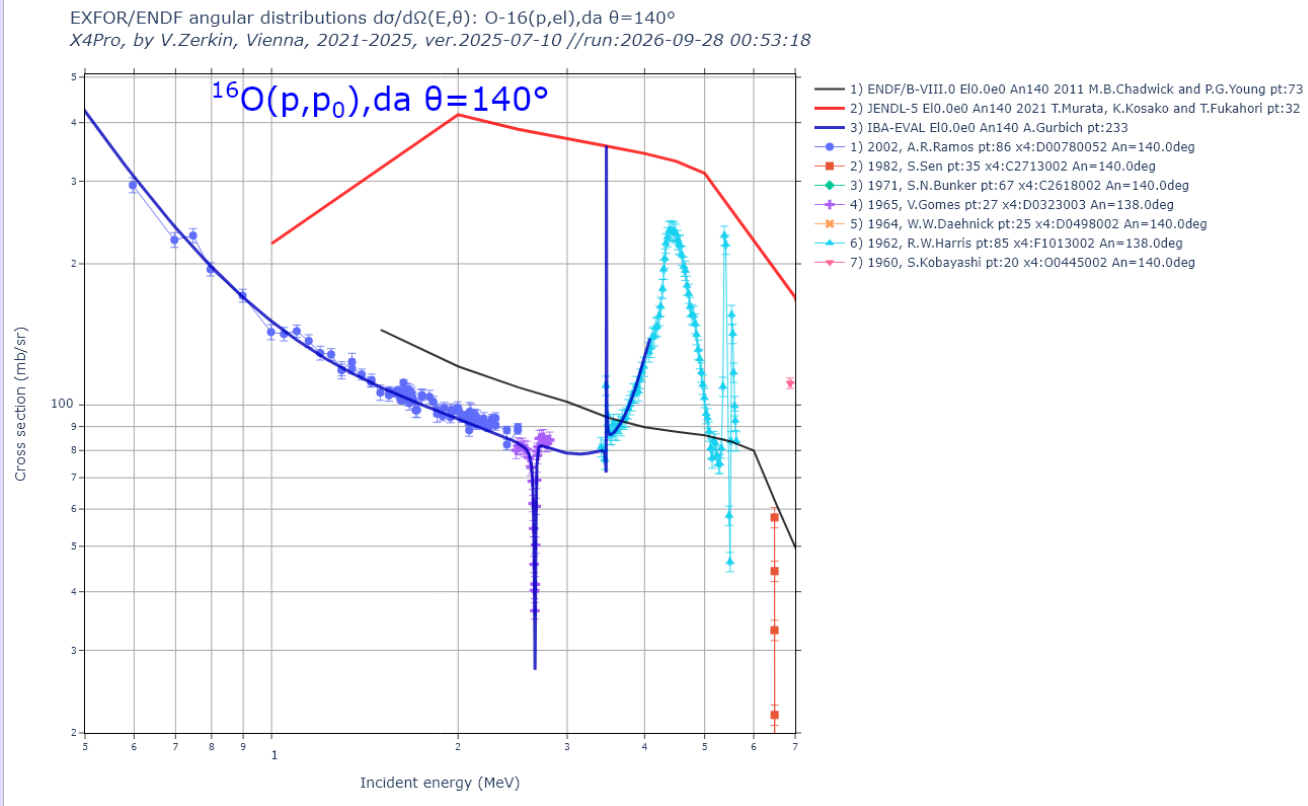
part2-2-dal an/ EXFOR/ENDF angular distributions (Ei:fixed)

11) part2-2-dal an/out00/dal an.html.png



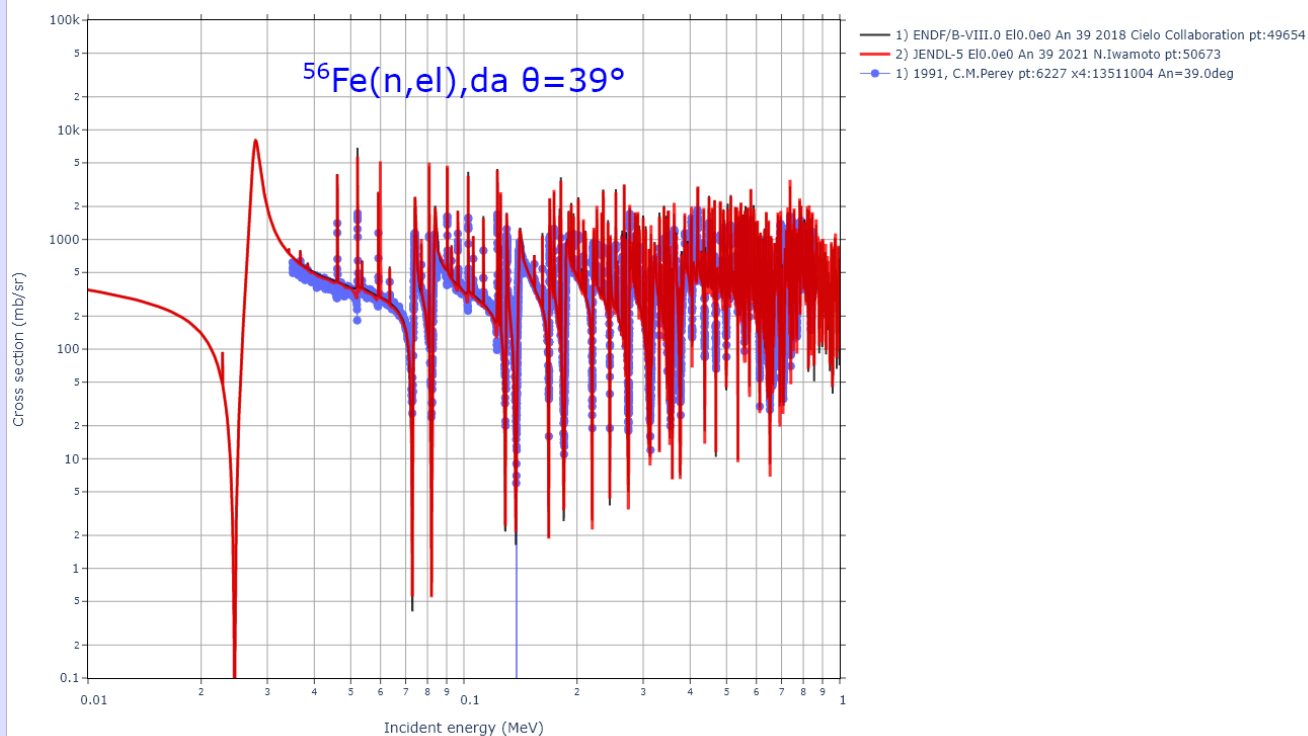
part2-3-dal ei/ EXFOR/ENDF angular distributions (angle:fixed)

12) part2-3-dal ei/out00/dal ei-ex1.html.png



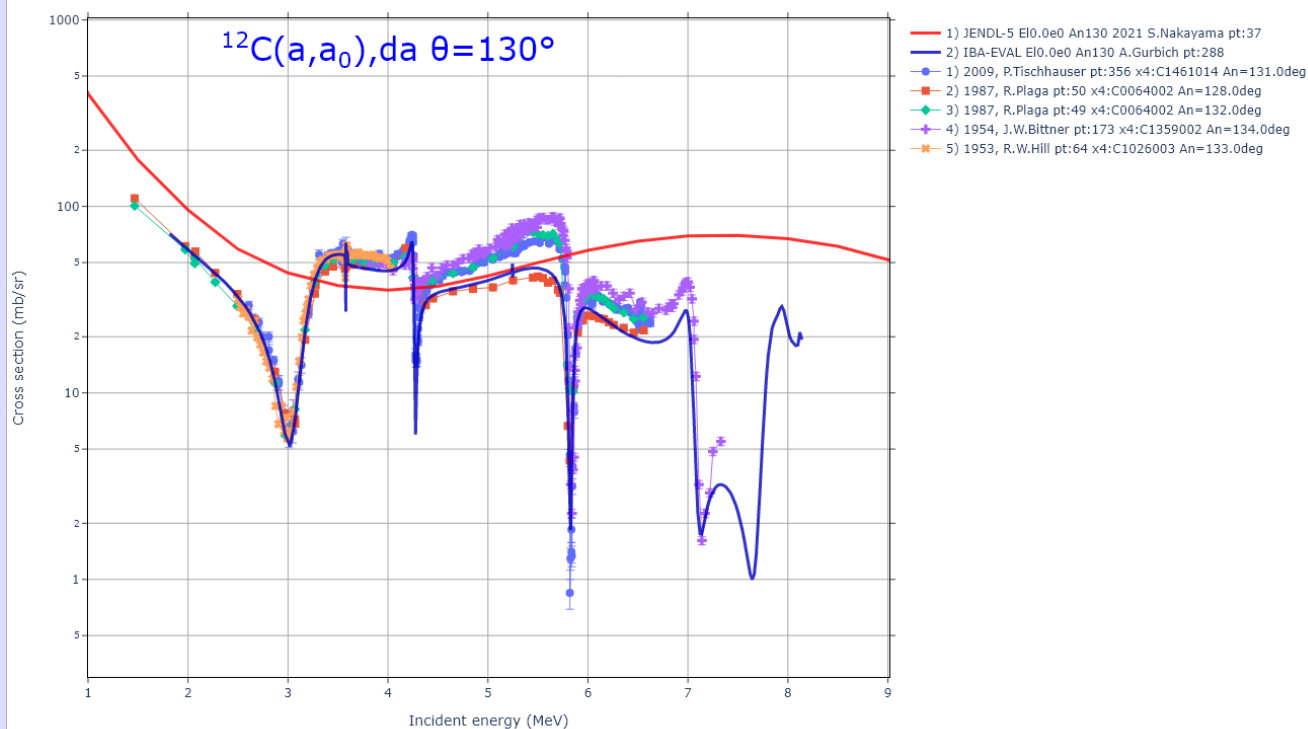
13) part2-3-da1ei/out00/da1ei-ex2.html.png

EXFOR/ENDF angular distributions $d\sigma/d\Omega(E,\theta)$: Fe-56(n,el), da $\theta=39^\circ$
 X4Pro, by V.Zerkin, Vienna, 2021-2025, ver.2025-07-10 //run:2026-09-28 00:53:20



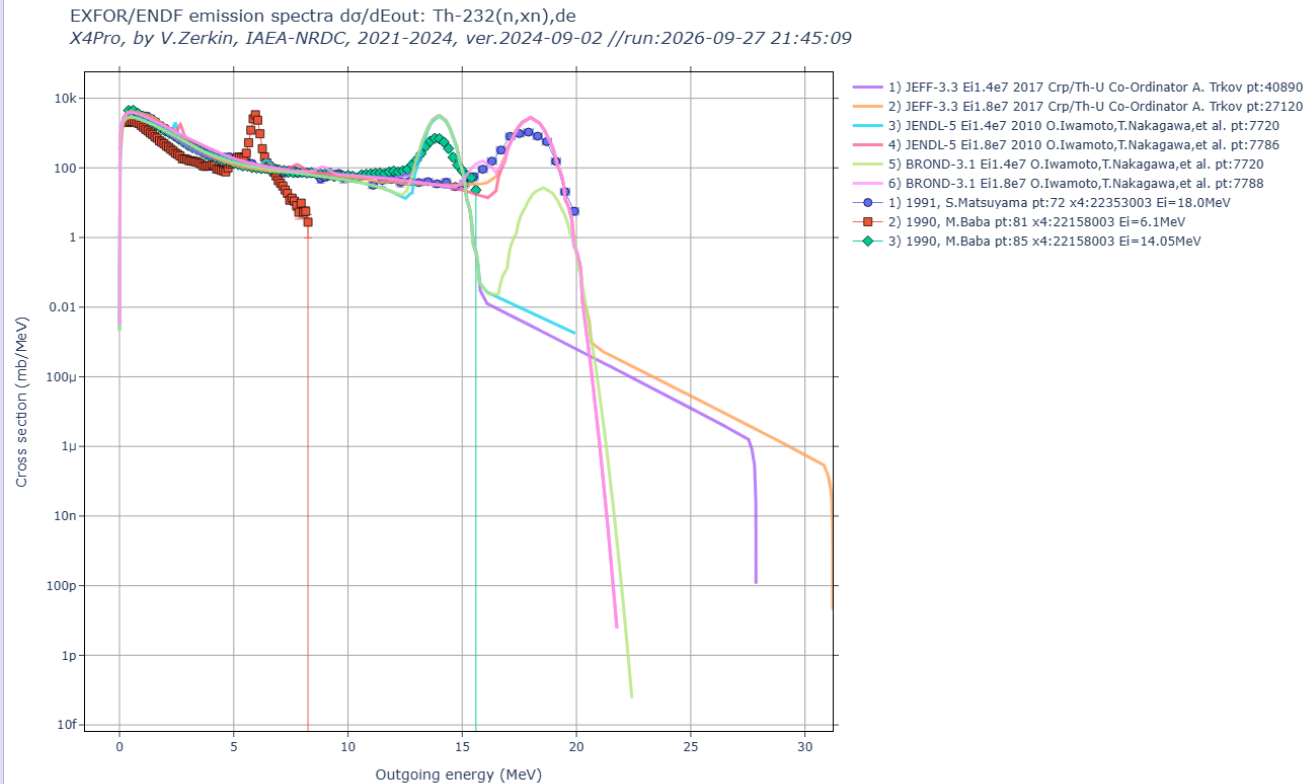
14) part2-3-da1ei/out00/da1ei.html.png

EXFOR/ENDF angular distributions $d\sigma/d\Omega(E,\theta)$: C-12(a,el), da $\theta=130^\circ$
 X4Pro, by V.Zerkin, Vienna, 2021-2025, ver.2025-07-10 //run:2026-09-28 00:53:13



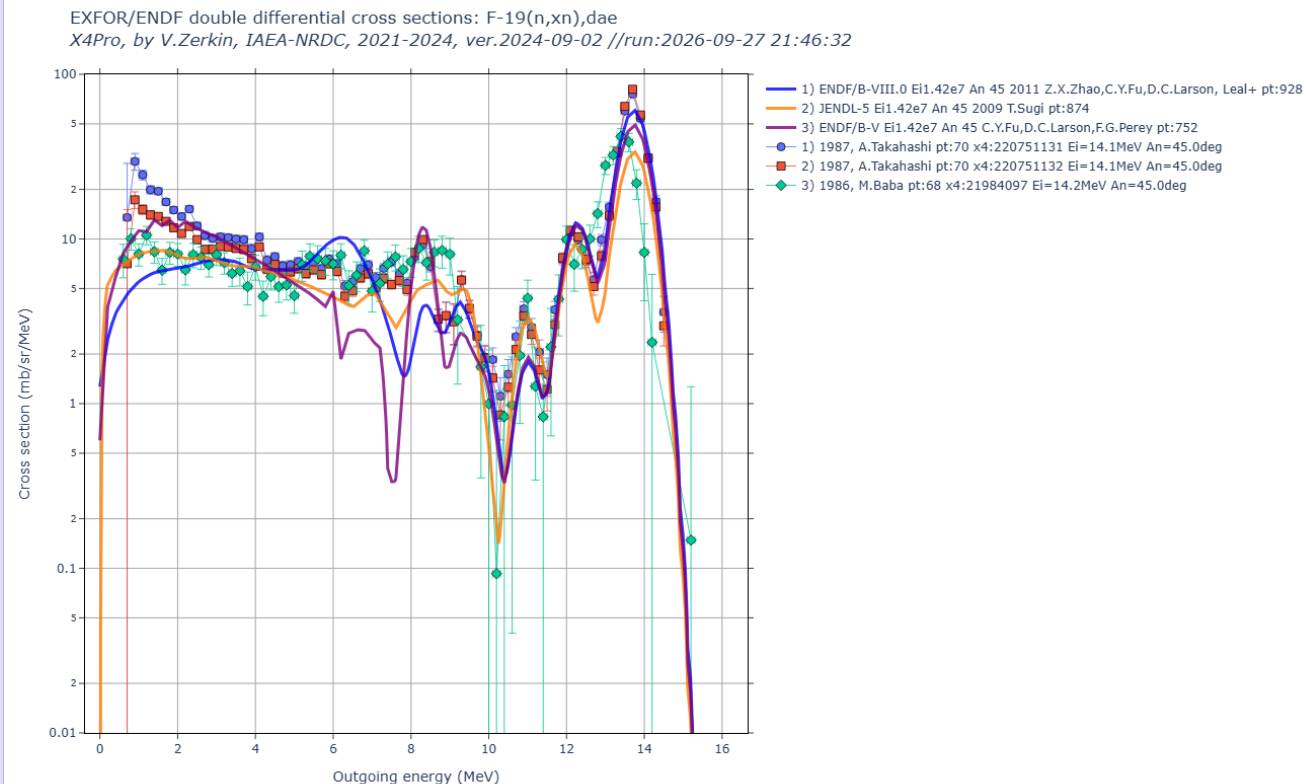
part2-4-de1eo/ EXFOR/ENDF energy spectra of outgoing particles

15) part2-4-de1eo/out00/de1eo.html.png



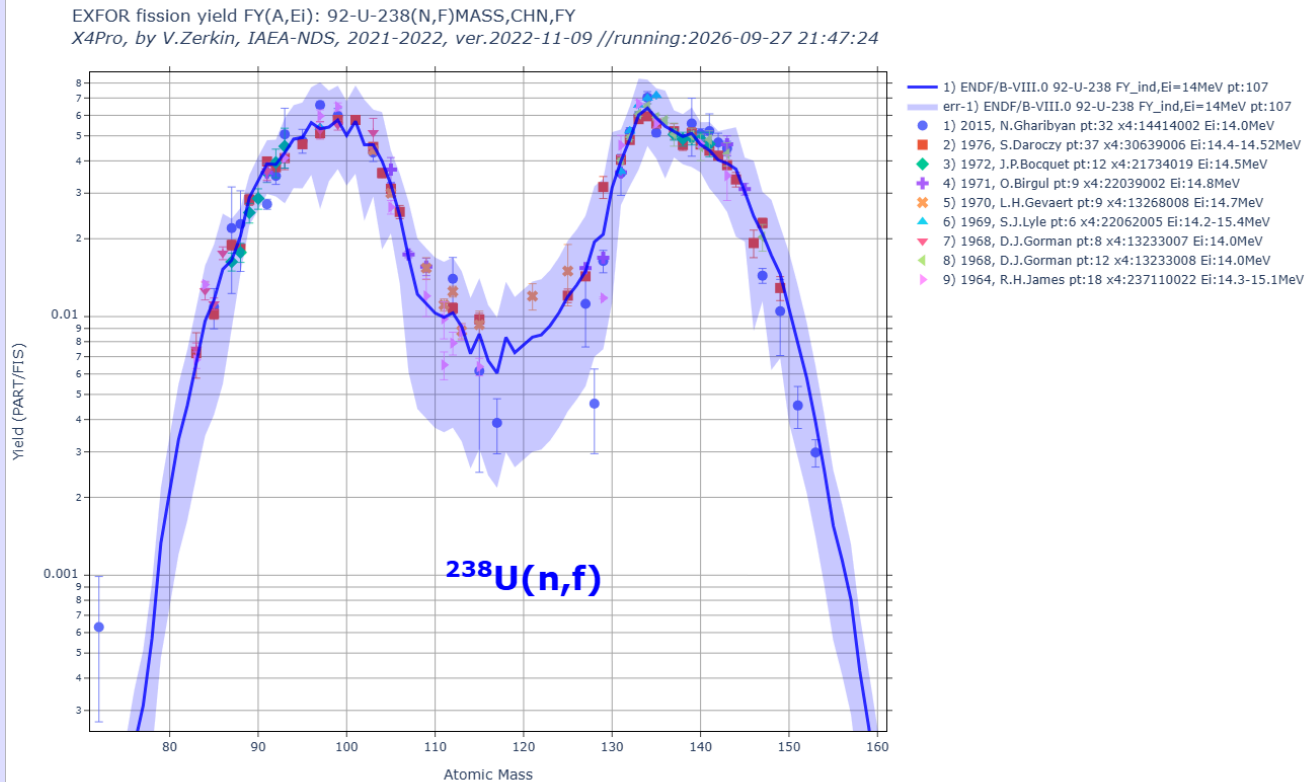
part2-5-dae1eo/ EXFOR/ENDF double differential cross sections

16) part2-5-dae1eo/out00/dae1eo.html.png



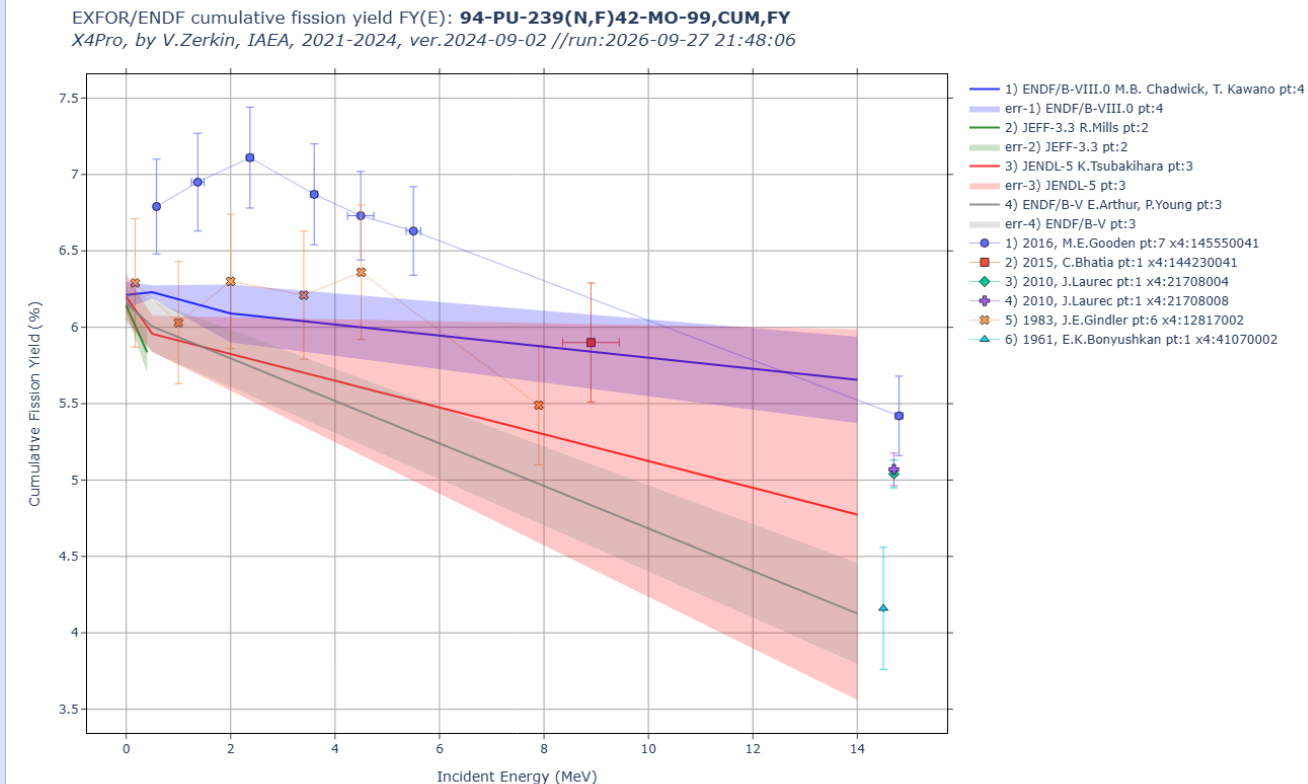
part2-6-fy/ EXFOR/ENDF fission yeild (mass distribution)

17) part2-6-fy/out00/U238nf-fy.html.png

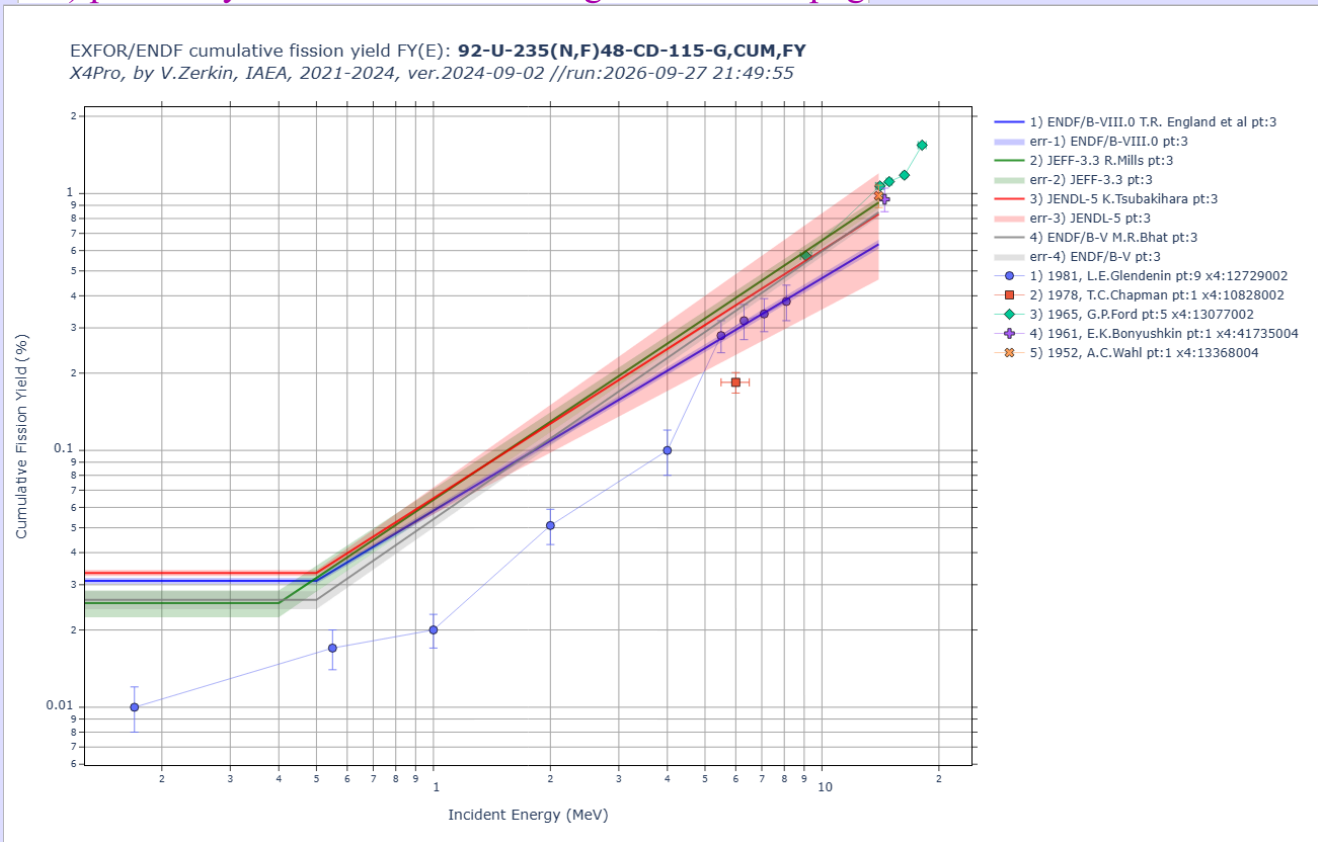


part2-7-fye/ EXFOR/ENDF fission yeild (energy distribution)

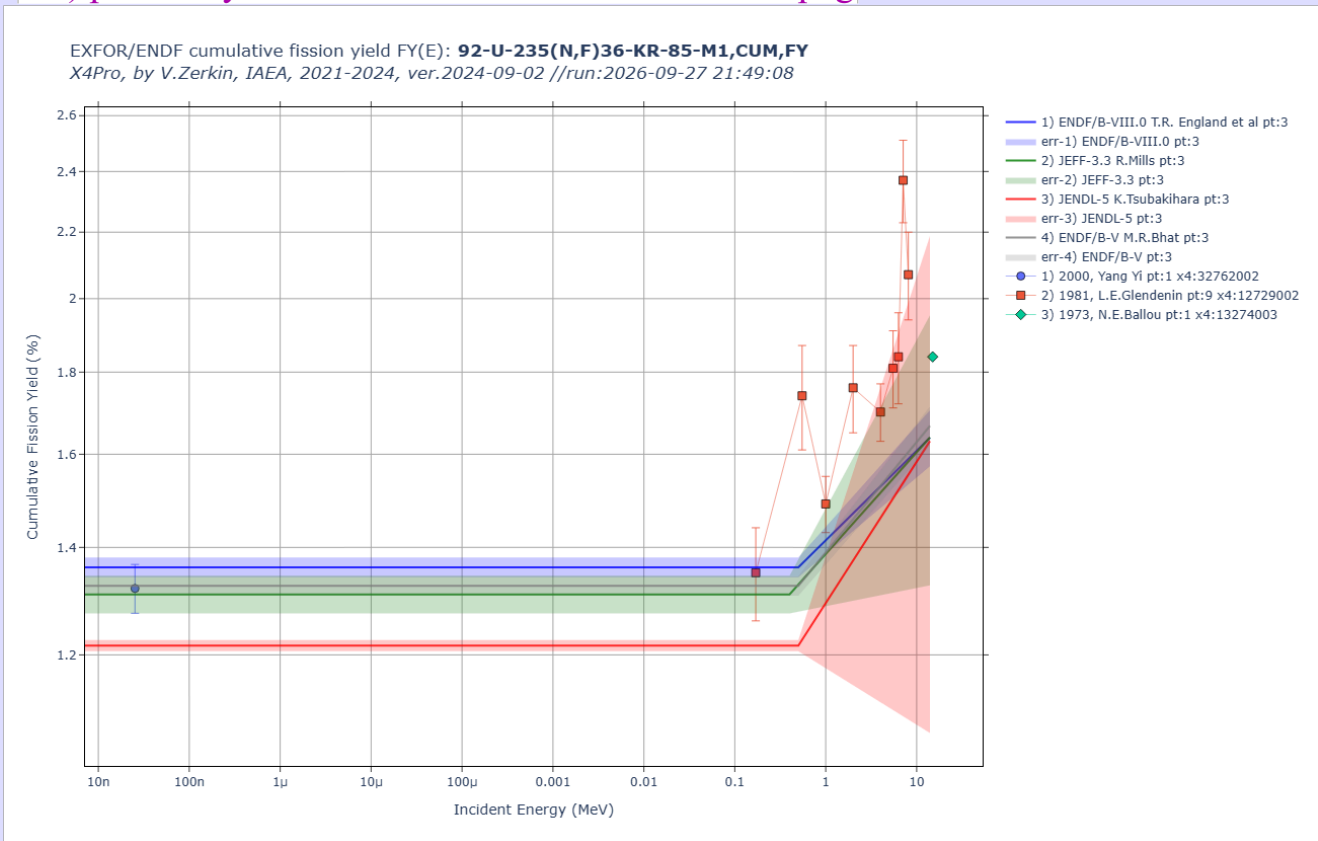
18) part2-7-fye/out00/Pu239-nf-Mo99-cumFY.html.png



19) part2-7-fye/out00/u235-nf-cd115g-cumFY.html.png

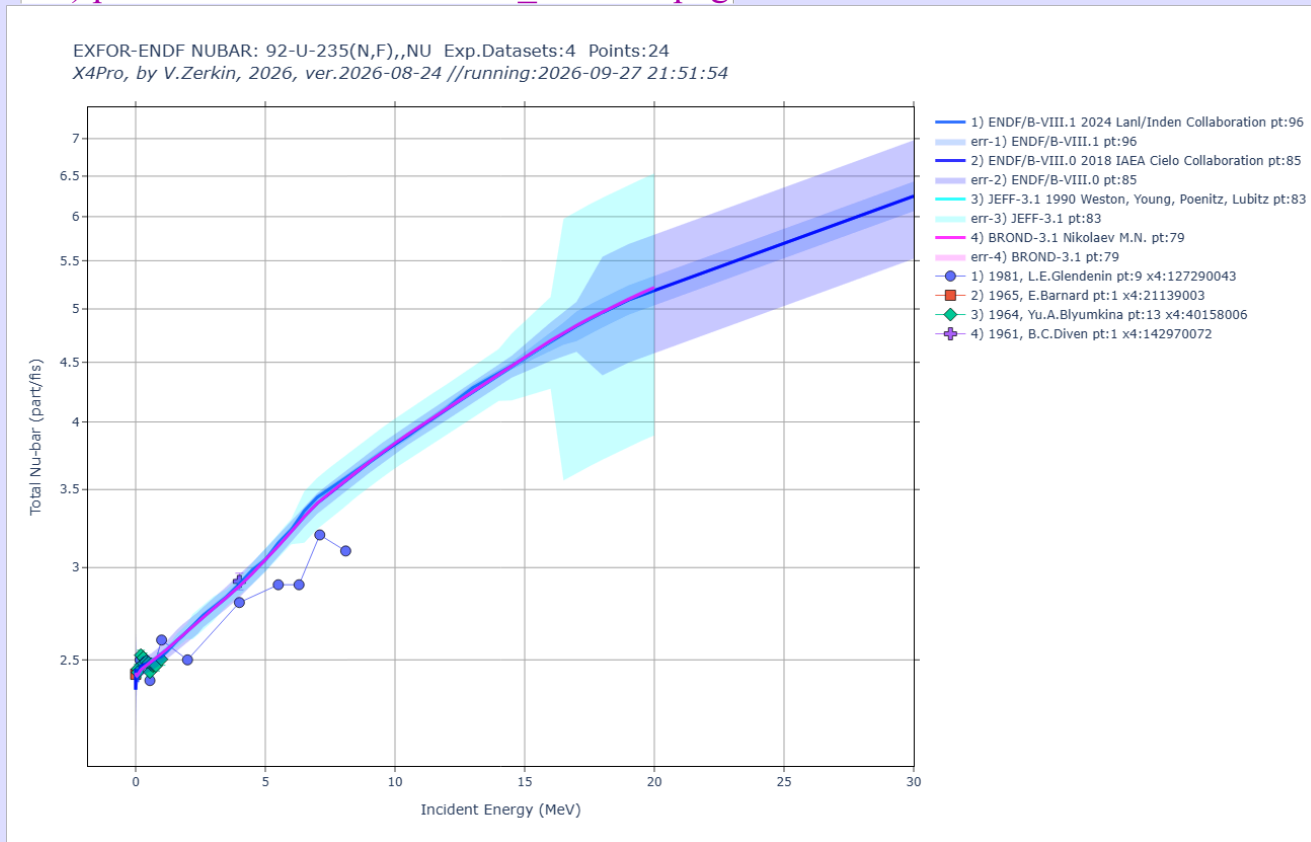


20) part2-7-fye/out00/u235-nf-kr85m-cumFY.html.png

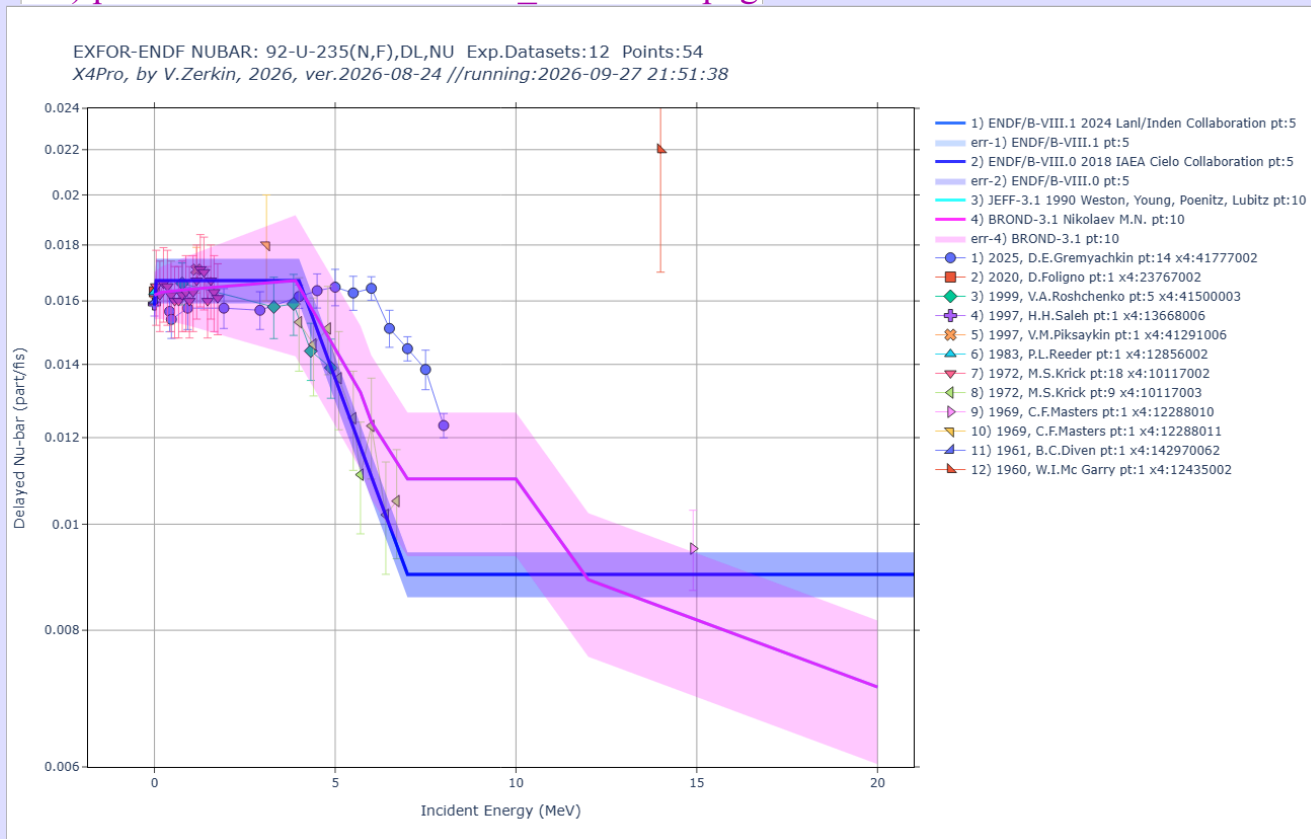


part2-9-nubar/ EXFOR-ENDF NUBAR - average number of neutrons per fission

21) part2-9-nubar/out00/U235nf_-nu.html.png

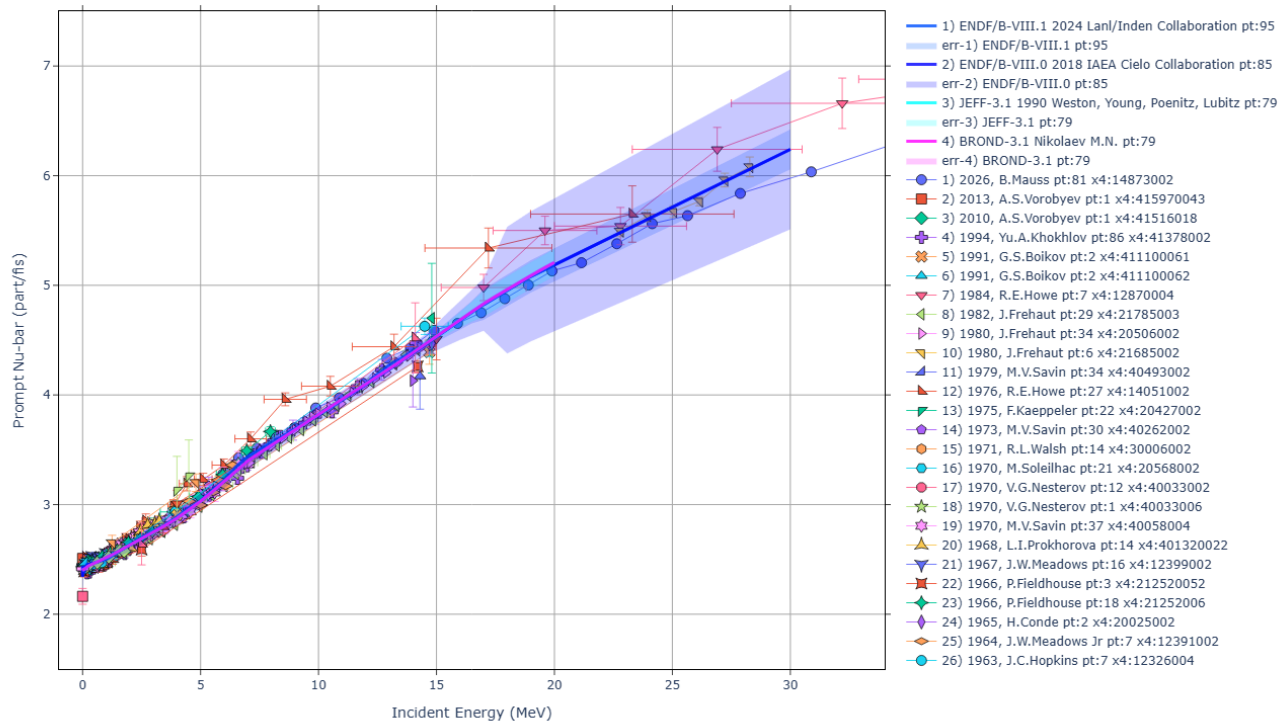


22) part2-9-nubar/out00/U235nf_dl-nu.html.png



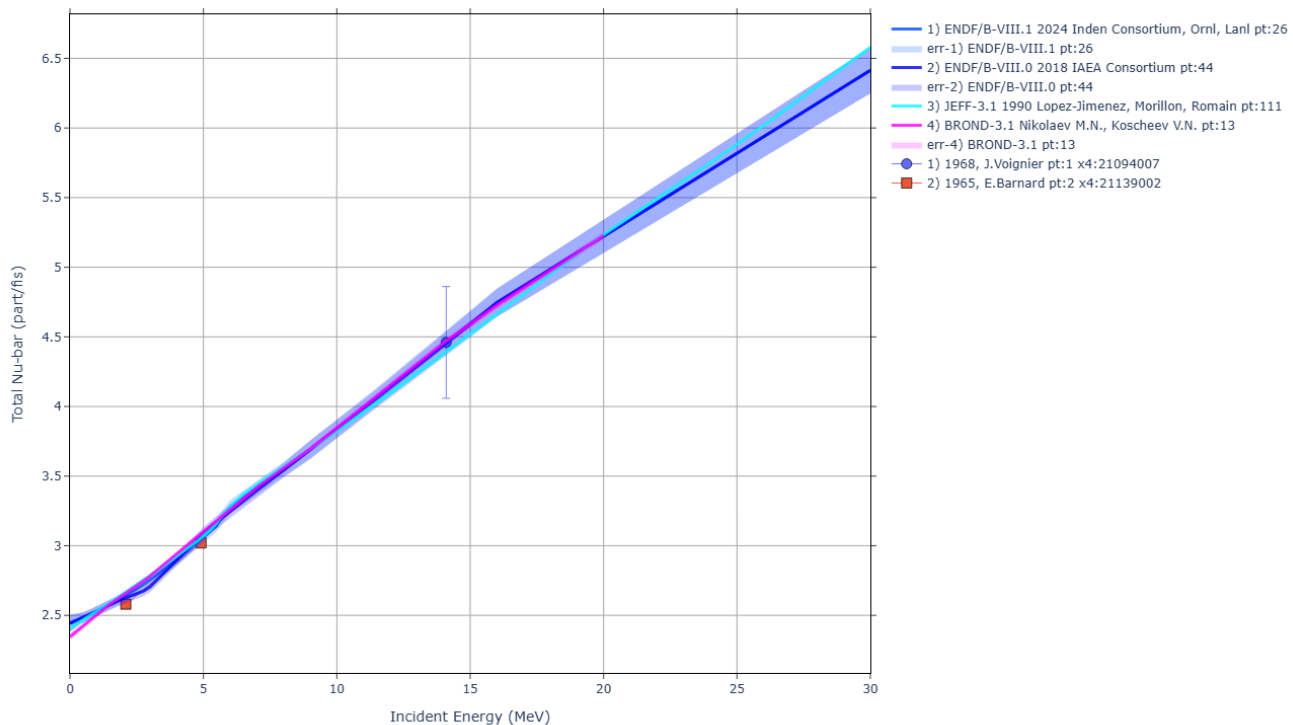
23) part2-9-nubar/out00/U235nf_pr-nu.html.png

EXFOR-ENDF NUBAR: 92-U-235(N,F),PR,NU Exp.Datasets:38 Points:535
X4Pro, by V.Zerkin, 2026, ver.2026-08-24 //running:2026-09-27 21:50:46

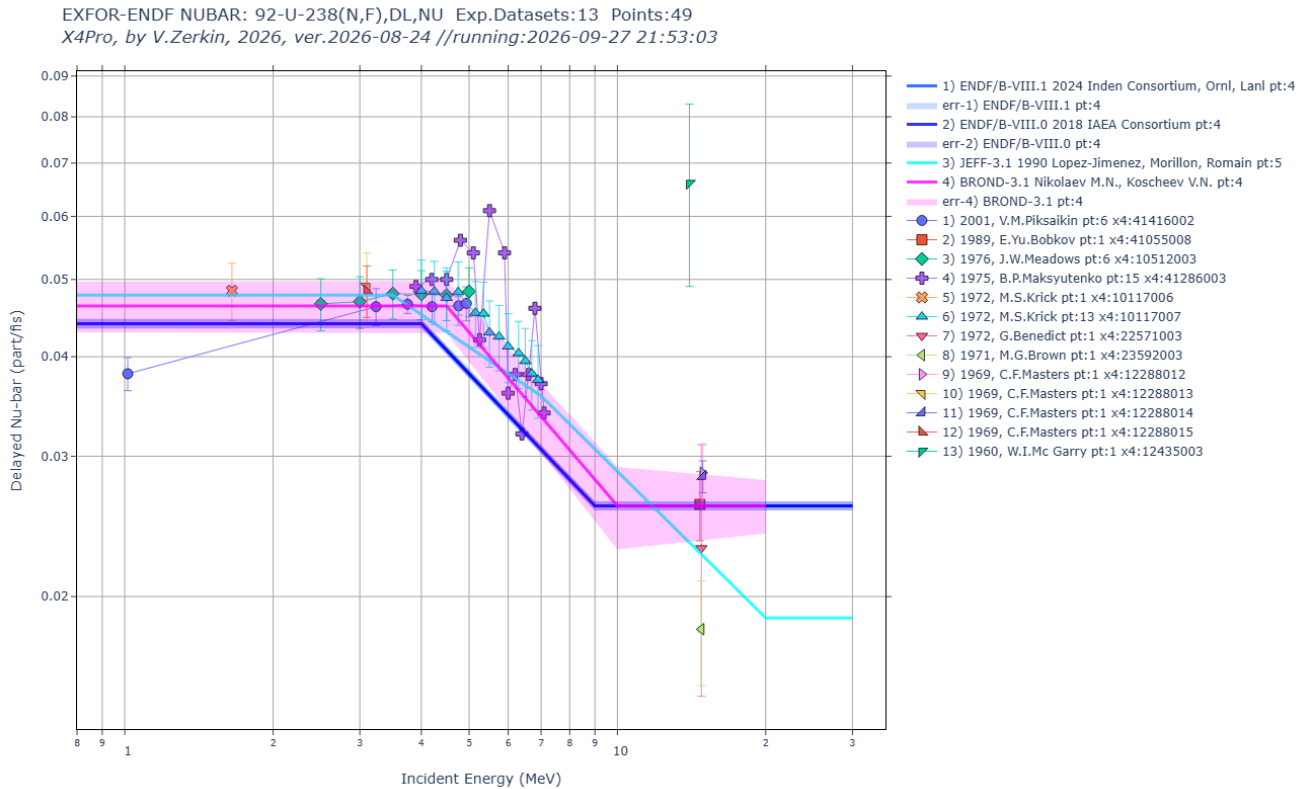


24) part2-9-nubar/out00/U238nf_-nu.html.png

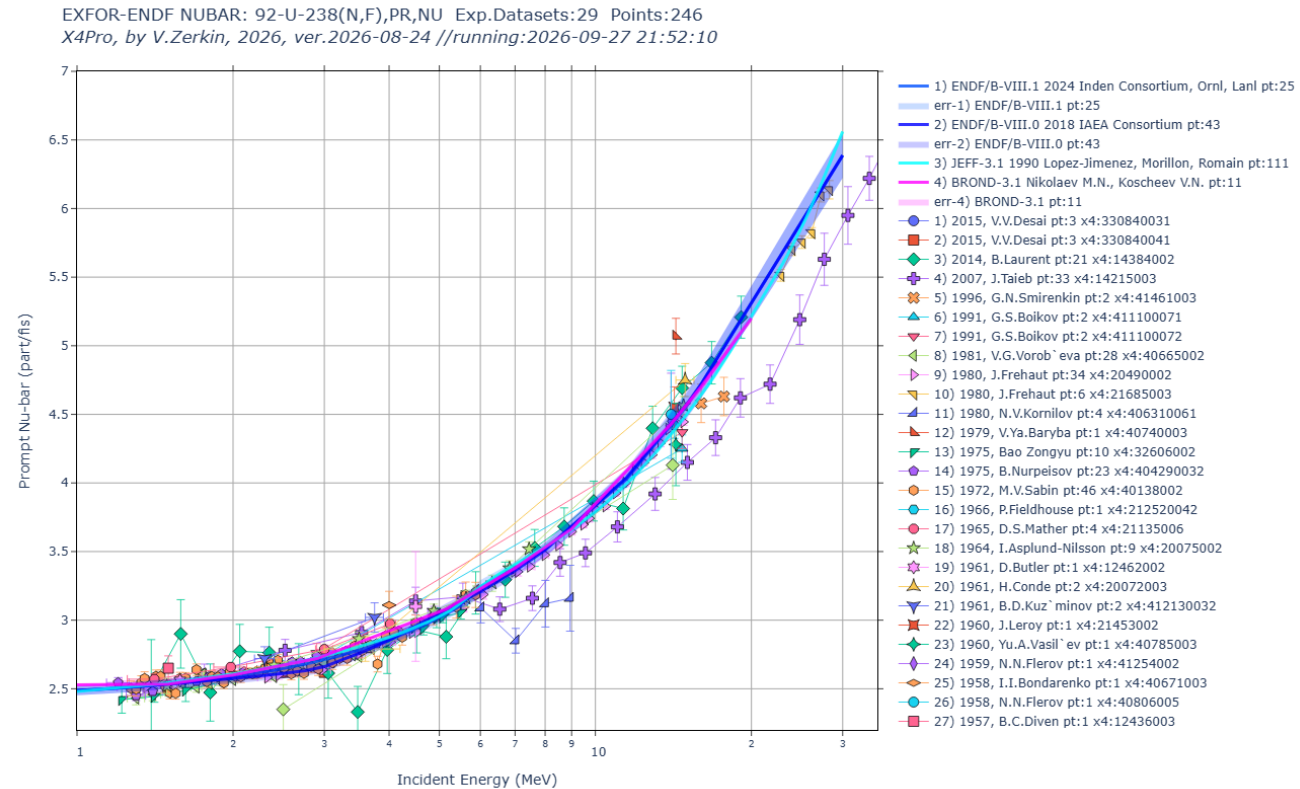
EXFOR-ENDF NUBAR: 92-U-238(N,F),,NU Exp.Datasets:2 Points:3
X4Pro, by V.Zerkin, 2026, ver.2026-08-24 //running:2026-09-27 21:53:19



25) part2-9-nubar/out00/U238nf_dl-nu.html.png

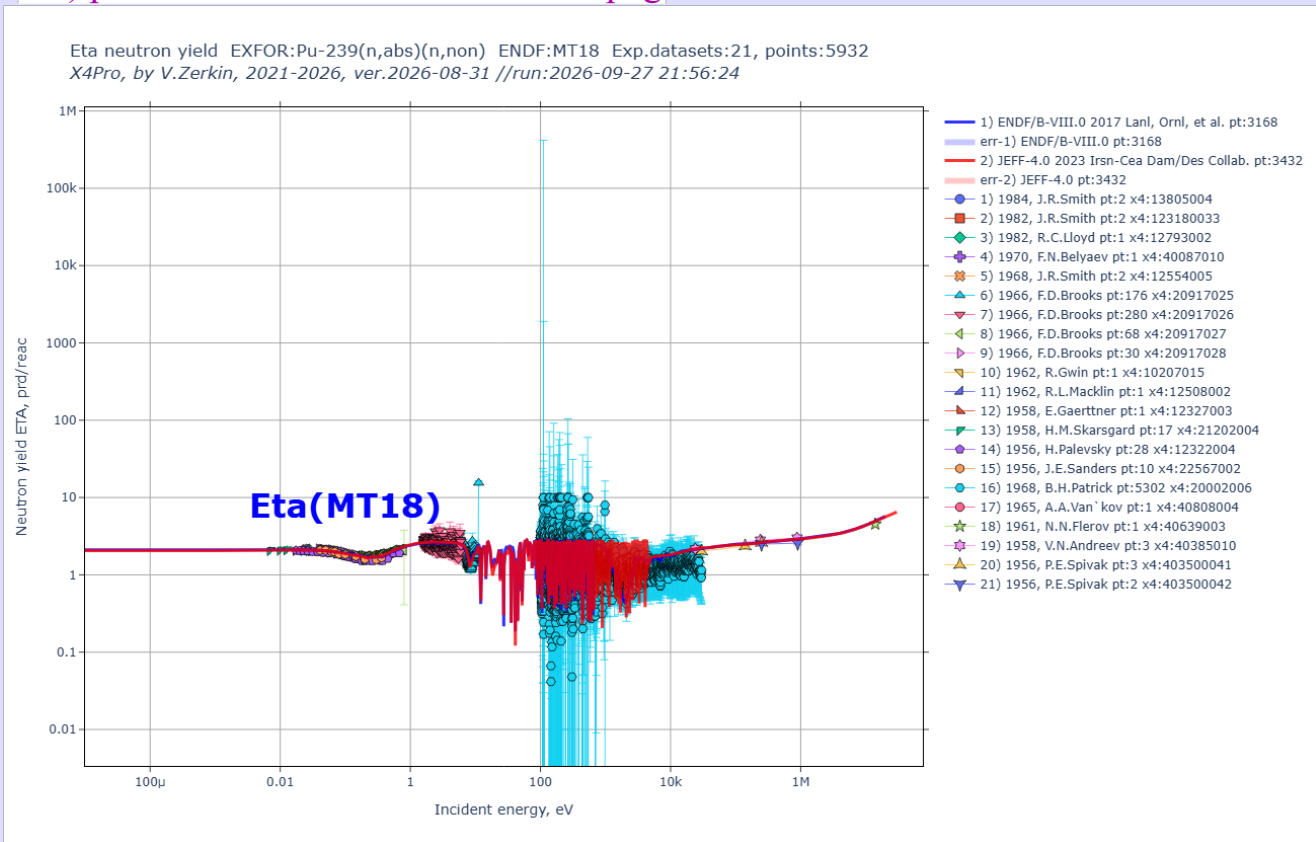


26) part2-9-nubar/out00/U238nf_pr-nu.html.png

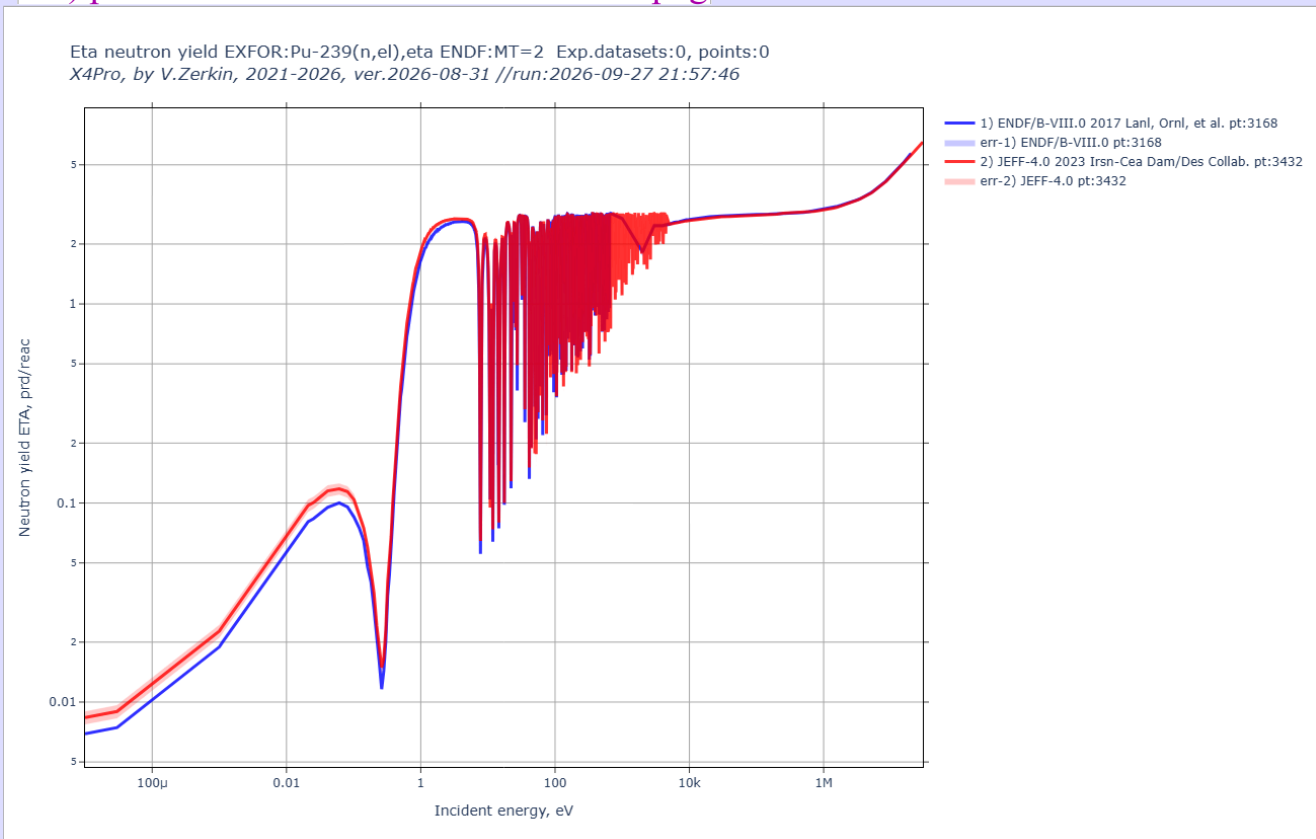


part2-A-eta/ EXFOR/ENDF Eta - average neutron yield per nonelastic event

27) part2-A-eta/out00/Pu239-eta.html.png

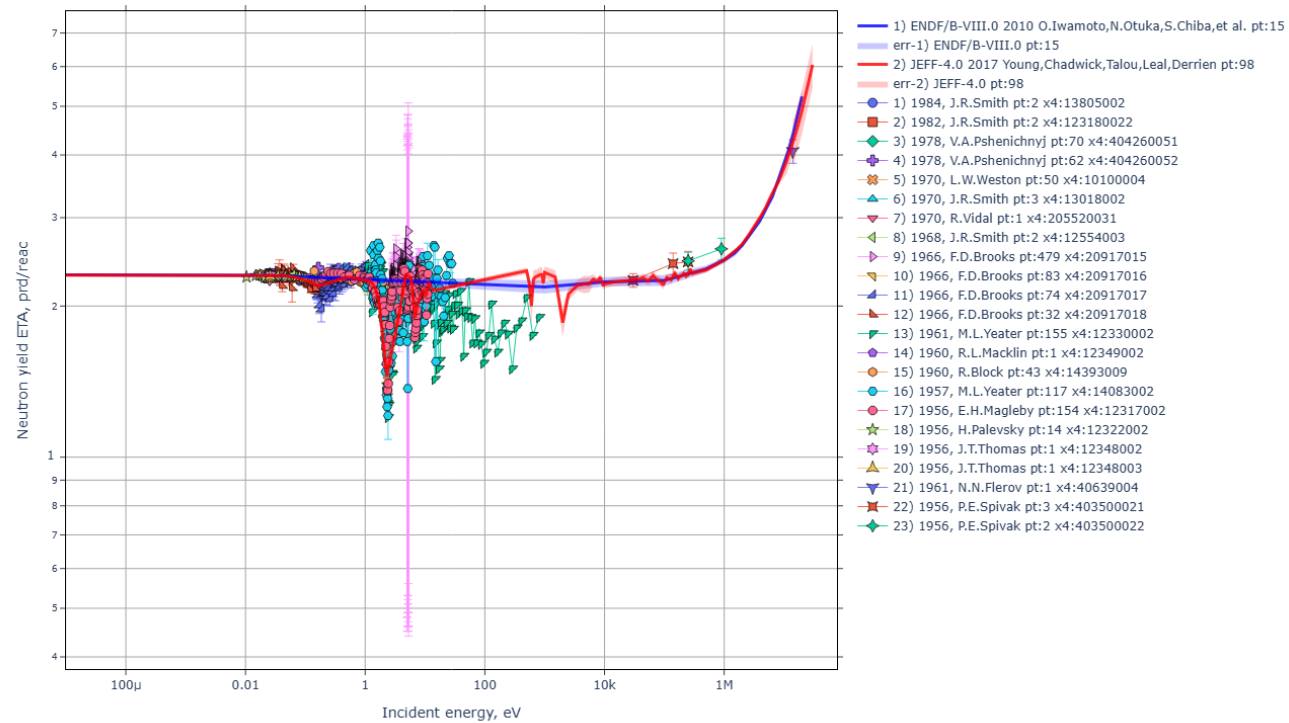


28) part2-A-eta/out00/Pu239nel-eta.html.png



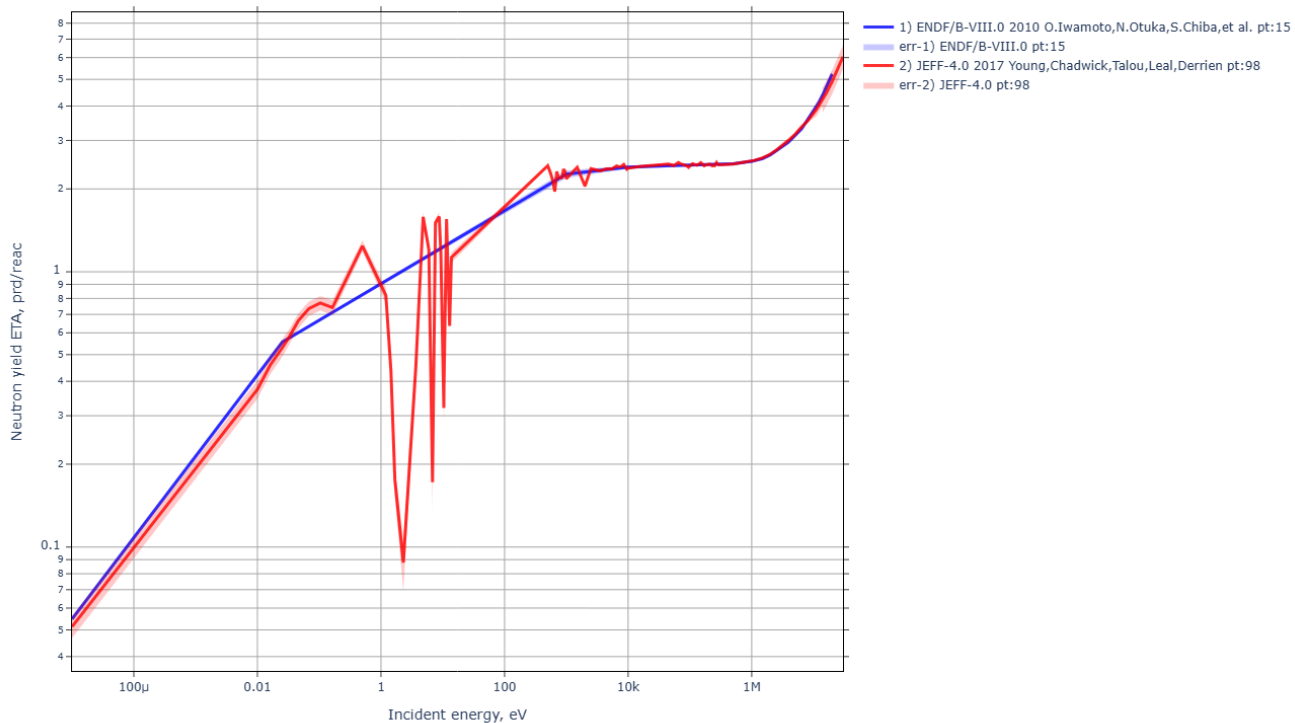
29) part2-A-eta/out00/U233-eta.html.png

Eta neutron yield EXFOR:U-233(n,abs)(n,non) ENDF:MT18 Exp.datasets:23, points:1352
X4Pro, by V.Zerkin, 2021-2026, ver.2026-08-31 //run:2026-09-27 21:55:37

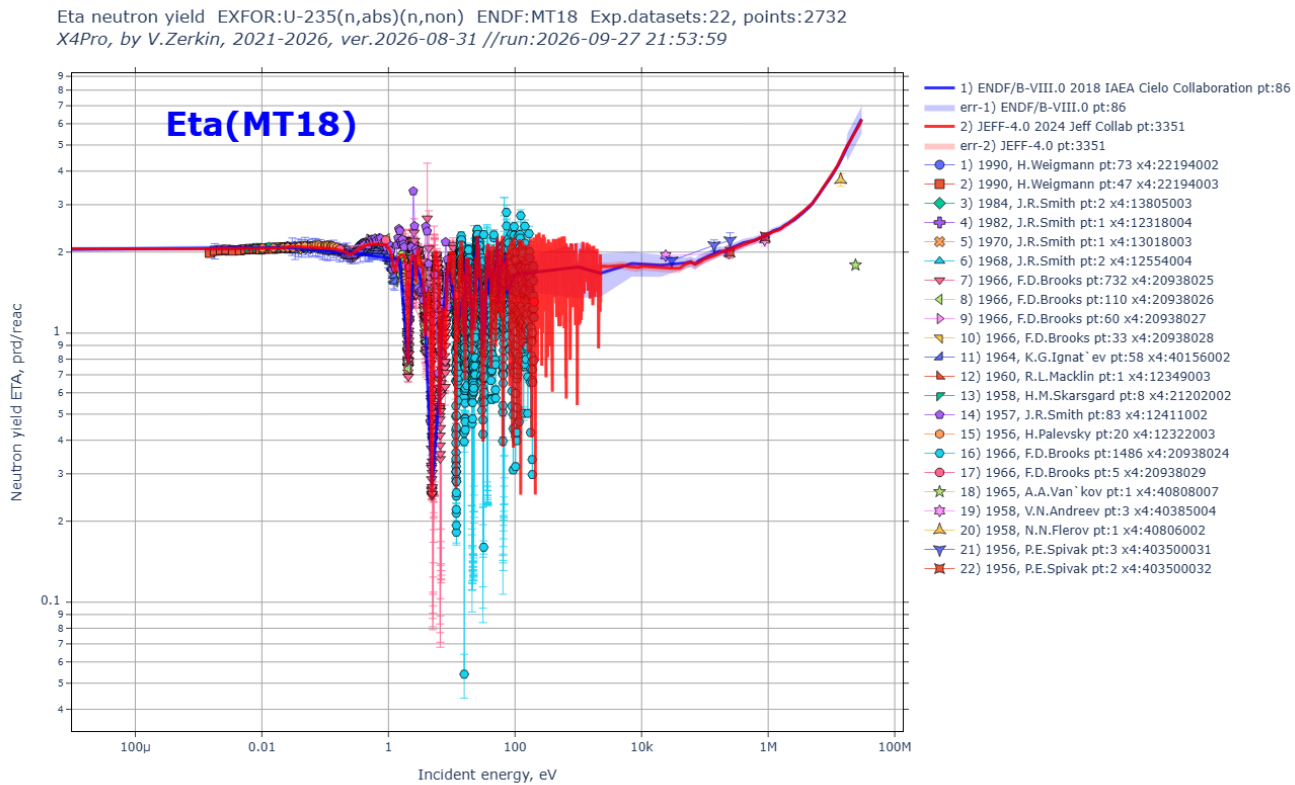


30) part2-A-eta/out00/U233nel-eta.html.png

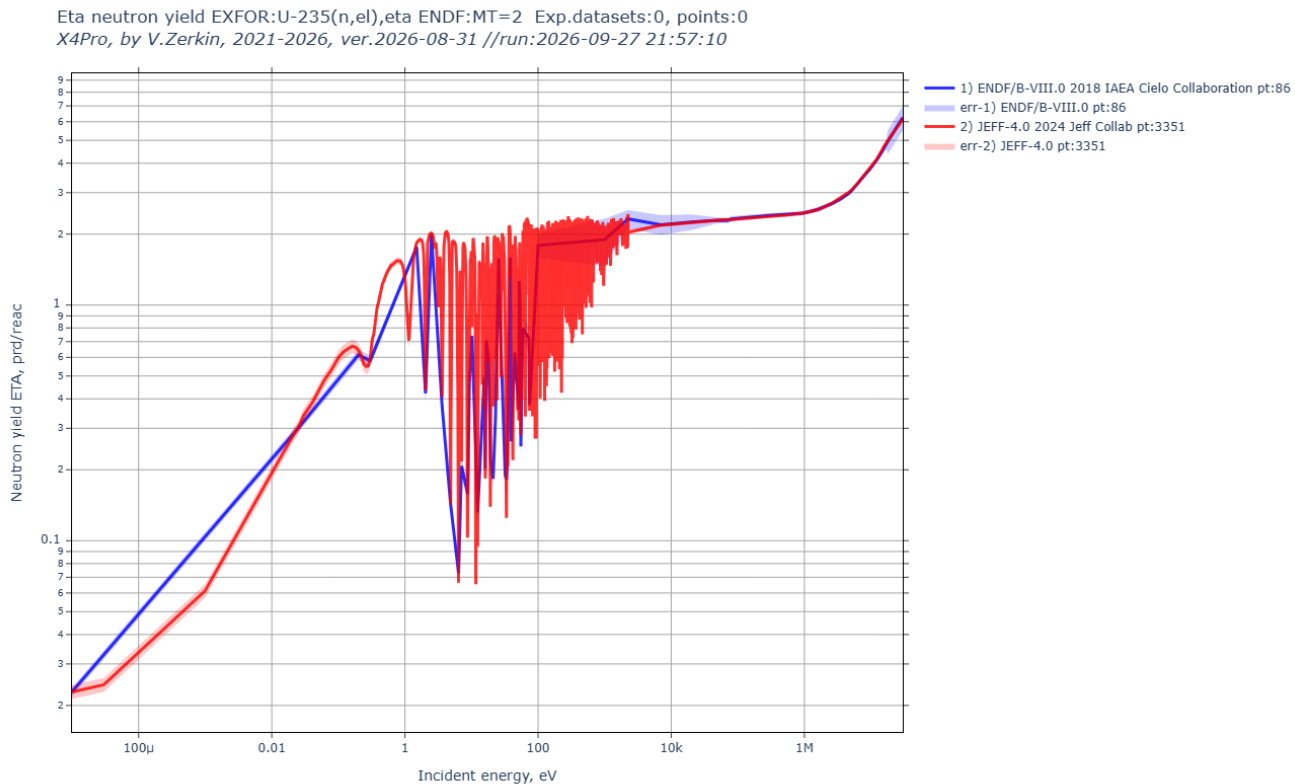
Eta neutron yield EXFOR:U-233(n,el),eta ENDF:MT=2 Exp.datasets:0, points:0
X4Pro, by V.Zerkin, 2021-2026, ver.2026-08-31 //run:2026-09-27 21:57:36



31) part2-A-eta/out00/U235-eta.html.png

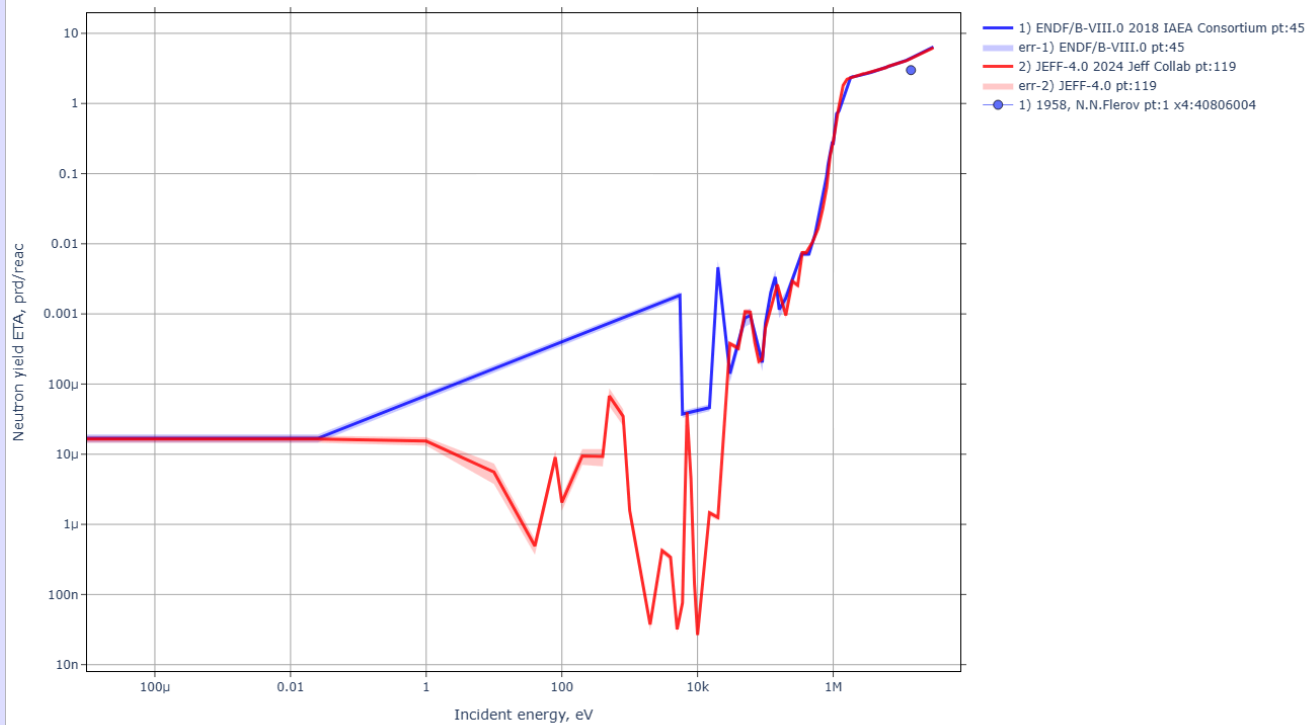


32) part2-A-eta/out00/U235nel-eta.html.png



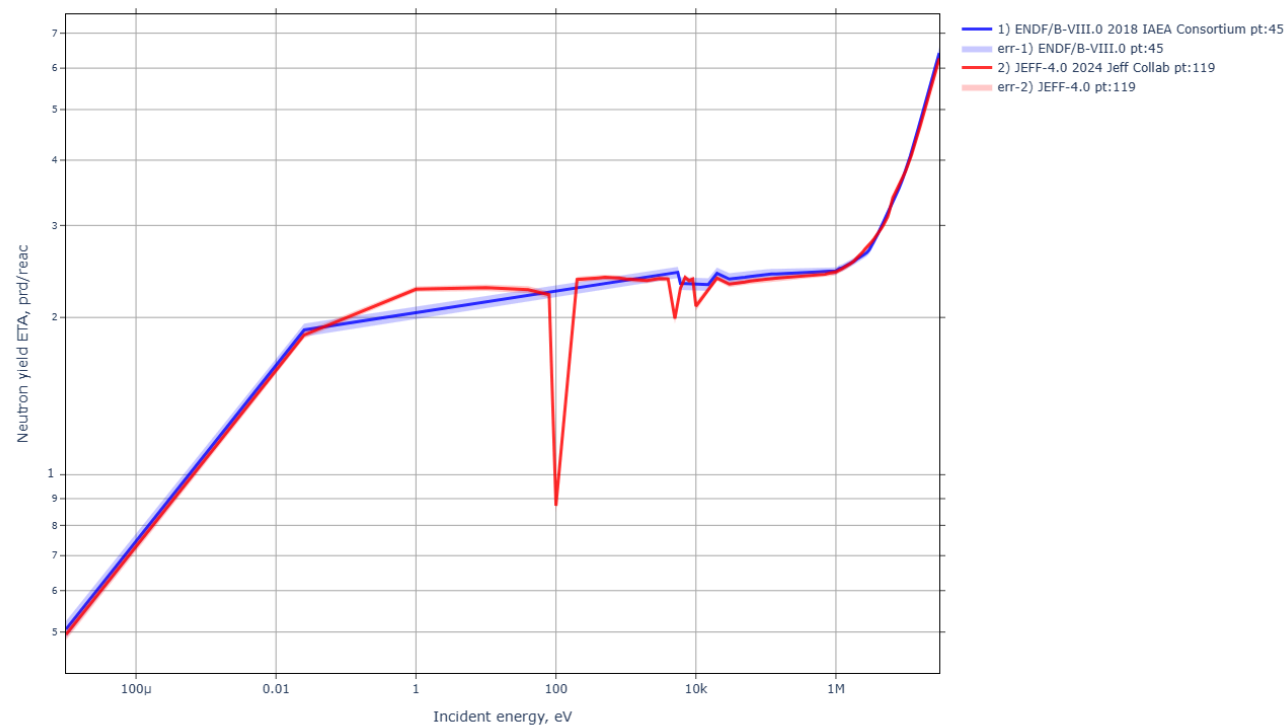
33) part2-A-eta/out00/U238-eta.html.png

Eta neutron yield EXFOR:U-238(n,non) ENDF:MT18 Exp.datasets:1, points:1
X4Pro, by V.Zerkin, 2021-2026, ver.2026-08-31 //run:2026-09-27 21:54:49



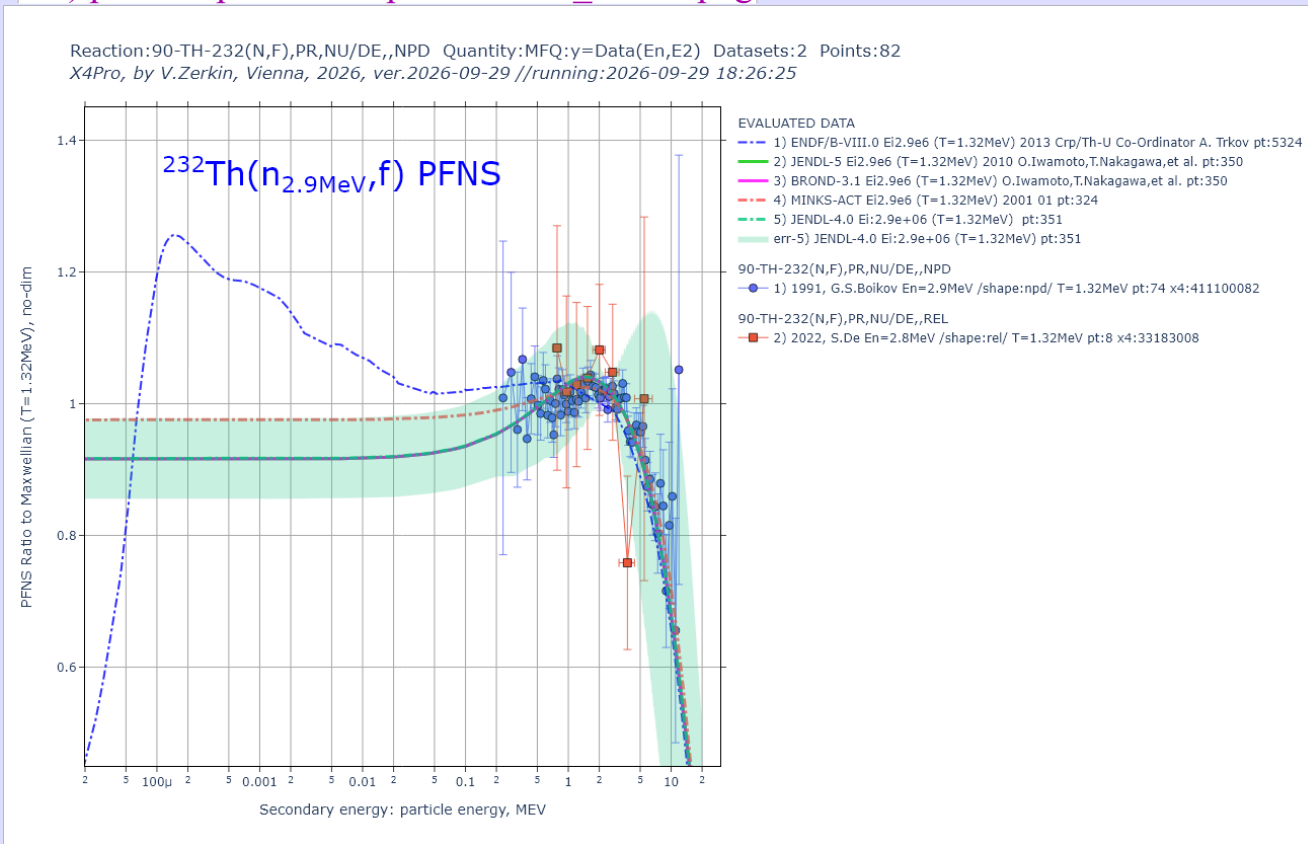
34) part2-A-eta/out00/U238nel-eta.html.png

Eta neutron yield EXFOR:U-238(n,el),eta ENDF:MT=2 Exp.datasets:0, points:0
X4Pro, by V.Zerkin, 2021-2026, ver.2026-08-31 //run:2026-09-27 21:57:23

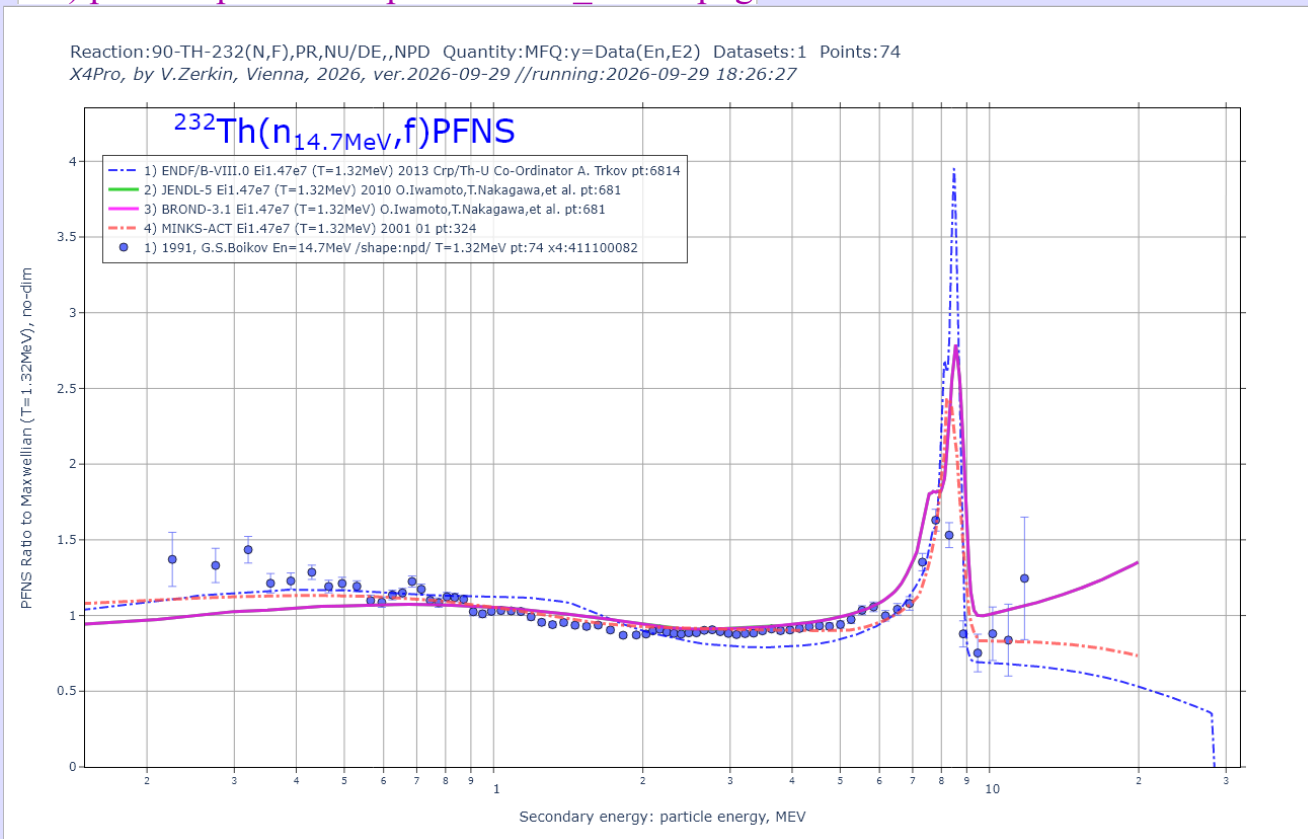


part2-B-pfns/ PFNS - energy spectrum of prompt fission neutrons

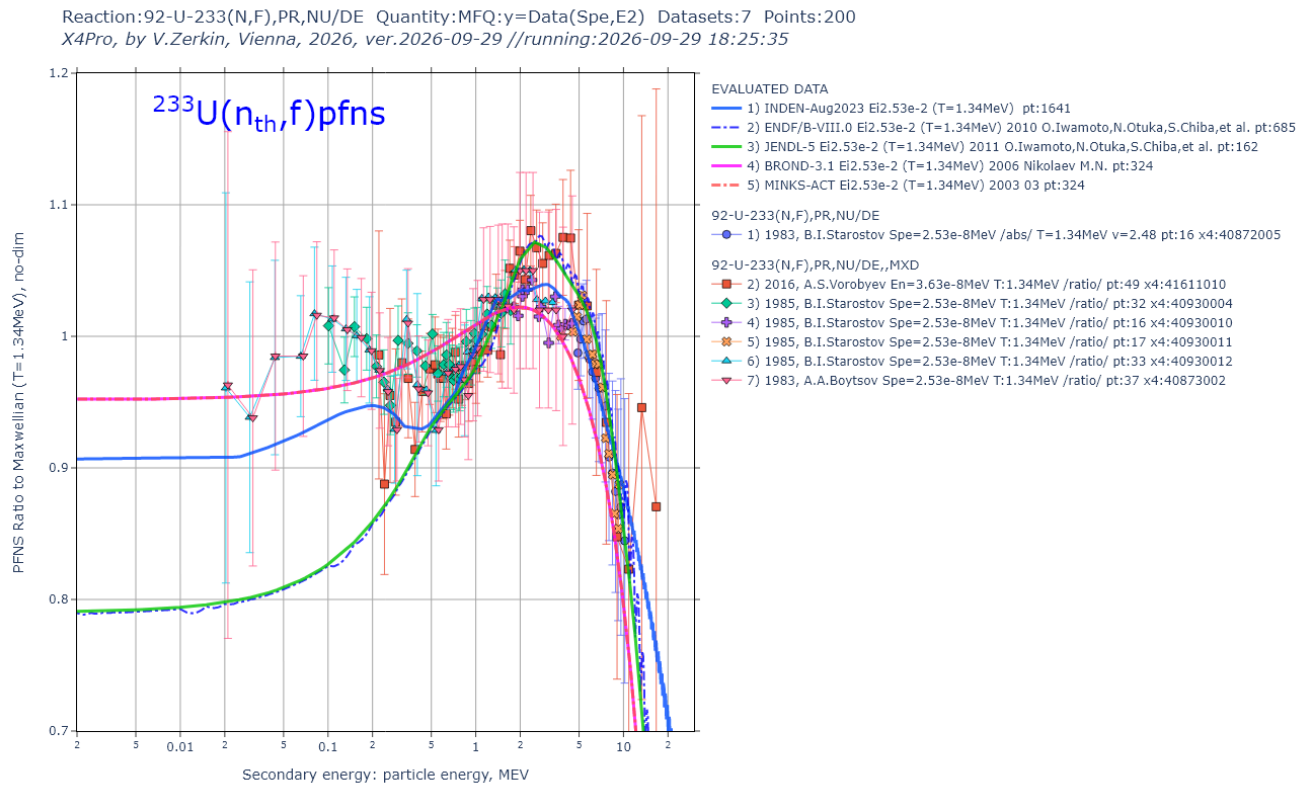
35) part2-B-pfns/out00/pfns232th02_9.html.png



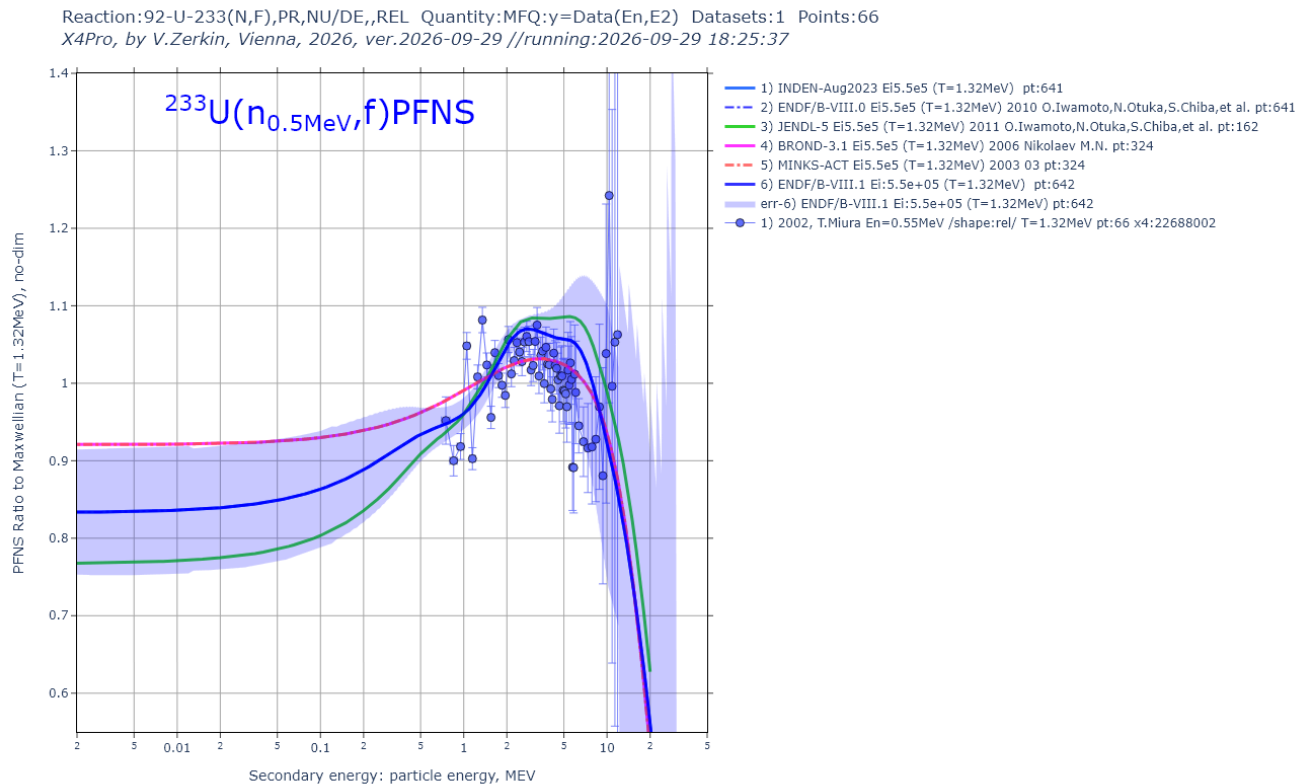
36) part2-B-pfns/out00/pfns232th14_7.html.png



37) part2-B-pfns/out00/pfns233u00_0th.html.png

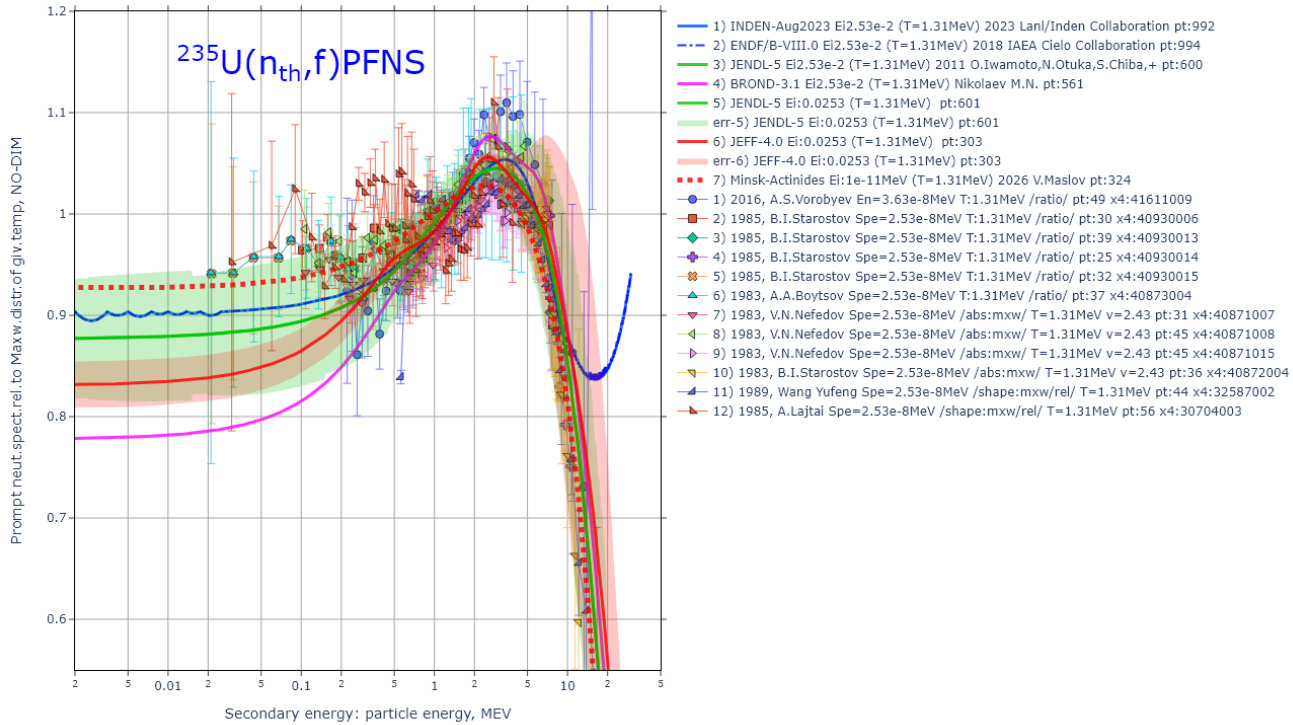


38) part2-B-pfns/out00/pfns233u00_5.html.png



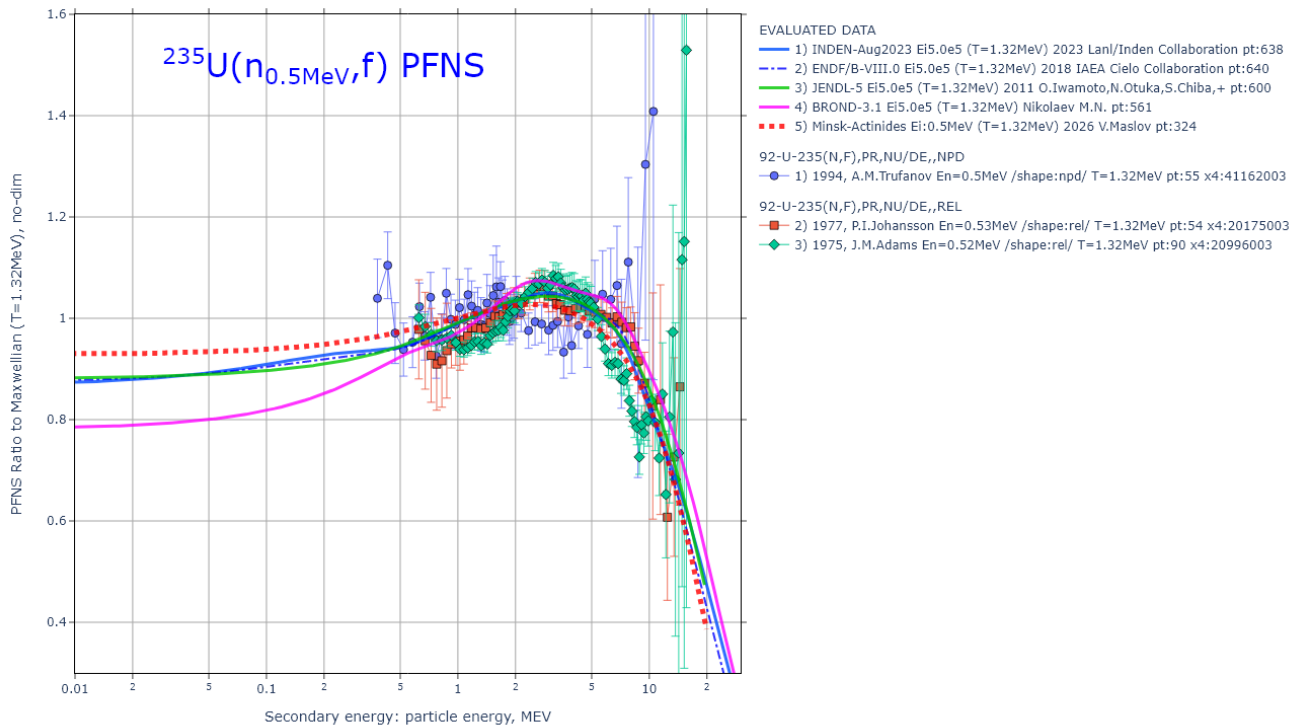
39) part2-B-pfns/out00/pfns235u00_0th.html.png

Reaction: 92-U-235(N,F),PR,NU/DE,,MXD Quantity: MFQ: y=Data(En,E2) Datasets: 12 Points: 469
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-29 //running: 2026-09-29 18:25:39



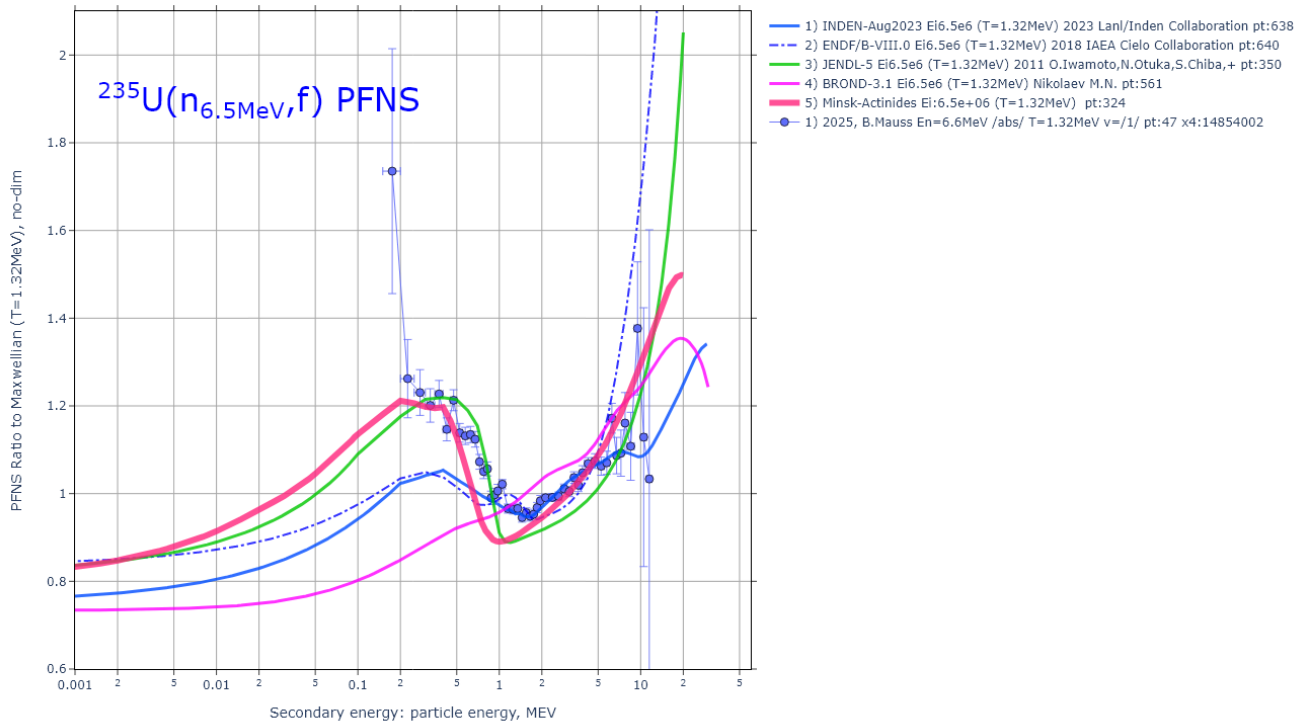
40) part2-B-pfns/out00/pfns235u00_5.html.png

Reaction: 92-U-235(N,F),PR,NU/DE,,NPD Quantity: MFQ: y=Data(En,E2) Datasets: 3 Points: 199
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-29 //running: 2026-09-29 18:25:41



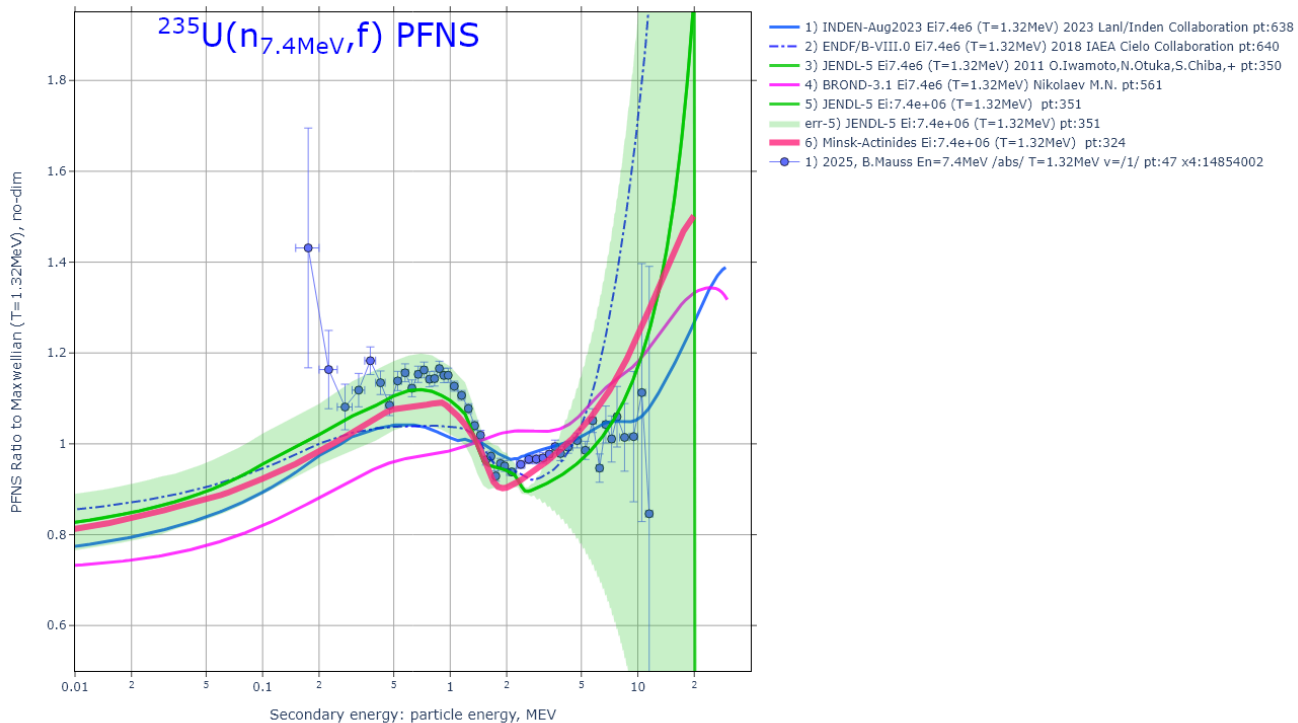
41) part2-B-pfns/out00/pfns235u06_6.html.png

Reaction:92-U-235(N,F),PR,NU/DE Quantity:MFQ:y=Data(En,E2) Datasets:1 Points:47
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-29 //running:2026-09-29 18:25:48

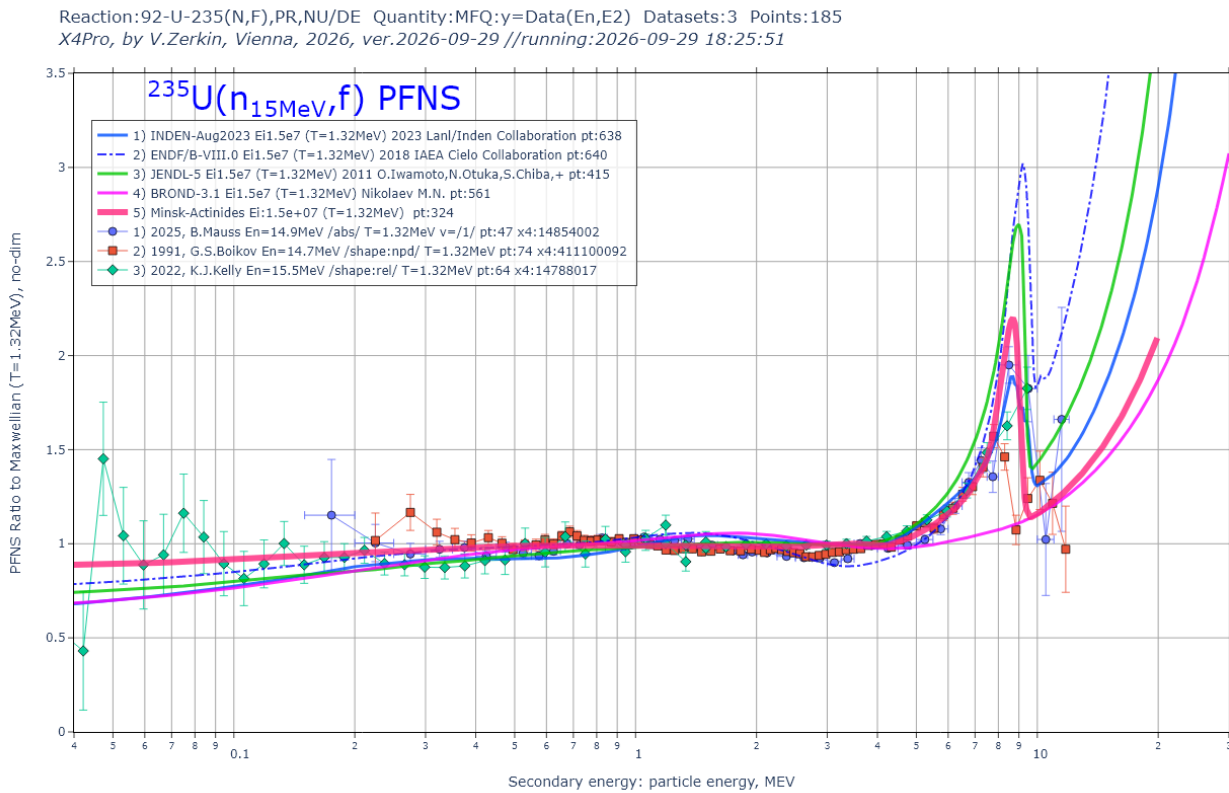


42) part2-B-pfns/out00/pfns235u07_4.html.png

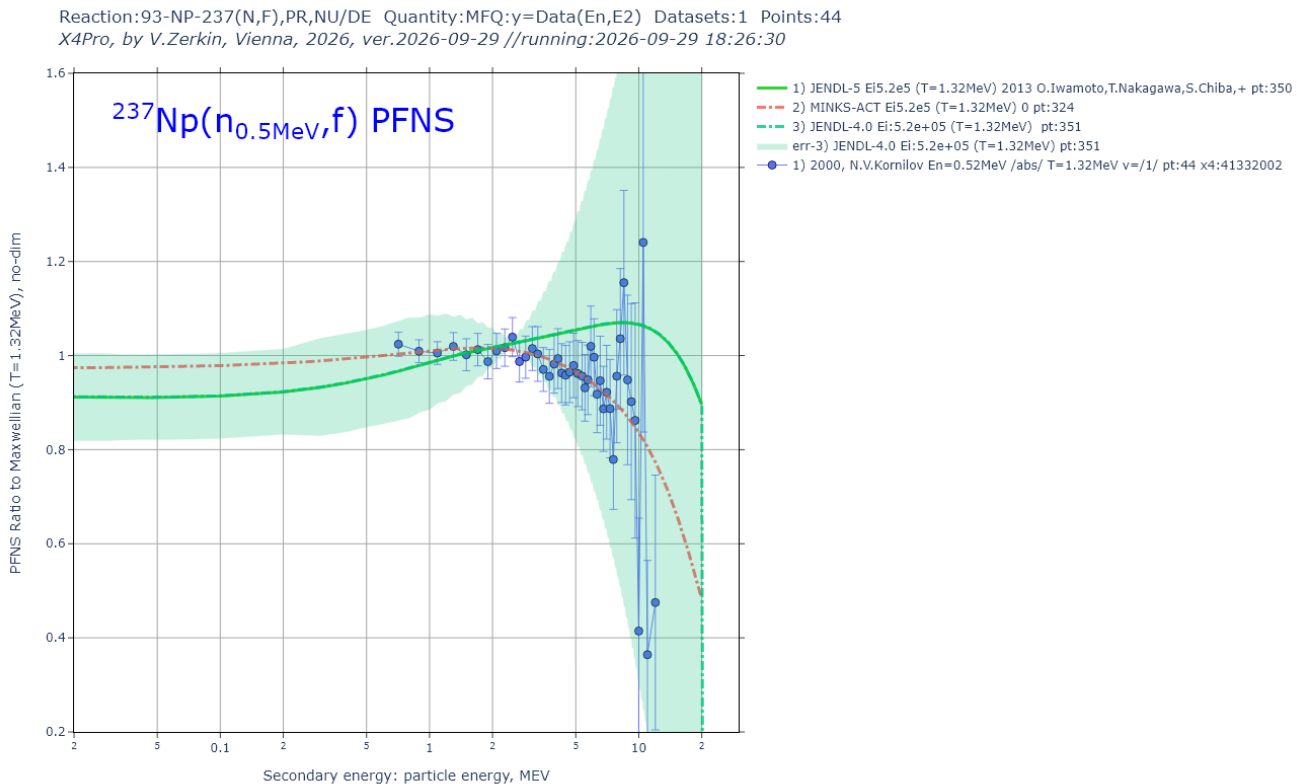
Reaction:92-U-235(N,F),PR,NU/DE Quantity:MFQ:y=Data(En,E2) Datasets:1 Points:47
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-29 //running:2026-09-29 18:25:44



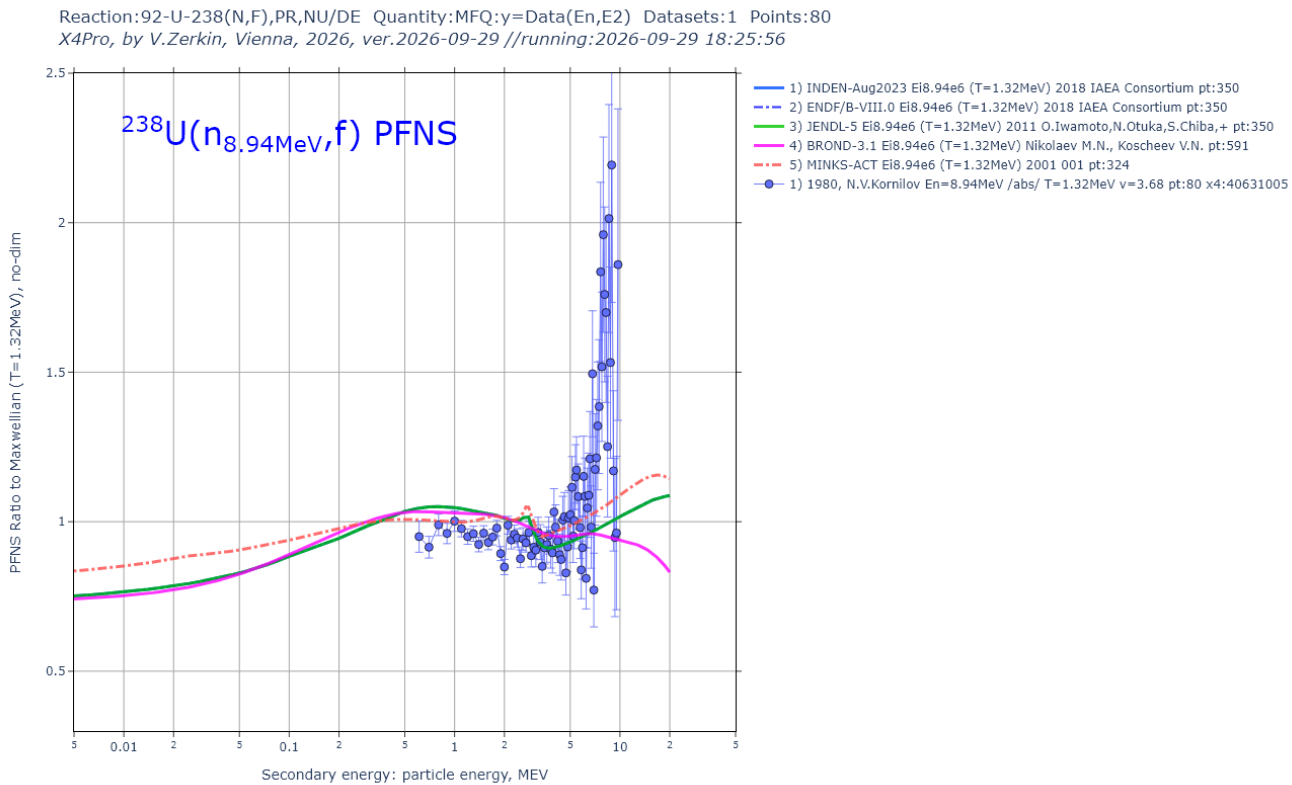
43) part2-B-pfns/out00/pfns235u15.html.png



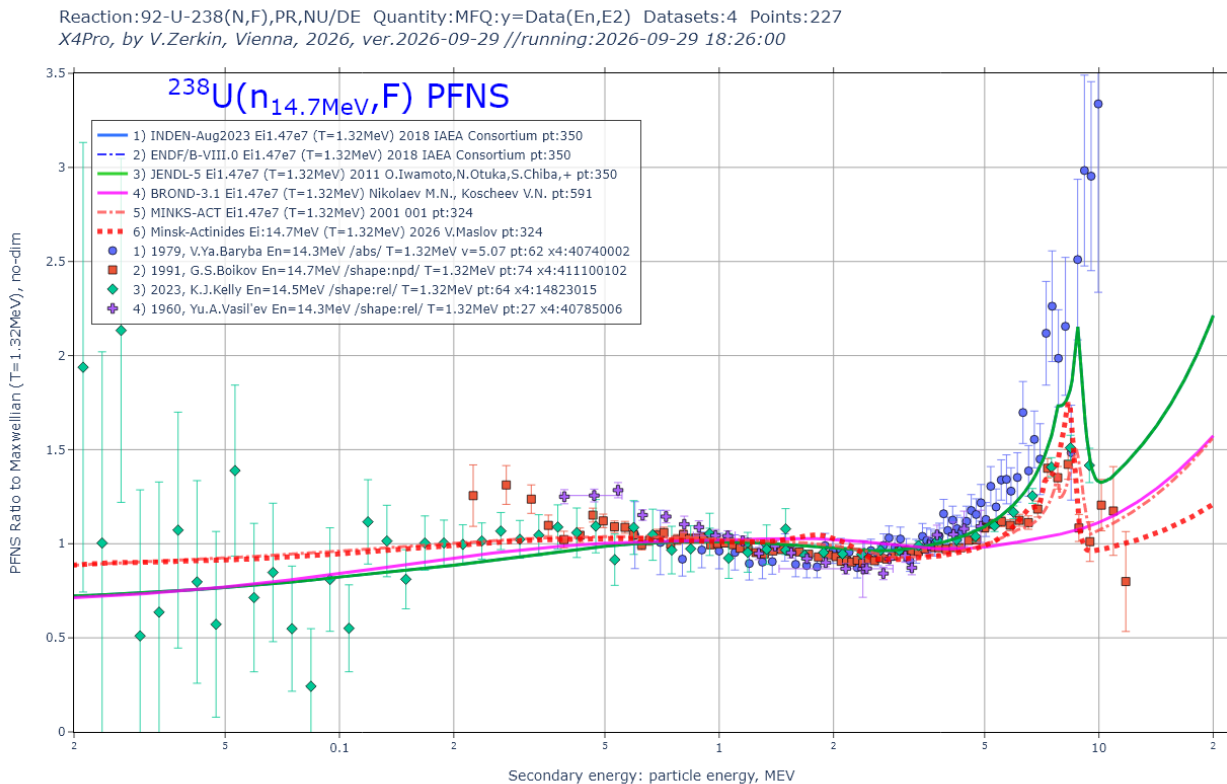
44) part2-B-pfns/out00/pfns237np00_5.html.png



45) part2-B-pfns/out00/pfns238u08_94.html.png

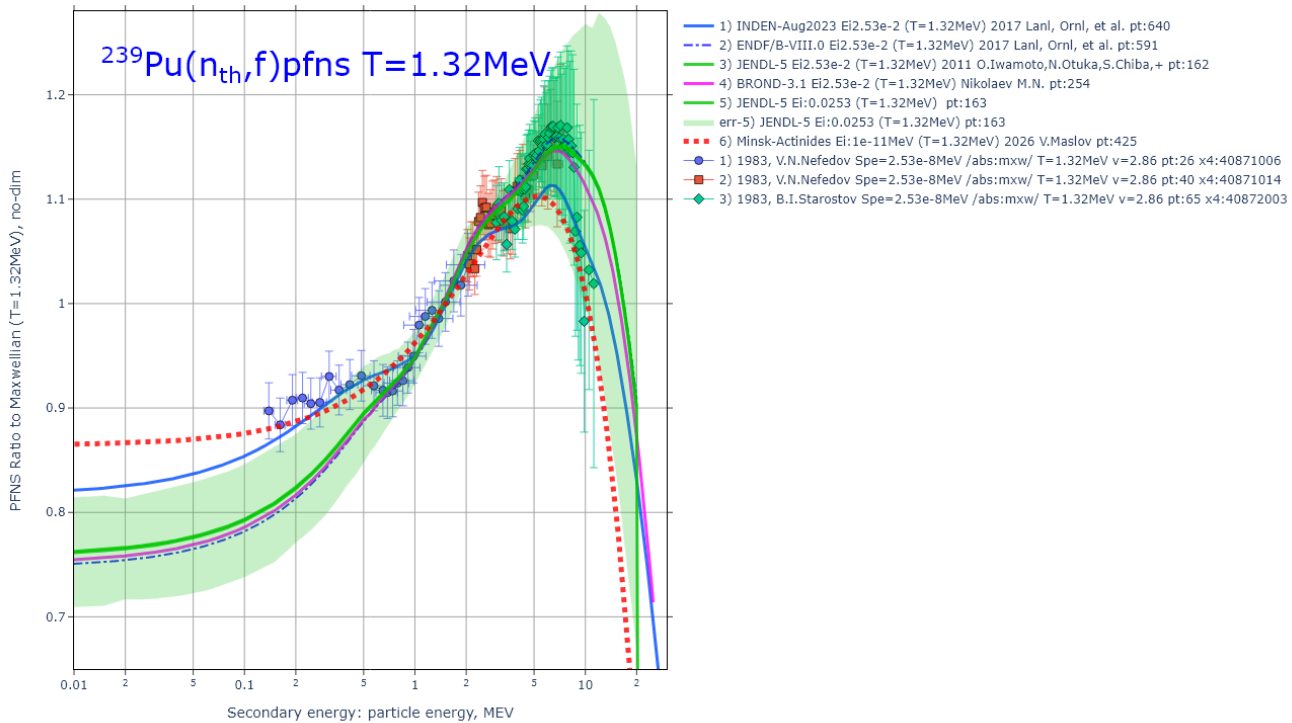


46) part2-B-pfns/out00/pfns238u14_7.html.png



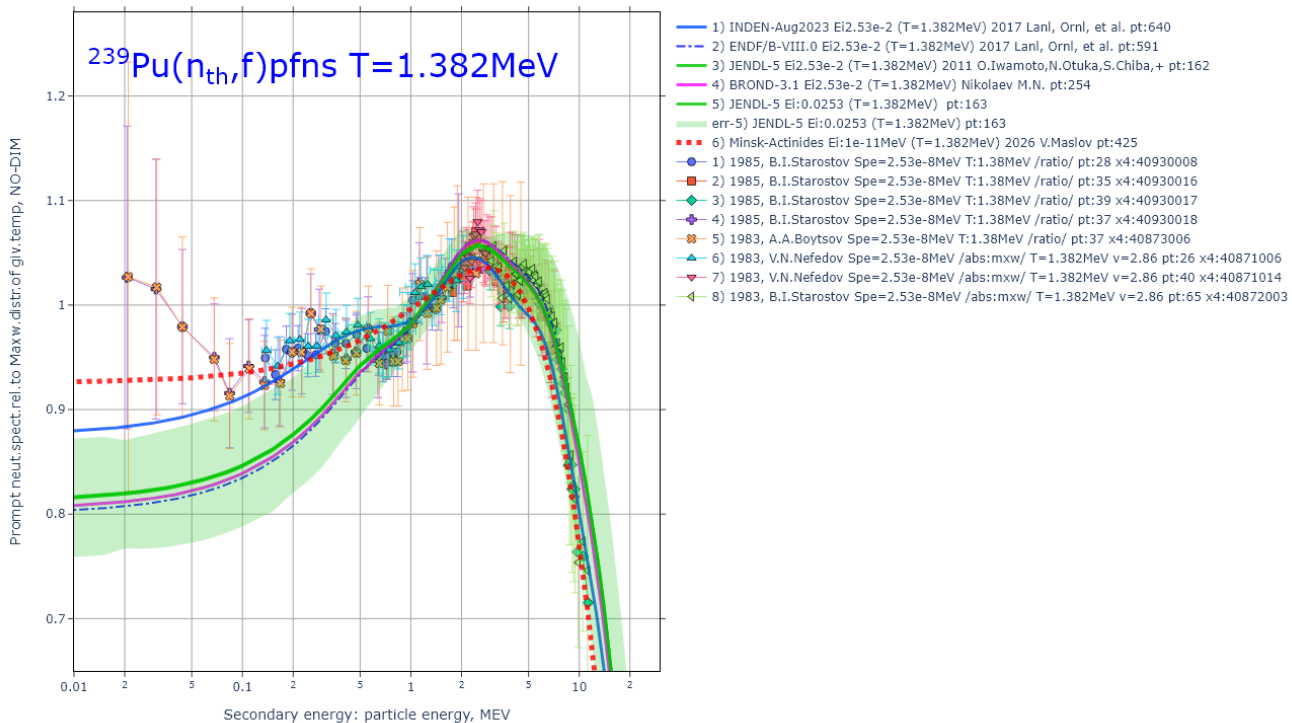
47) part2-B-pfns/out00/pfns239pu00_0th.html.png

Reaction: 94-PU-239(N,F),PR,NU/DE,,MXW Quantity: MFQ: y=Data(Spe,E2) Datasets: 3 Points: 131
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-29 //running: 2026-09-29 18:26:02



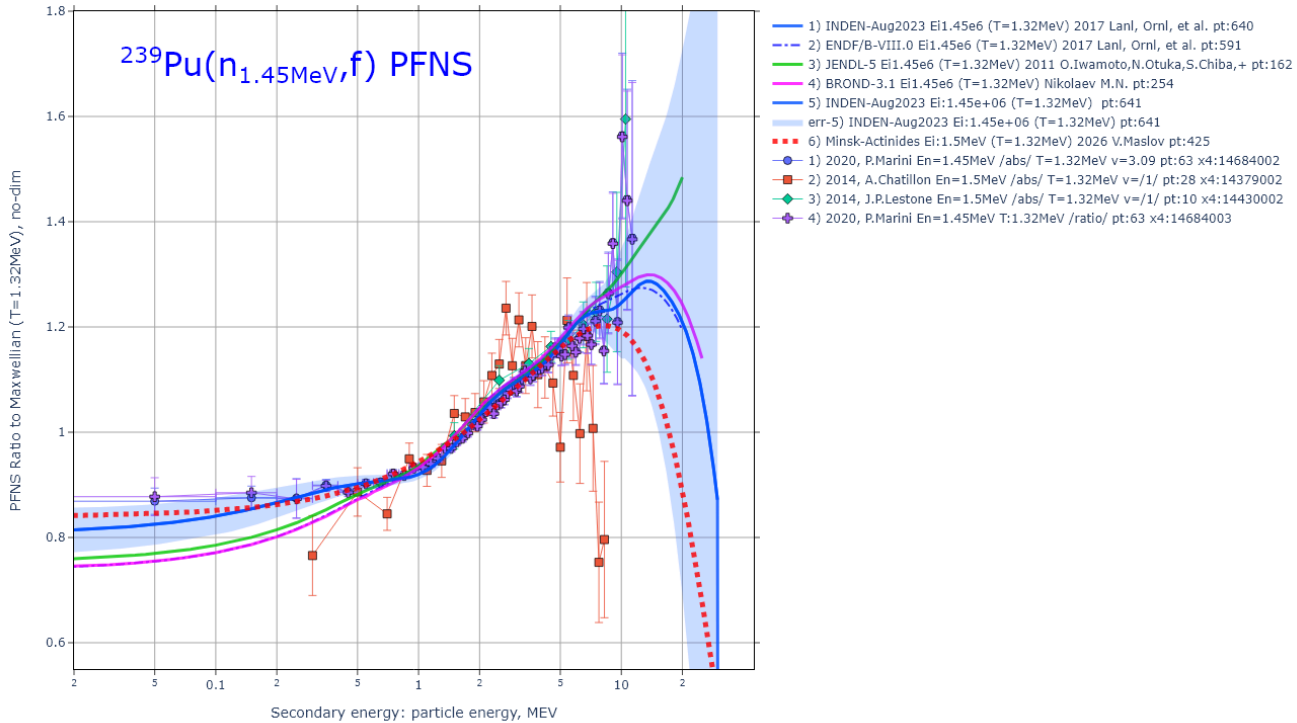
48) part2-B-pfns/out00/pfns239pu00_0th138.html.png

Reaction: 94-PU-239(N,F),PR,NU/DE,,MXD Quantity: MFQ: y=Data(Spe,E2) Datasets: 8 Points: 307
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-29 //running: 2026-09-29 18:26:06



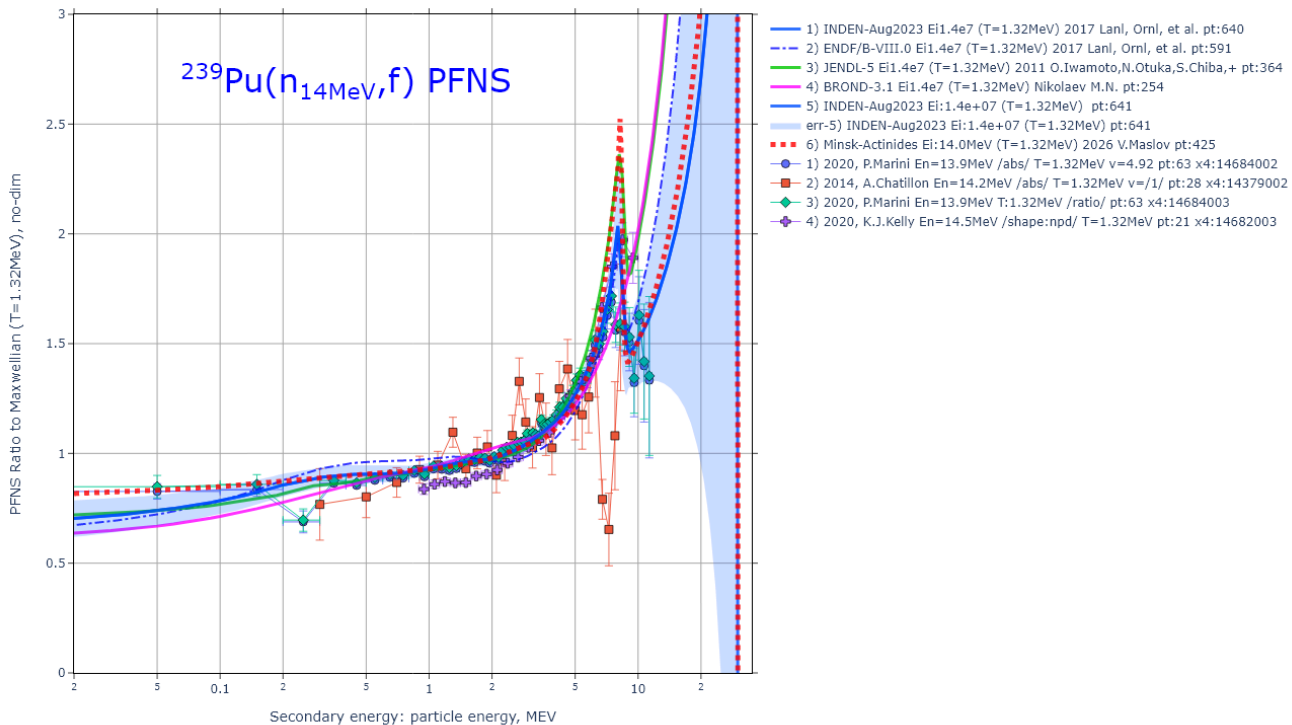
49) part2-B-pfns/out00/pfns239pu01_45.html.png

Reaction:94-PU-239(N,F),PR,NU/DE Quantity:MFQ:y=Data(En,E2) Datasets:4 Points:164/165
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-29 //running:2026-09-29 18:26:10



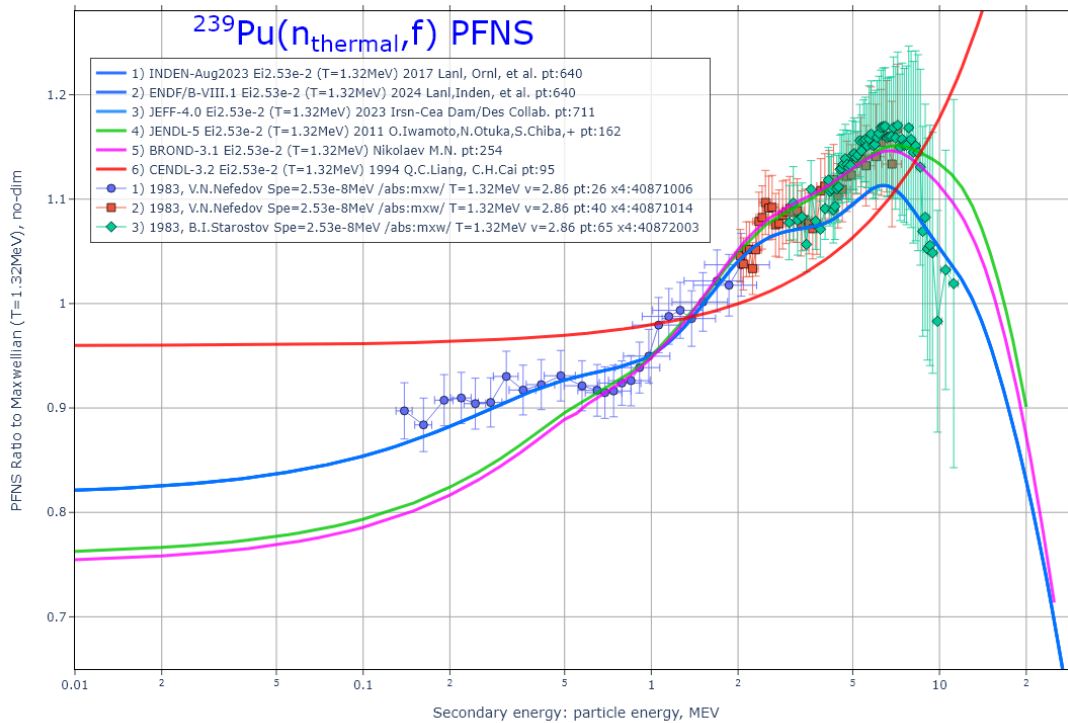
50) part2-B-pfns/out00/pfns239pu14.html.png

Reaction:94-PU-239(N,F),PR,NU/DE Quantity:MFQ:y=Data(En,E2) Datasets:4/5 Points:175/215
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-29 //running:2026-09-29 18:26:17



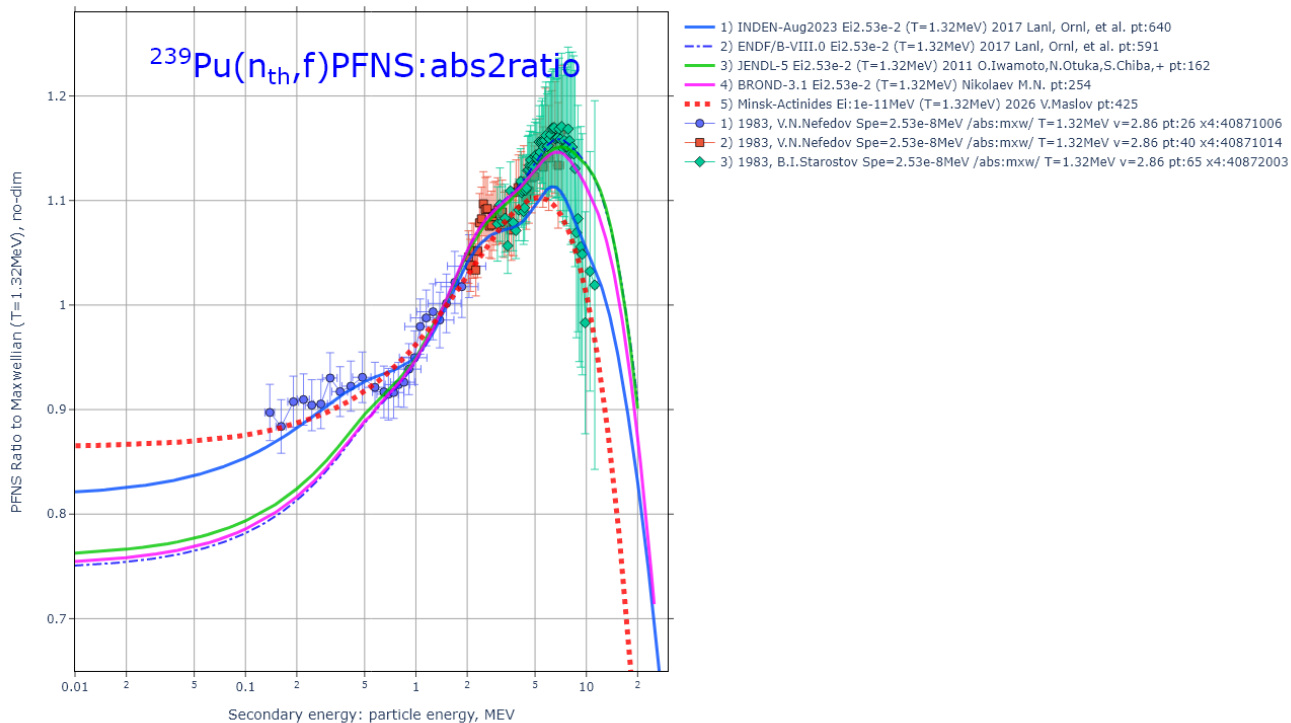
51) part2-B-pfns/out00/pfns239pu_t2.html.png

Reaction:94-PU-239(N,F),PR,NU/DE,,MXW Quantity:MFQ:y=Data(Spe,E2) Datasets:3 Points:131
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-20 //running:2026-09-22 19:02:32



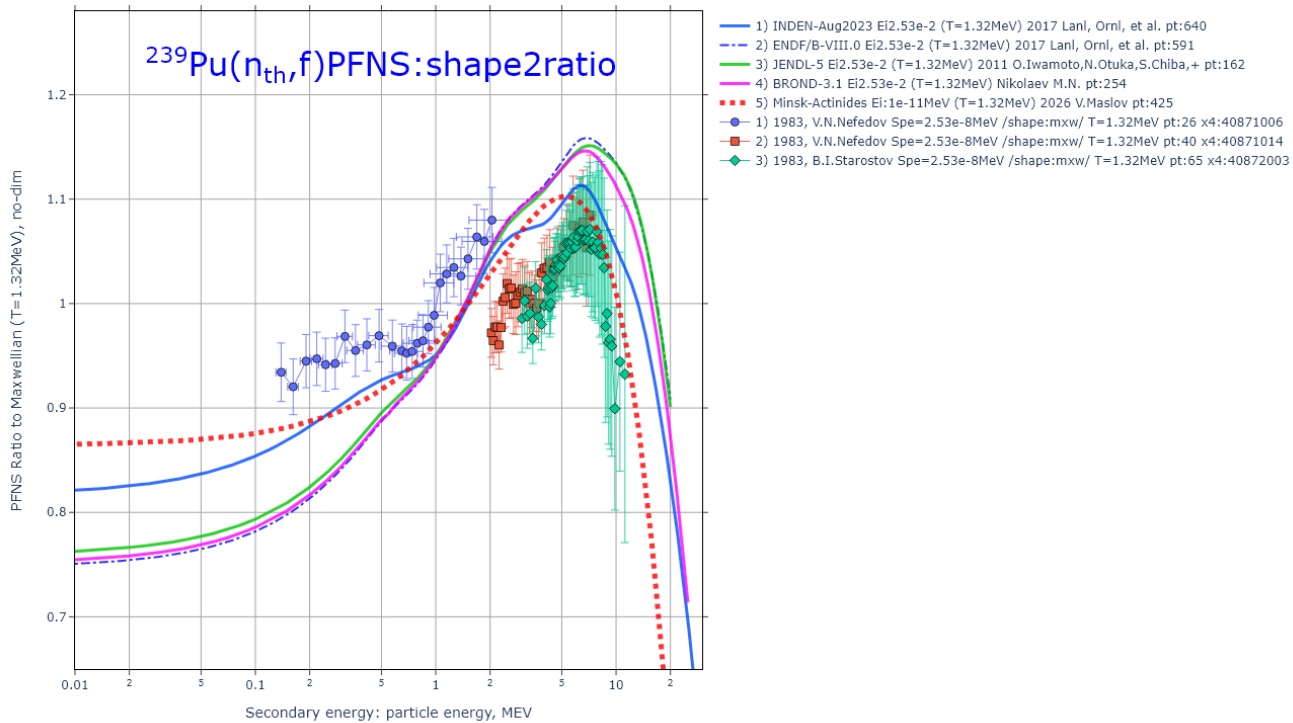
52) part2-B-pfns/out00/pfns239pu_t_abs.html.png

Reaction:94-PU-239(N,F),PR,NU/DE,,MXW Quantity:MFQ:y=Data(Spe,E2) Datasets:3 Points:131
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-29 //running:2026-09-29 18:26:34



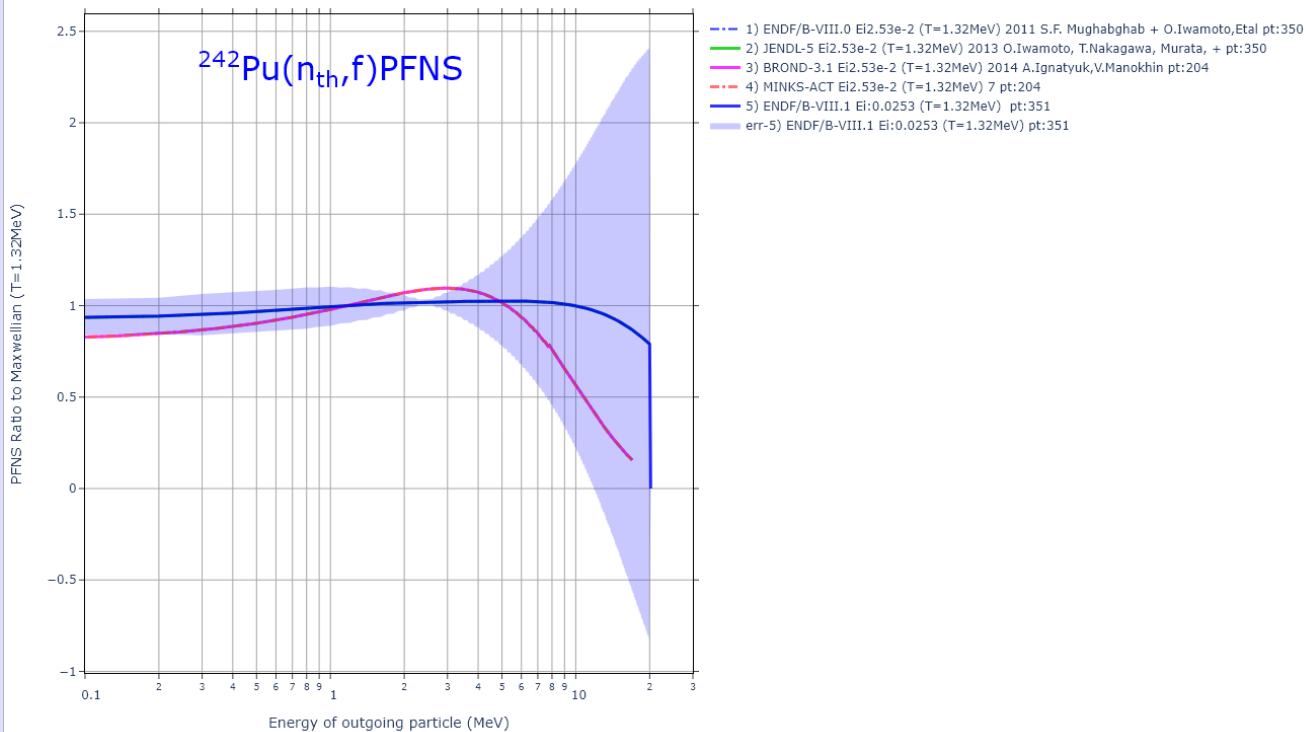
53) part2-B-pfns/out00/pfns239pu_t_shp.html.png

Reaction: 94-PU-239(N,F),PR,NU/DE,,MXW Quantity: MFQ:y=Data(Spe,E2) Datasets: 3 Points: 131
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-29 //running:2026-09-29 18:26:38



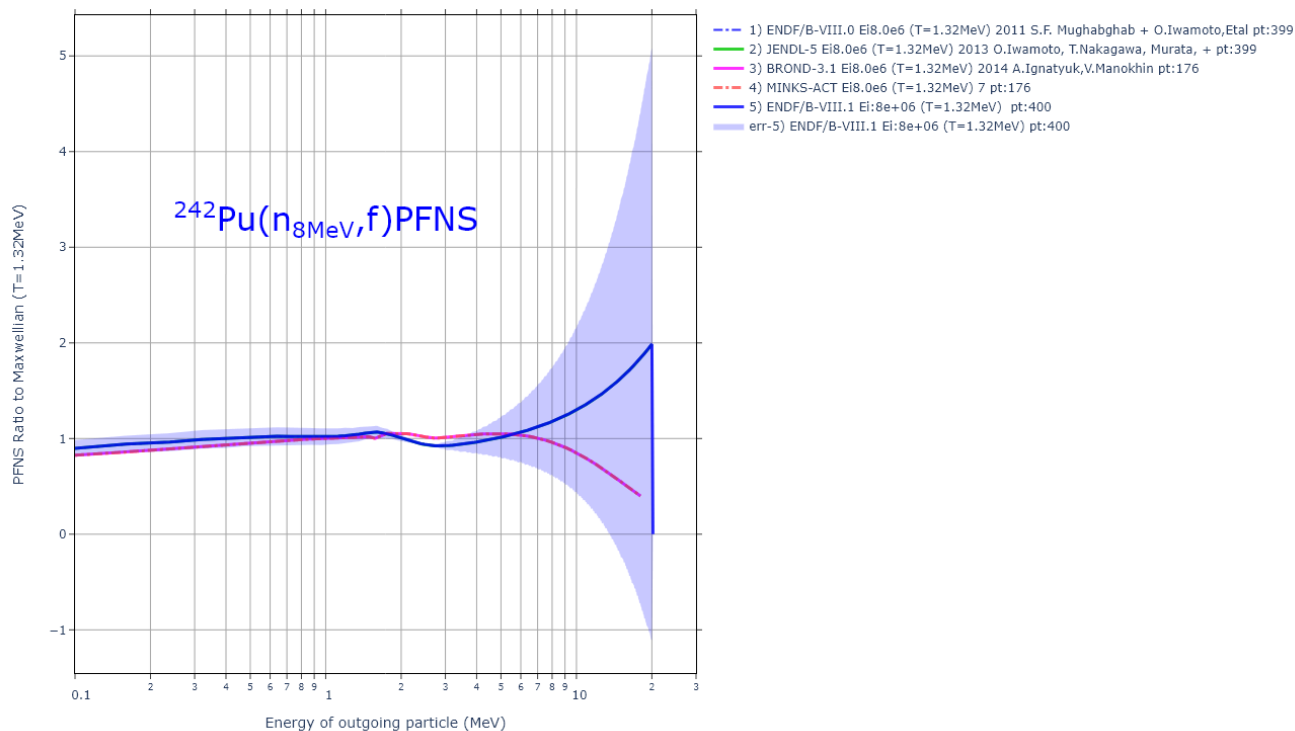
54) part2-B-pfns/out00/pfns242pu00_0th.html.png

Reaction: Pu-242(n,f),pfns ENDF: MF=5, MT=18, thermal, $T=1.32\text{MeV}$
X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-29 //running:2026-09-29 18:26:42



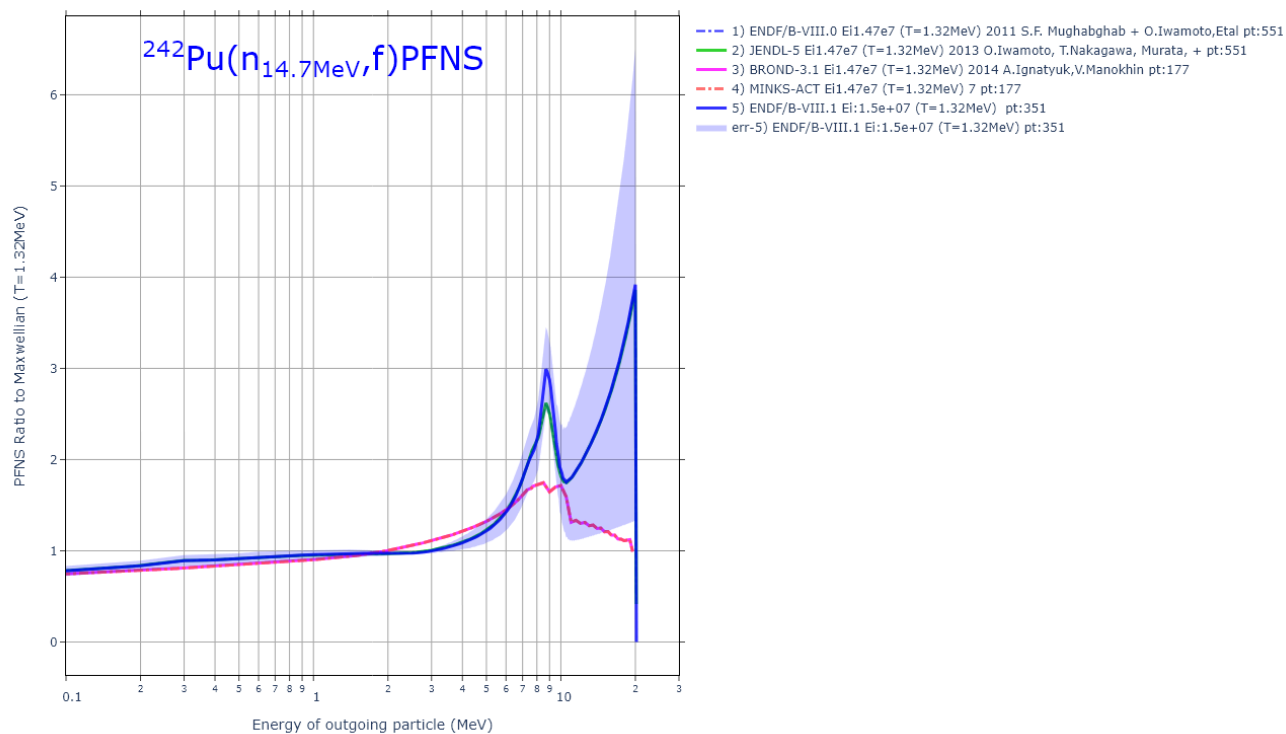
55) part2-B-pfns/out00/pfns242pu08.html.png

Reaction: Pu-242(n,f),pfns ENDF: MF=5, MT=18, Einc=8MeV, T=1.32MeV
 X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-29 //running:2026-09-29 18:26:46

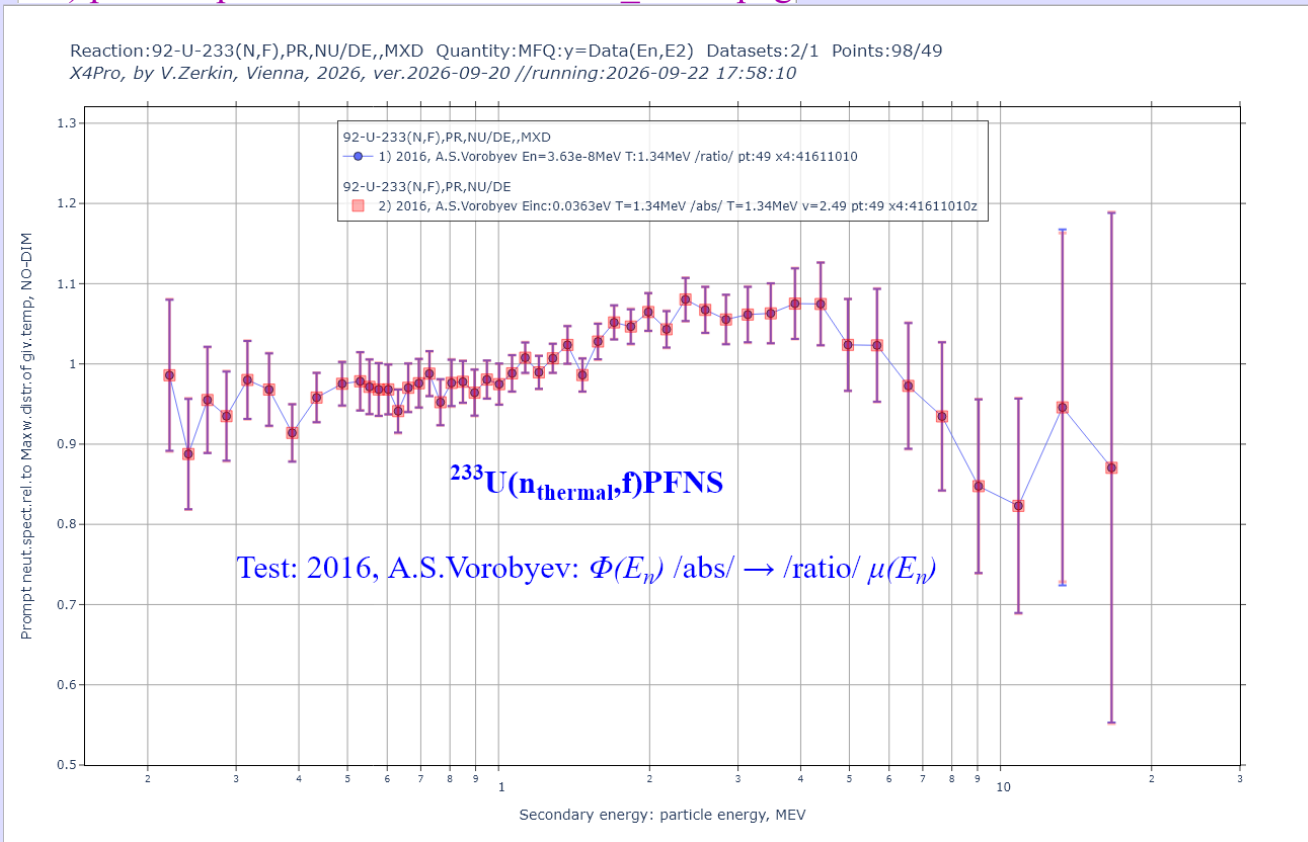


56) part2-B-pfns/out00/pfns242pu14_7.html.png

Reaction: Pu-242(n,f),pfns ENDF: MF=5, MT=18, Einc=14.7MeV, T=1.32MeV
 X4Pro, by V.Zerkin, Vienna, 2026, ver.2026-09-29 //running:2026-09-29 18:26:49

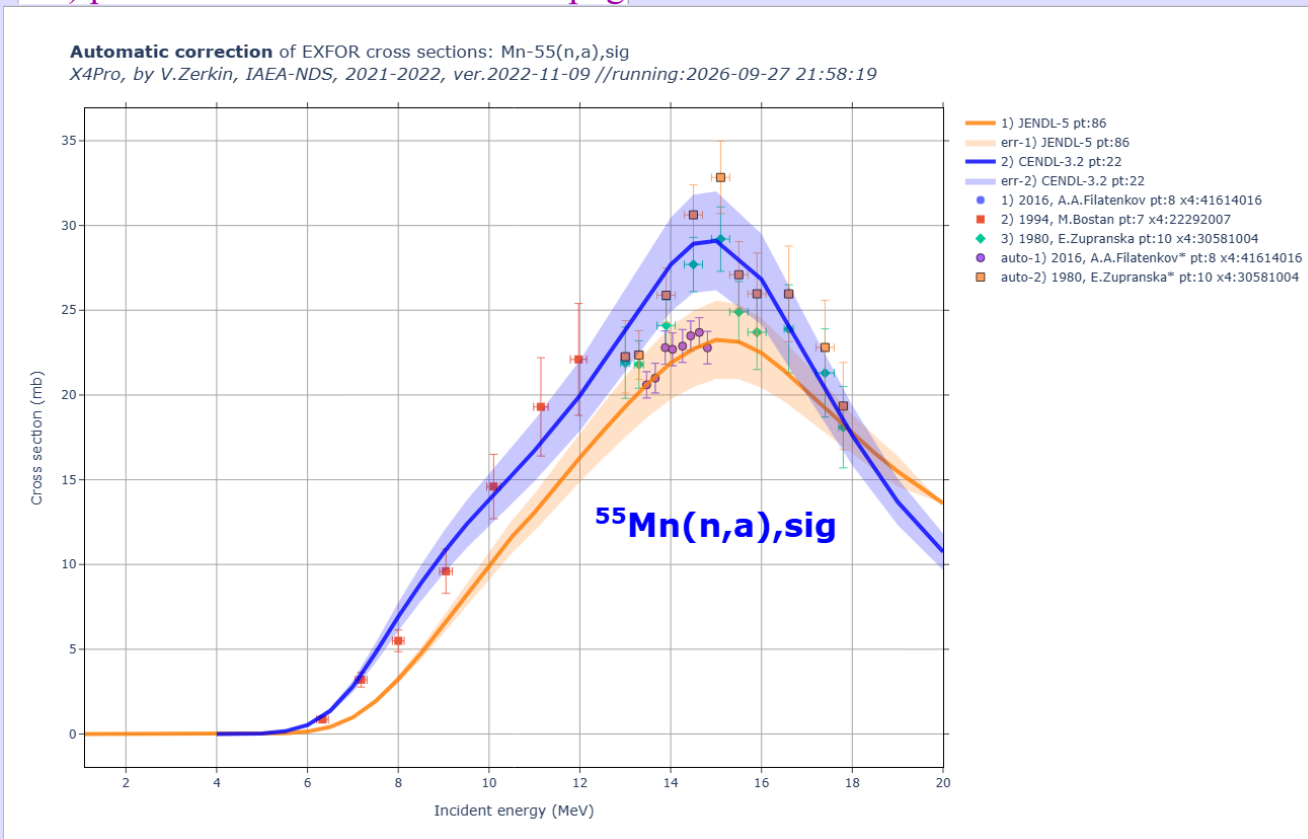


57) part2-B-pfns/out00/zabs2mxw233u_t.html.png



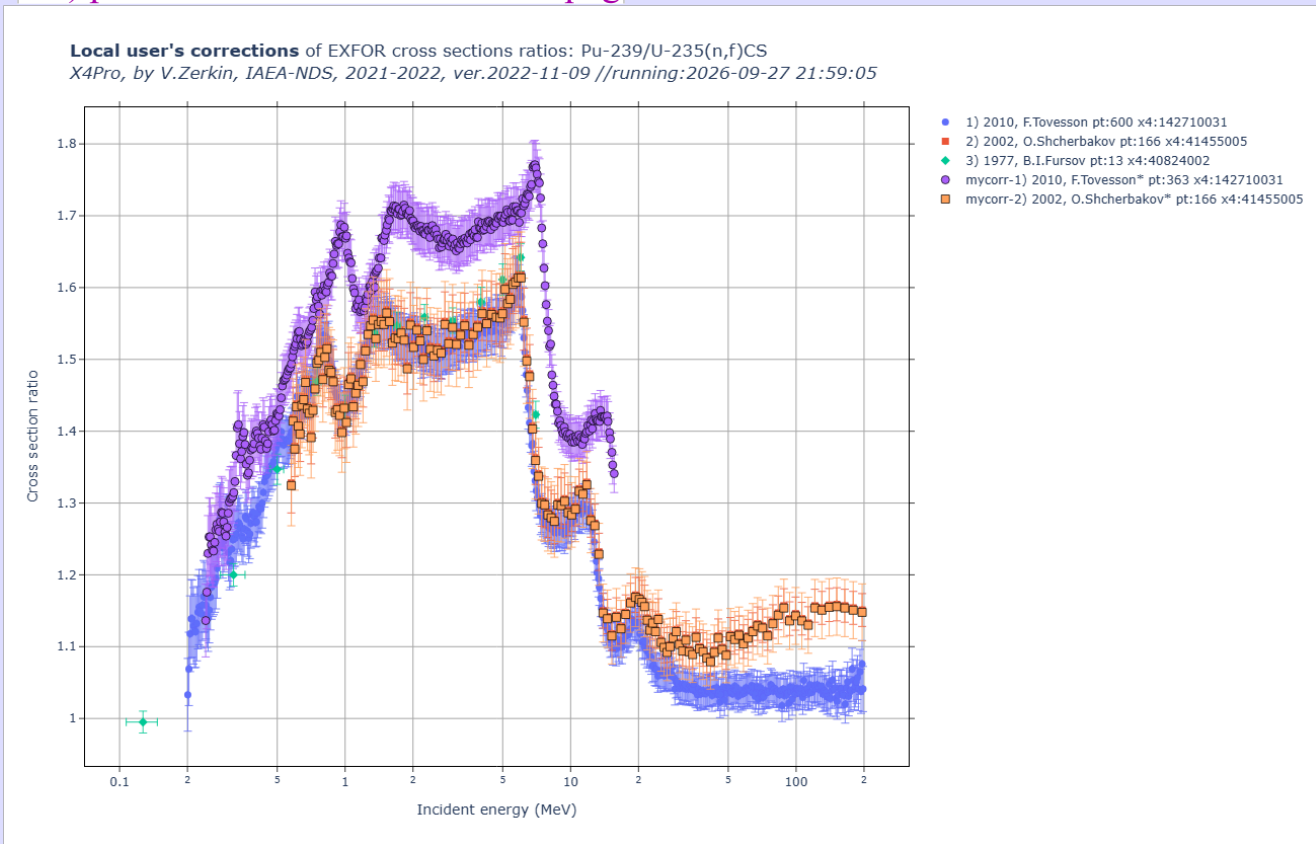
part3-1-auto1/ Automatically renormalize EXFOR CS using old monitor CS and new standards

58) part3-1-auto1/out00/auto1.html.png



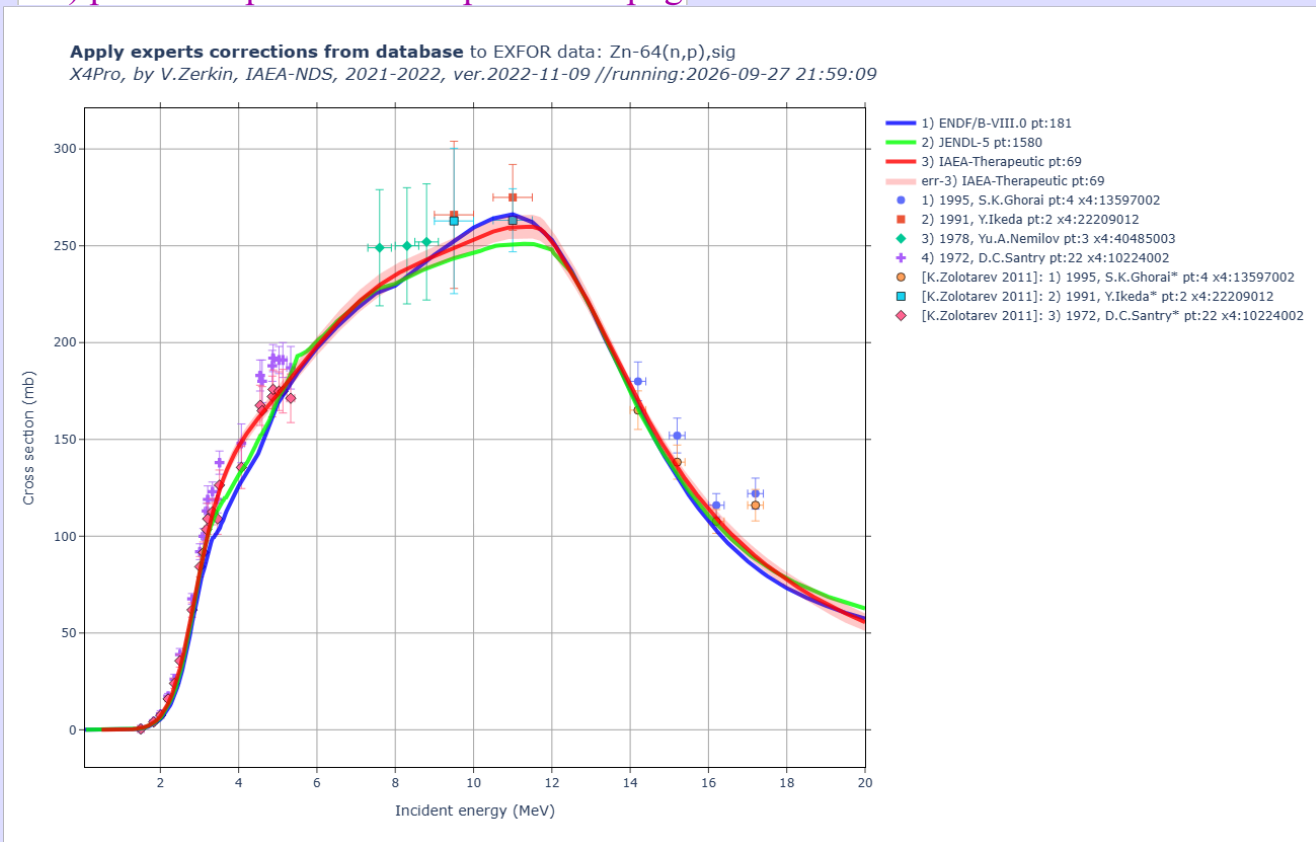
part3-2-user1/ *Correct data from EXFOR by user's program*

59) part3-2-user1/out00/user1.html.png



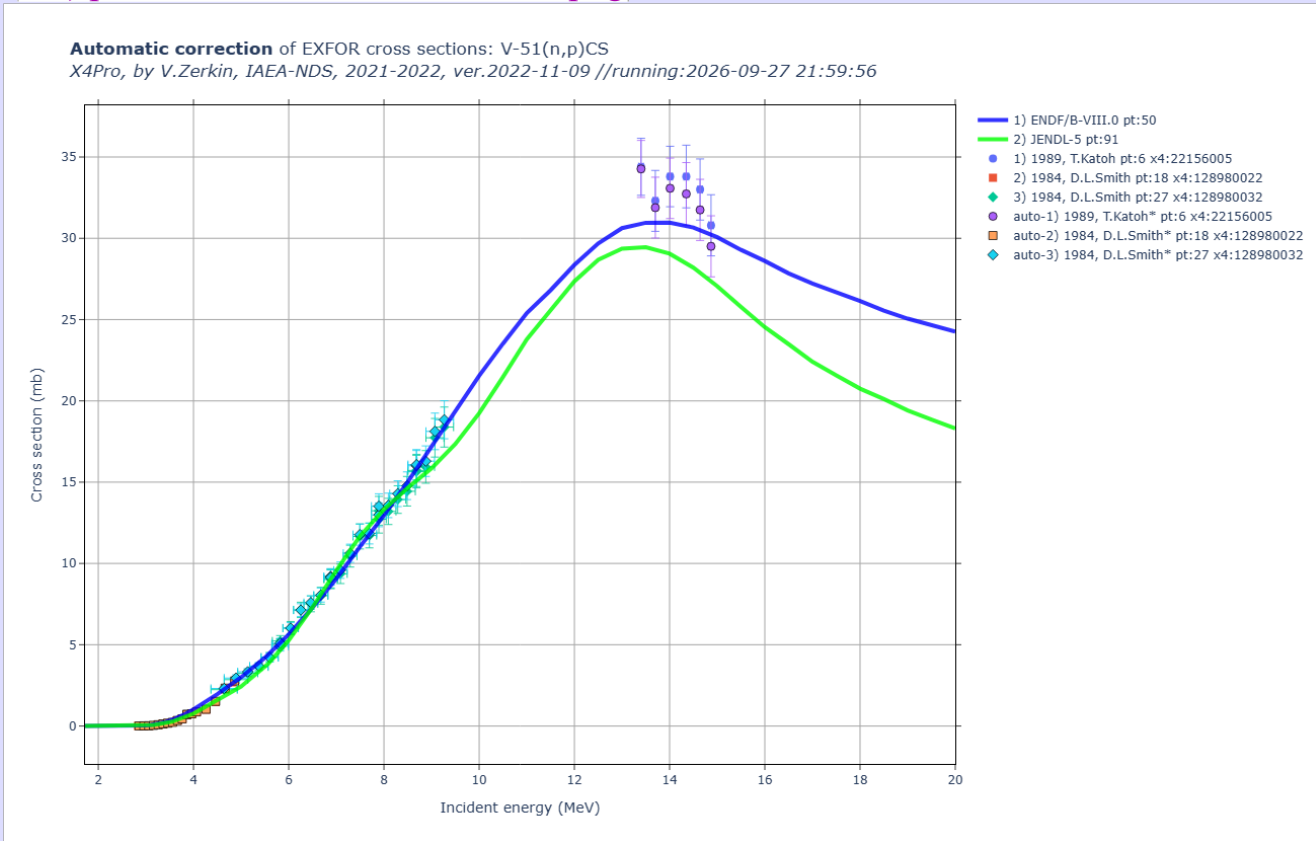
part3-3-expert1/ *Apply experts corrections stored in X4Pro database*

60) part3-3-expert1/out00/expert1.html.png



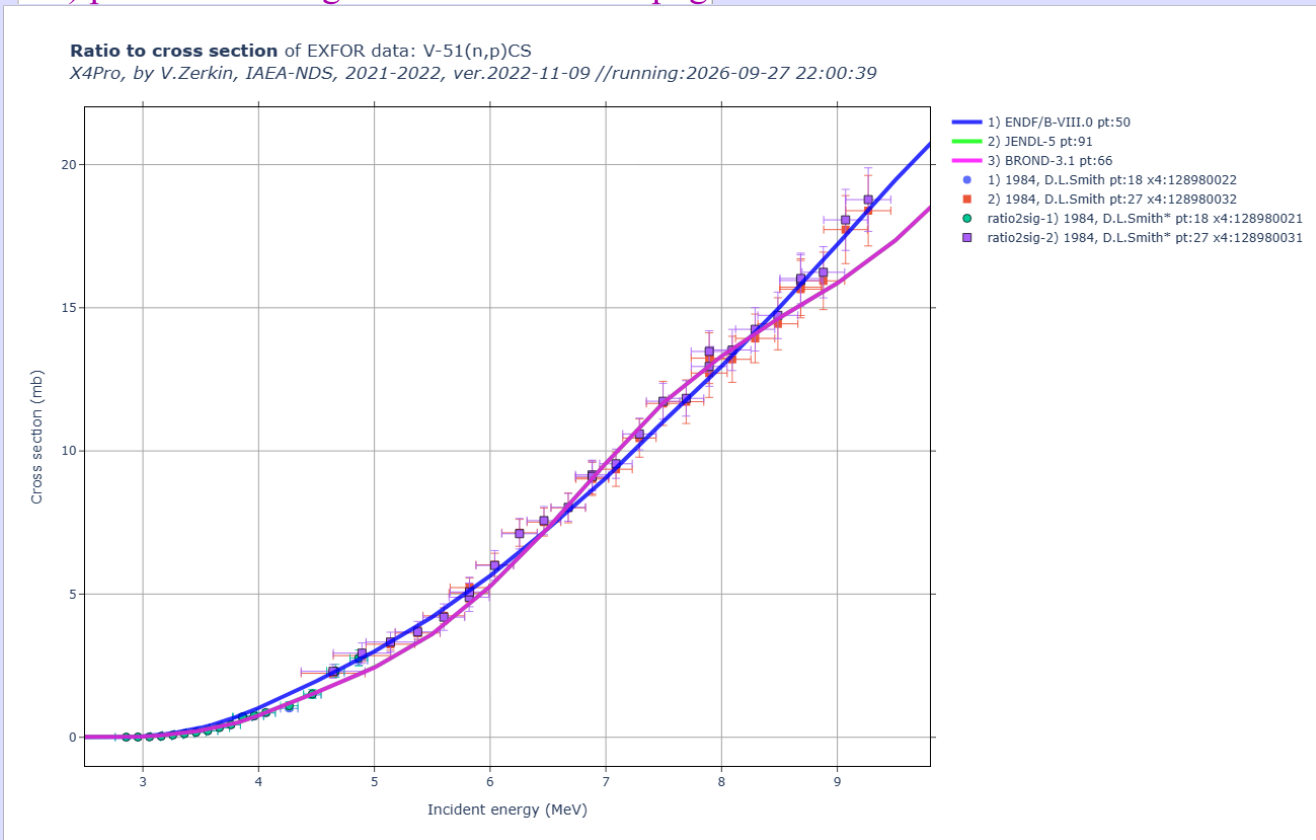
part3-4-auto2/ Automatically renormalize EXFOR CS using old monitor CS and new standards, old and new decay data

61) part3-4-auto2/out00/auto2.html.png



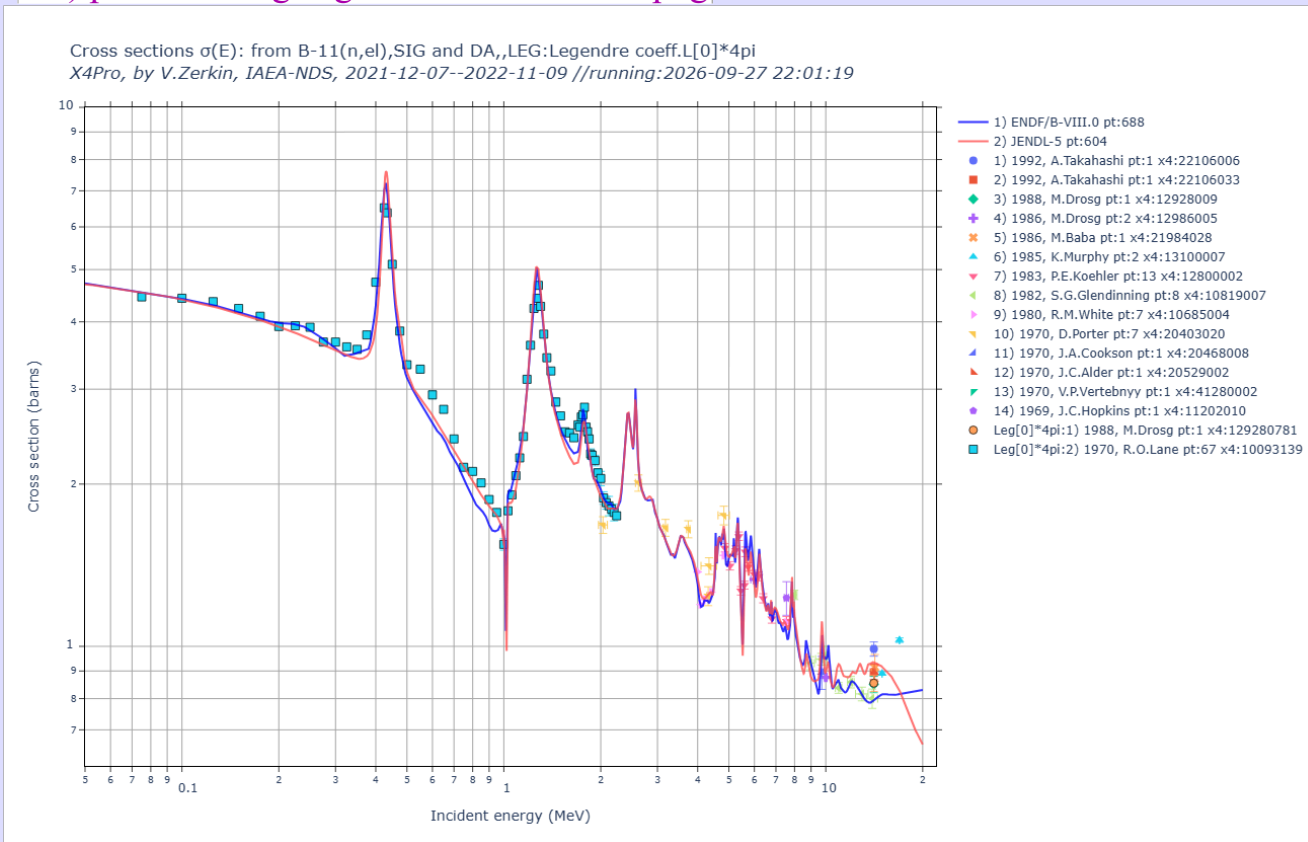
part3-5-ratio2sig/ Retrieve ratios, calculate CS and uncertainties using latest standards

62) part3-5-ratio2sig/out00/ratio2cs.html.png



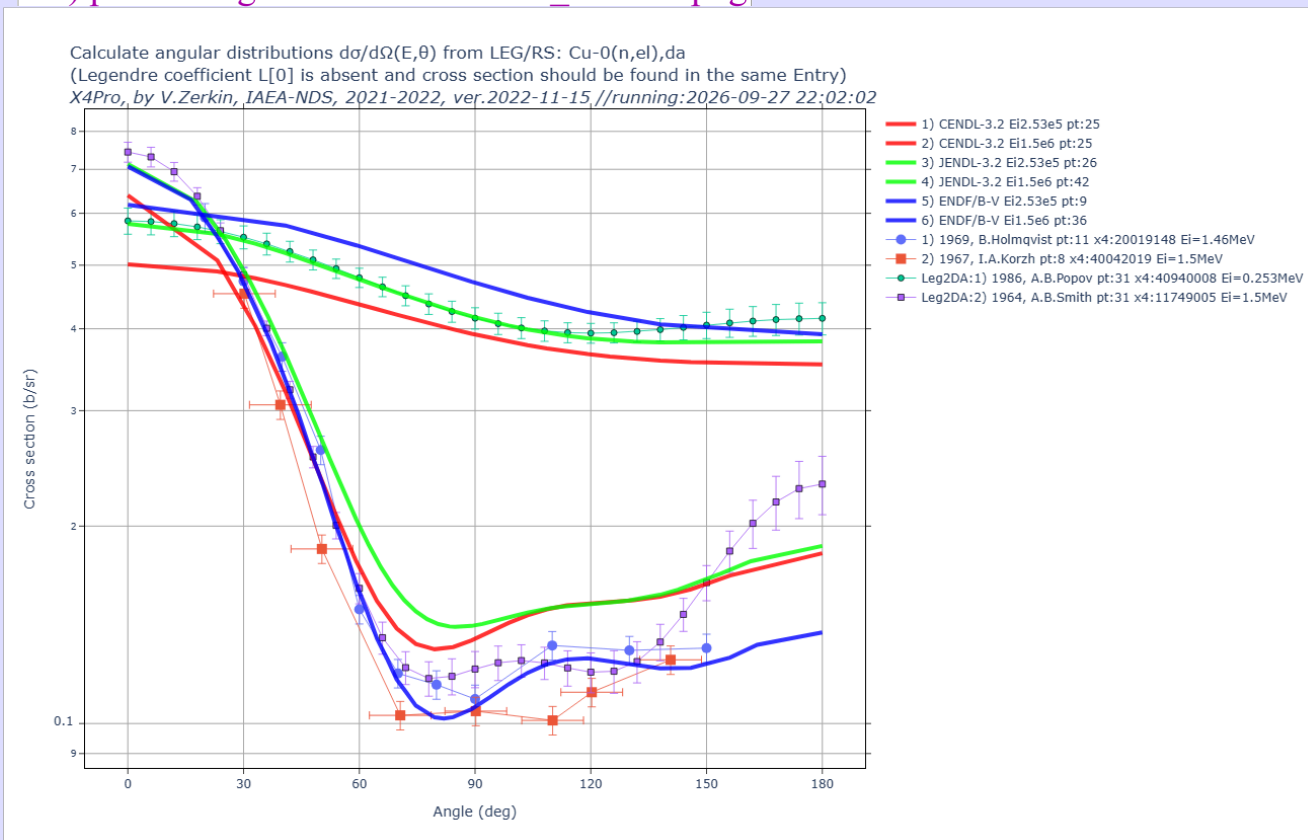
part3-6-daleg2sig/ Retrieve from EXFOR: Legendre coefficient $L[0]$, calculate cross sections and errors

63) part3-6-daleg2sig/out00/B11nel.html.png



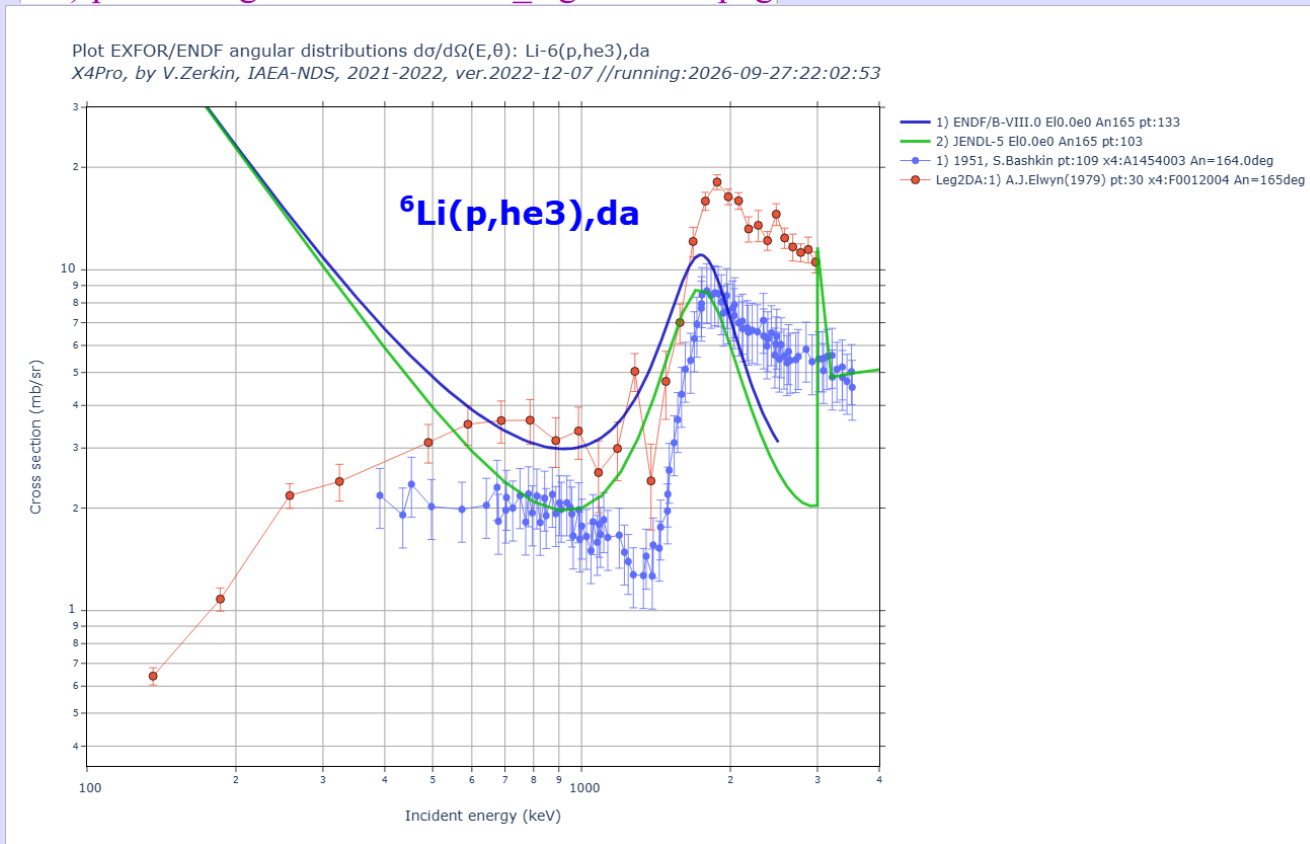
part3-7-legrs2da/ EXFOR LEG/RS and SIG from the same Entry; calc. missing Legendre coefficient $L[0]$; calc. angular distributions

64) part3-7-legrs2da/out00/Cu0nel_da.html.png



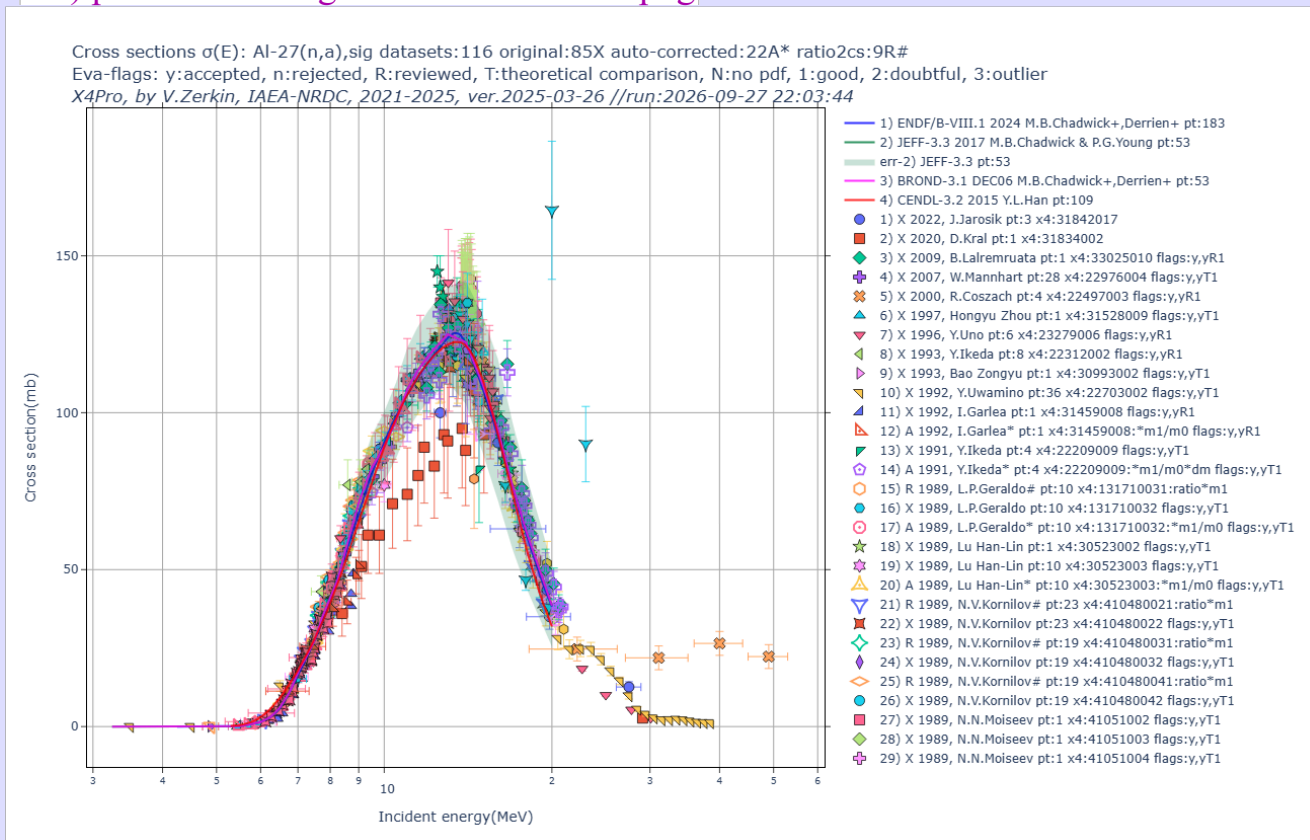
part3-8-leg2r33/ Retrieve Legendre coefficient calculate angular distributions store R33dat (draft of R33)

65) part3-8-leg2r33/out00/da1ei_leg2da.html.png

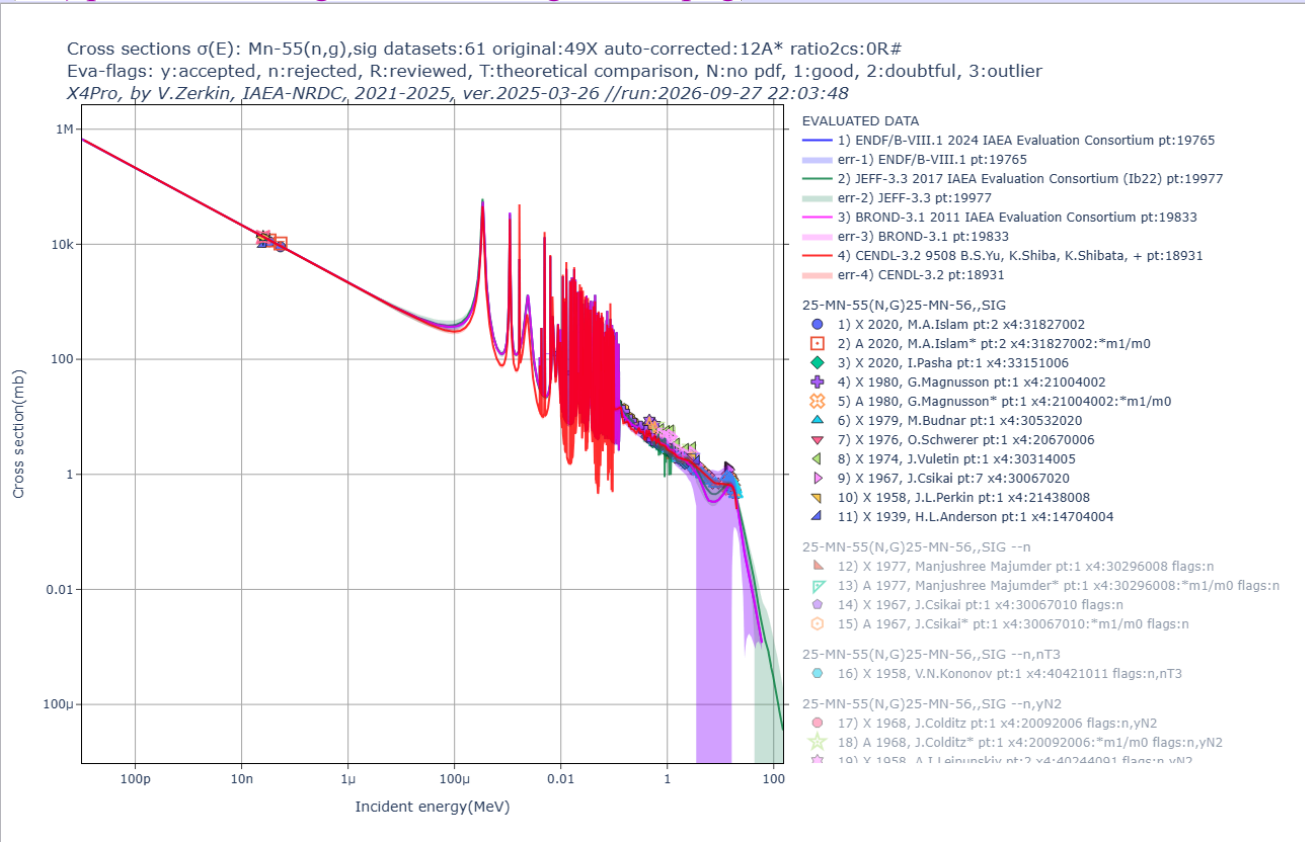


part6-1-eval2sig/ Retrieve CS, ratios, monit-data; renormalize to new standards and decay data; convert ratios to CS

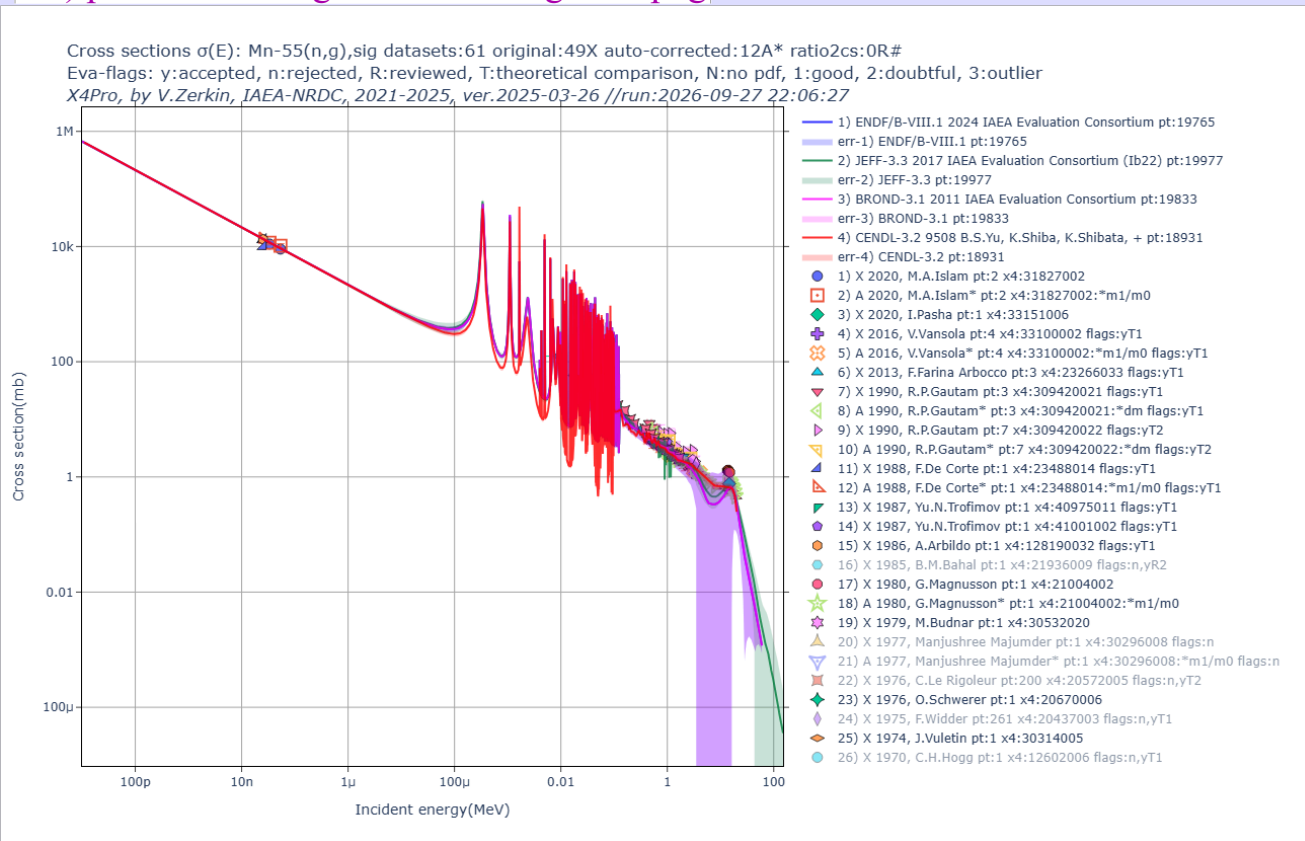
66) part6-1-eval2sig/out00/Al27na.html.png



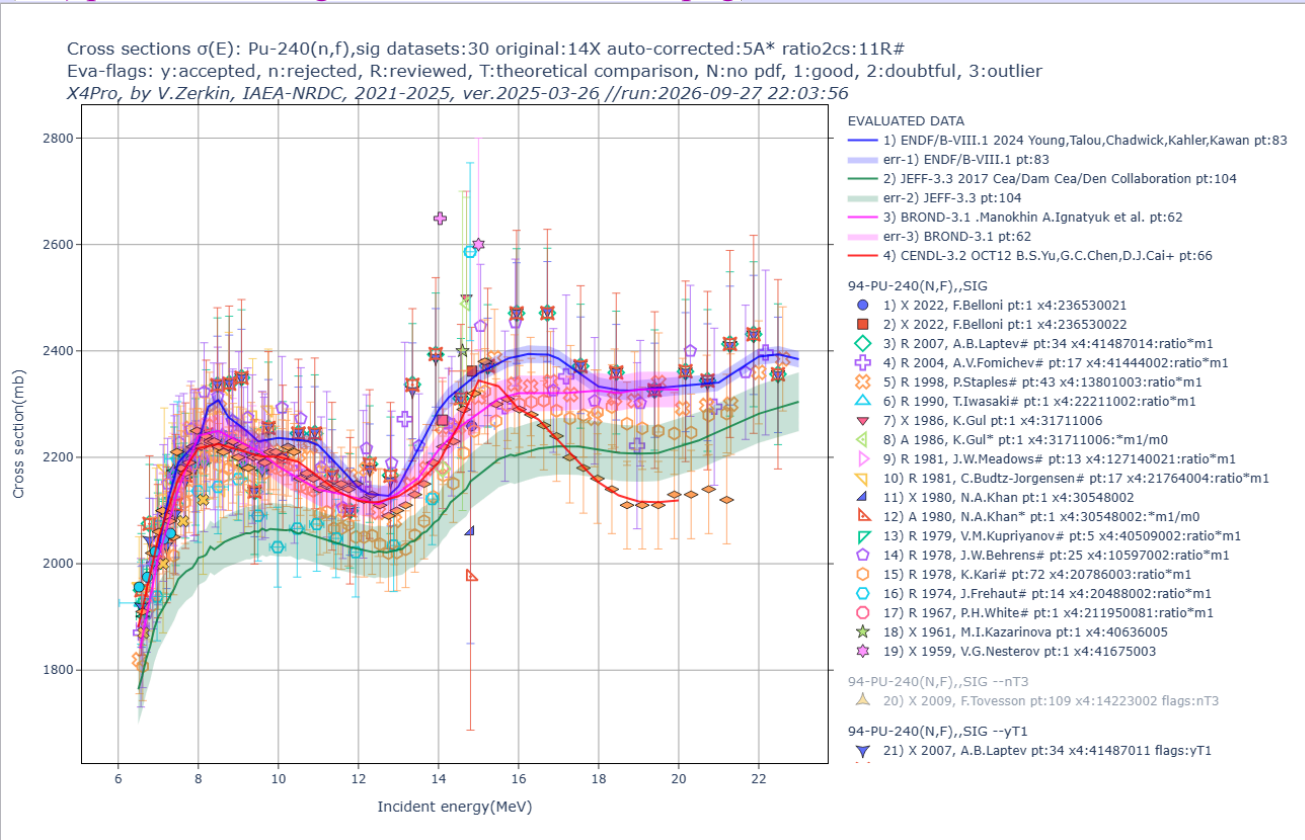
67) part6-1-eval2sig/out00/Mn55ng-n.html.png



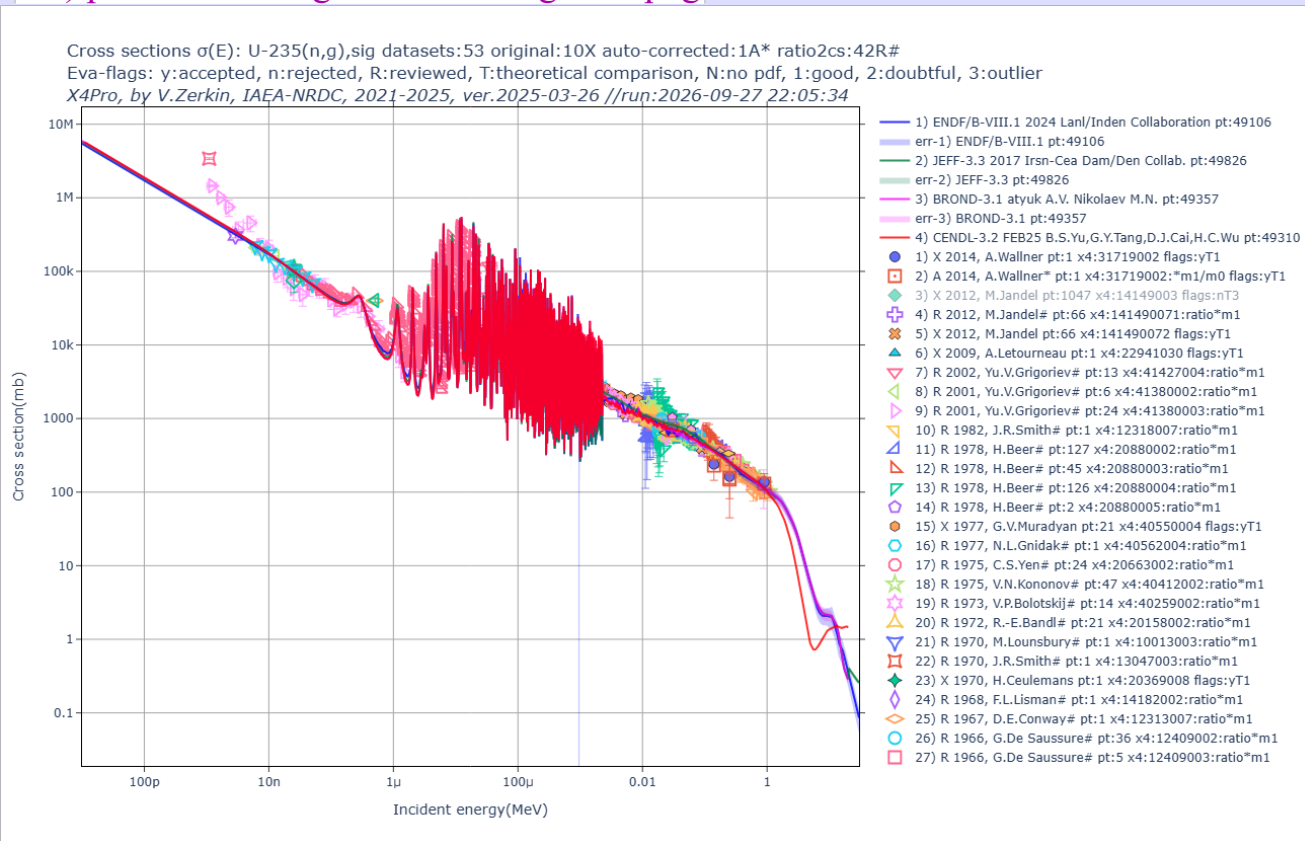
68) part6-1-eval2sig/out00/Mn55ng.html.png



69) part6-1-eval2sig/out00/Pu240nf-n.html.png



70) part6-1-eval2sig/out00/U235ng.html.png

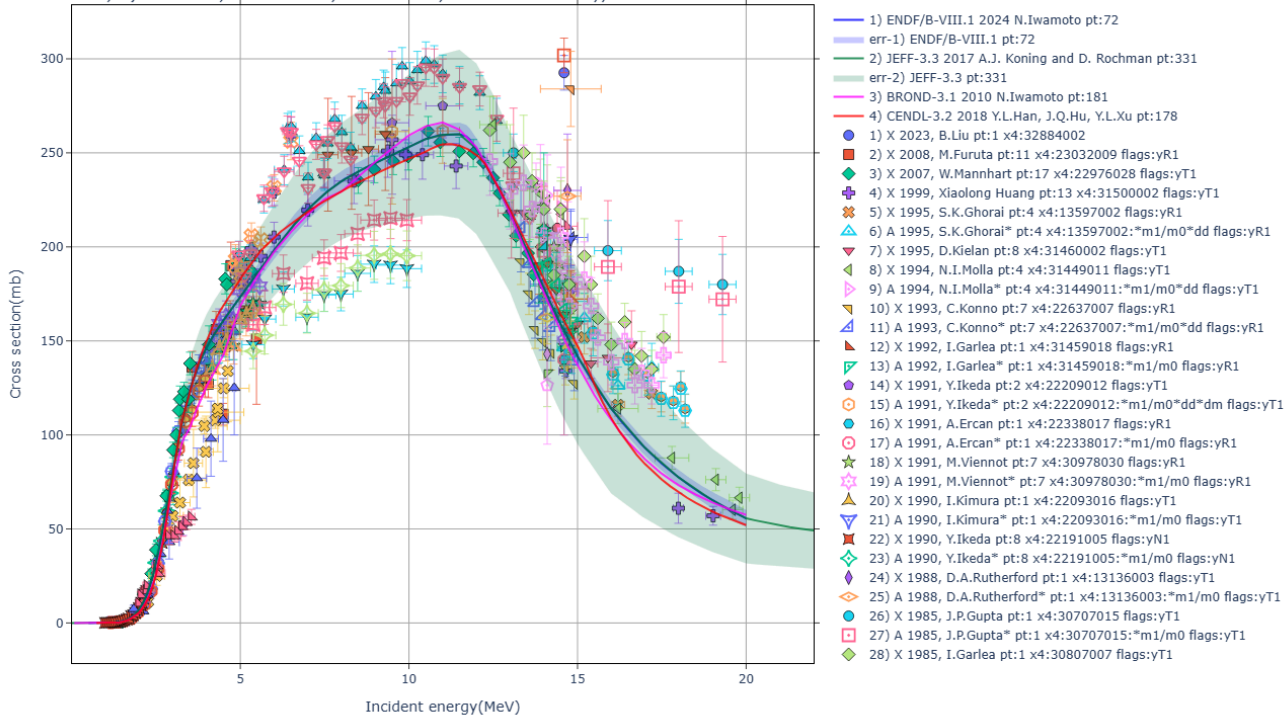


71) part6-1-eval2sig/out00/Zn64np.html.png

Cross sections $\sigma(E)$: Zn-64(n,p),sig datasets:87 original:51X auto-corrected:30A* ratio2cs:6R#

Eva-flags: y:accepted, n:rejected, R:reviewed, T:theoretical comparison, N:no pdf, 1:good, 2:doubtful, 3:outlier

X4Pro, by V.Zerkin, IAEA-NRDC, 2021-2025, ver.2025-03-26 //run:2026-09-27 22:04:48



72) myplot1.png Interactive plot: EXFOR:ETA + ENDF:MT452,18,102,MF1,3,31,33

X4Pro - advanced plot

by V.Zerkin, 2025-2026

